

Physics-Informed Neural Quantum States for Quantum Dots

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Computational Science: Physics
60 ECTS study points

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Abstract

A neural network used as a variational wavefunction is judged by its energy. When that energy stalls, or two networks reach it by different routes, the number cannot say why: the representation may be wrong for the problem, the optimiser may be failing, the samples that define the loss may no longer describe the state, or the physics may have become harder. This thesis tries to tell those explanations apart in the two-dimensional harmonic quantum dot, for $N \in \{2, 6, 12, 20\}$ electrons, from the compact droplet at $\omega = 1$ to the Wigner molecule at $\omega = 10^{-3}$.

The ansatz is a Slater determinant dressed by a neural Jastrow correlator, which reshapes the amplitude, and a coordinate backflow, which reshapes the nodes. Trained by residual pretraining and stochastic reconfiguration, it agrees to within 0.05% with published reference energies at the confinements where we compare with them, $\omega = 1, 0.5$ and 0.1 ; below that, for $N \geq 6$, its energies are predictions.

The correlator can be *separable*, encoding each particle and each pair independently before pooling them, or *message-passing*, letting the particles exchange information first. The implemented families reach the same energy but not the same state. The message-passing models carry a local-energy variance lower by a factor of 1.2–3.4 at every confinement, and their leading tangent directions are better described by a selected dictionary of collective operators. Because the learned features are not otherwise matched, these differences belong to the tested implementations rather than to message passing alone. Stochastic reconfiguration and Adam agree to four decimals on models whose metric has condition numbers of 10^7 – 10^{10} ; the failures suggest that natural-gradient preconditioning pays when the metric can be estimated from a batch, not merely when it is ill-conditioned.

On its own, the strong-form residual could not train the backflow in any tested variant and stalled near 0.3%, consistent with the nodal heavy tails derived in Appendix C. A weak form, the variational energy gradient estimated by importance sampling from an analytic proposal, keeps the Markov chain out of the update and has no such observed floor. With a proposal refitted during training it trains the six-electron dot, on every seed and across the whole range of confinement, to within 0.35% of the reference or, where there is none, of our own energies. The observed failures coincide with severe sample degeneracy: a fixed proposal falls to 1.5 effective samples of 4096 in the Wigner regime.

Read as instruments, the trained states resolve the crossover shell by shell. Radial shells form and stiffen across the whole range, and the six-electron ring approaches its classical radius to within 1.7%. Angular order does not keep pace: n -fold order within the shells stays at its random-angle value down to $\omega = 0.1$ and becomes large only at $\omega = 10^{-3}$, where the twelve-electron dot is predominantly a (3, 9) molecule and the twenty-electron dot a (1, 7, 12) one. Those measurements return to the solver. A proposal built from the measured shell geometry raises the effective sample size fifteen- to thirtyfold over the fixed one, and carries a cascade with MCMC-free updates to a stable six-electron checkpoint at $\omega = 0.0035$. Below that, the checkpoint does not survive the step to a weaker trap.

Every state is the lowest found at fixed spin projection, the architecture and optimiser comparisons rest mainly on $N=6$, and the benchmark references also guide pretraining. The archived backflow implementations are not matched in capacity, input scaling or centre-of-mass handling.

*To go wrong in one's own way is better than to go right in
someone else's.*

FYODOR DOSTOEVSKY
Crime and Punishment

Preface

*How easy it is
to look out across the sea
while you rest on the white sand.*

*Perhaps you never noticed,
but you swam so swiftly
that no one saw you leave.*

*The waves carried you
while you simply lay there.
You reached for air
as your vision faded.*

*You gasp to call out
and that was when your chest burned.*

When I came to UiO, Morten read my background and said: “You know PINNs, you know many-body quantum systems. I want you to do PINNs on quantum dots.” I said “sure.” I knew enough about each of those things separately to imagine that putting them together would mostly be a technical problem.

It was not.

The difficult part was rarely producing a result. The difficult part was knowing what a result meant. When something stopped improving, there were always several possible explanations. Perhaps the model was wrong for the problem, perhaps the optimisation was failing, perhaps the physics itself had become more difficult, or perhaps I had simply made a mistake somewhere along the way. Much of this thesis grew out of trying to tell those possibilities apart.

The trouble with having so many explanations was that, as one after another failed to account for what I was seeing, a much simpler one began to acquire weight: perhaps it simply could not be done. At first it was only another possibility. Gradually, it became difficult to see past it.

For a while, I stopped expecting the project to succeed.

I kept working, but for a different reason. If it could not be done, I wanted at least to understand why. In some ways, that became a more difficult problem than the original one. The worst periods were not those in which the method failed, but those in which I knew that it had failed and could not say what the failure meant.

I was fortunate to spend part of that period in Rome. The city’s ethereal quality lifted my spirits and lent the days a sense of something larger than the problem I happened to be working on. Yet it was my girlfriend Elida, more than anything, who gave me a steadiness I could not have supplied for myself. Her patience, warmth, and faith in me carried me through more of this work than I know how to put into words.

Eventually, some of the mystery began to give way. Not every obstacle turned out to be profound. Some were almost embarrassingly ordinary. Things I had taken for limitations of the method occasionally turned out to be limitations of my own implementation; other failures, once understood properly, pointed toward more interesting questions than the ones I had originally been asking.

That changed the way I looked at the project. A disappointing result was no longer merely something to overcome. Sometimes it contained information about the model, sometimes about the physics, and sometimes about the question itself. What began as an attempt to obtain a sufficiently good answer gradually became an attempt to understand what the answers, and the failures, were telling me.

I owe a warm and sincere thanks to my family, who have always made sure there was a home for me to return to, whenever I needed one. And to my friends, for countless conversations in which we tried to work out what we actually understood, and usually discovered that it was less than we thought.

To Morten, again, and properly: thank you for the assignment, for the two years we have had together, and above all for a genuine and undisguised interest in what I was finding at every stage, including the stages where what I found was that I had been wrong. That interest is most of what encouragement actually consists of.

When I look back, it is easy to make a catalogue of what I would now do differently. Most of that knowledge, however, was purchased by the failures and could not have been read off in advance.

I began this project thinking that the difficult part would be finding an answer to the many-body problem. More often, it was discovering which question a disappointing result was actually answering.

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Part I

Introduction

Introduction

A system of many interacting electrons is described completely by its wavefunction, and the equation that fixes it is known exactly; the entire difficulty is solving it. For N electrons in two dimensions the wavefunction is a function of $2N$ coordinates at once. Stored on a grid of M points per coordinate it would require M^{2N} numbers, which for twenty electrons and a modest ten points per axis is 10^{40} . The competition that makes this function hard to approximate is physical. Kinetic energy tends to delocalise the particles, while the interactions and the external potential drive localisation and correlation.

Established methods make one of two trades to remain tractable: they accept a cost that grows combinatorially, or they decide in advance which correlations may be described. Full configuration interaction makes the first. It is exact within a finite basis, but the number of determinants it requires grows so quickly with the basis and with N that a modest two-dimensional dot of twenty electrons is already far beyond reach. Hartree–Fock and coupled-cluster theory make the second. Hartree–Fock keeps only an average field, and coupled cluster truncates the excitations it includes, which works well when correlation is weak and less well when it is strong and several configurations contribute comparably. Fixed-node diffusion Monte Carlo makes the same kind of decision through the nodes of its trial wavefunction. It provides accurate benchmarks for continuum systems, but its accuracy is bounded by those nodes.

Variational Monte Carlo, the route this thesis takes, places the whole of that decision in the trial wavefunction. The energy of any trial wavefunction bounds the ground-state energy from above and can be written as an average of the local energy $\hat{H}\Psi/\Psi$ over configurations distributed as $|\Psi|^2$. The trial state therefore never has to be tabulated. It only has to be evaluated where the samples fall, and improving it becomes an optimisation problem. Because $|\Psi|^2$ is positive, the method has no sign problem; it accepts any trial wavefunction that can be evaluated; and its cost grows polynomially, as N^3 to evaluate a determinant-based ansatz and one further power of N for the Laplacian in the local energy. Its accuracy is set by the trial wavefunction, and whether that accuracy is reached depends on the optimiser and on the samples. In practice, then, much of the many-body problem reduces to a narrower one: choosing an expressive, well-conditioned ansatz, and optimising and sampling it reliably.

Quantum dots as a controlled testbed. Two-dimensional parabolic quantum dots are a clean and tunable place to study that narrower problem. As the trap frequency ω decreases, the single-particle level spacing collapses while Coulomb repulsion becomes dominant, and the electrons localise into rings and shells, so that a single system spans the range from Fermi-liquid-like behaviour to Wigner-molecule physics. In the lab frame, rotational symmetry and quantum fluctuations smear these patterns into smooth densities; in a co-rotating frame the crystalline structure reappears, with a topology that depends on particle number. For $N=6$ and $N=12$, classical and semiclassical studies have identified preferred shellings such as (1, 5) and (3, 9) and mapped the radial and orientational development of the Wigner regime [1–7].

Quantum dots suit this thesis for two reasons. They are genuinely non-trivial: Coulomb singularities, strong correlation and a real crossover appear already at moderate N . And they are simple enough that accurate DMC references exist in many regimes and that detailed structural diagnostics are computationally feasible. For fully quantum dots, quantitative, null-calibrated and topology-conditioned diagnostics remain less common, and it is not well understood how modern neural ansätze represent these states internally.

Neural wavefunctions. The narrower problem of choosing and optimising an ansatz is where machine learning enters. As a tool for representing complex, high-dimensional functions, it has taken two broad forms in physics. Neural networks act as *surrogates*, learning maps from parameters to observables or approximating the solution operators of partial differential equations, as physics-informed neural networks and operator learning do. Or they act as *variational ansätze* for quantum states, where the network outputs the amplitude of a many-body configuration and is trained by variational Monte Carlo. This thesis belongs to the second tradition and borrows a residual objective from the first: that objective serves as the pretraining stage of the variational method and, in weak form, as the collocation route with MCMC-free gradient updates in Chapter 7, a training method in its own right.

Carleo and Troyer introduced the modern neural-quantum-state formulation for lattice systems [8]. For continuum fermions the idea has since matured into highly accurate *ab initio* solvers such as FermiNet [9] and PauliNet [10], and into explicit neural *backflow* transformations [11], the direct ancestor of the coordinate backflow used here. Message passing has recently been built into neural quantum states for extended fermion systems [12, 13]. Against that backdrop the present work is deliberately narrow. It does not push a new architecture to ever larger systems. It asks why particular architectural and training choices succeed or fail on a problem where the answer can be checked, and what the trained state can then say about the physics. Accuracy is the precondition for that, not its point.

Why a neural ansatz. Universal approximation establishes representational capacity, not trainability. The theorems guarantee that a sufficiently large network can *represent* any suitably regular function to arbitrary accuracy; they say nothing about whether gradient-based optimisation will *find* that representation, nor at what sample cost, and it is the second question that limits practice. The case for a neural ansatz therefore cannot rest on expressiveness, and has to be made on cost. A variational neural ansatz makes neither of the two trades above: it requires no enumeration of a determinant space, and it fixes no prior truncation of which correlations may be described. It pays instead in optimisation and sampling, and its capacity can still have to grow with N or with the accuracy demanded (Chapter 7). What it gains is the freedom to concentrate capacity wherever the solution actually varies. Whether it does so is an empirical question, and this thesis treats it as one.

The sampling cost has two parts. Every sample needs the local energy, and with it a Laplacian of the network, however the samples are produced. In variational Monte Carlo the samples also come from a Markov chain: Metropolis steps are sequential, must equilibrate, and produce correlated configurations, and local moves mix slowly between well-separated regions of configuration space. That cost is highest in the Wigner regime, where the electrons localise into shell arrangements that the chain has to move between. Drawing configurations instead from an analytic proposal and reweighting them removes the chain, and with it the equilibration and the correlation between samples. It does not remove the Laplacian, and its price is the variance of the importance weights. Chapter 7 maps where that trade holds.

The problem also has three properties that make a neural ansatz difficult to train, and each shapes the ansatz used here. The Coulomb interaction diverges where two electrons meet, and the local energy diverges with it unless the wavefunction cancels the singularity, so the electron–electron cusp is built in analytically rather than learned. The kinetic energy is a second derivative, which amplifies any error in the curvature of the network, so the network’s inputs are smooth, scaled to the trap and soft at short range. At a generic simple node of an approximate state, the local energy has a pole; a strong-form objective that differentiates through it can therefore produce heavy-tailed gradients for backflow parameters that move the node. None of the variants tested in Appendix C closed the resulting accuracy gap, which motivated the weak-form route. Beyond these, the ansatz builds in antisymmetry and permutation symmetry and leaves the network to learn the remaining correlation. On top of a Slater determinant of harmonic-oscillator orbitals, two learnable objects carry that correlation: a Jastrow correlator, which reshapes the amplitude, and a coordinate backflow, which reshapes the nodes. Chapter 3 gives the construction.

Beyond the energy. A further choice concerns not *what* structure is imposed but *how* the network represents the electrons. Correlation is relational: what one electron should do depends on where the others are. Architectures that let particles exchange information are therefore natural candidates for both the correlator and the backflow. Each can be built in two forms. A separable network encodes per-particle and pairwise features and pools them in a single pass, as a DeepSet does; a *message-passing* network lets the particles exchange and update information through explicit edge states before a quantity is read out. In the implemented comparison the two correlator families reach the same energy to four figures, although their learned pair features are not otherwise identical. The energy therefore does not resolve their difference, which has to be sought in other quantities.

Such quantities are available because a trained ansatz can be inspected. Its physical structure is imposed, as in any method, but the correlations it develops within those constraints are not, and the trained network is a concrete object that can be examined directly, which the walker population of diffusion Monte Carlo is not. A twenty-electron dot is a function of forty coordinates, far too many to tabulate, but it does not follow that the solution itself is forty-dimensional in any useful sense. How many directions does a given parametrisation actually use to represent the solved state? That is a question about the solution and its representation, not about the equation, and the answer depends on how the state is parametrised (§2.2.3).

We make that question quantitative through the geometry of the wavefunction’s log-derivatives. Collecting the parameter sensitivities of a trained ansatz over an ensemble of configurations into a covariance matrix gives, in the simplest reading, a principal-component analysis of the state: the eigenvalue distribution counts how many directions are doing genuine work. The physically appropriate version of that covariance is the quantum geometric tensor, and the corresponding count is the participation ratio of its spectrum. The same tensor is the metric a natural-gradient optimiser inverts, and it is estimated from whatever samples define the loss, so it connects the architecture to the optimiser and to the sampling measure. Chapter 7 develops the instrument and applies it to all three.

Questions. When a variational calculation stalls, or two models reach the same energy by different routes, several explanations remain open: the representation may be wrong for the problem, the optimisation may be failing, the samples that define the loss may no longer describe the state, or the physics itself may have become harder. This thesis is organised as an attempt to separate those explanations. Its first three questions follow them; the fourth uses the resulting states for the physics they were meant to describe.

1. **Architecture.** How do the trained message-passing and separable implementations differ in the correlator and the backflow, and which differences survive state-level diagnostics?
2. **Optimiser.** When does natural-gradient (stochastic reconfiguration) preconditioning change the result, and when is it unnecessary or harmful?
3. **Paradigm.** Removing the Markov chain from the gradient makes the role of the proposal explicit. Can an accurate wavefunction be trained by importance-sampled collocation instead, and how far into the correlated regime does that reach before the samples stop describing the state?
4. **Physics.** Can the trained wavefunction serve as an instrument for Wigner-molecule physics (shell topologies, angular order, melting), and what does it show about the crossover?

The results take these in a different order, because a state has to be shown accurate enough to trust before either its physics or the solver that produced it can be interpreted. Chapter 6 presents the energy comparisons and reads the physics from the states; Chapter 7 turns to the solver.

The questions are answered with three kinds of measurement: the tangent geometry above, structural diagnostics of the trained state, which read shell topology and angular order against a random-angle baseline and a held-out polygon shell model, and sampling diagnostics for the MCMC-free update route. The structural diagnostics act only on configuration samples and can therefore be transferred to other many-body distributions; the tangent diagnostics also require derivatives of the parametrised state. Chapter 5 defines them.

Main contributions.

1. A compact Slater–Jastrow–backflow ansatz that agrees to within 0.05% with the published fixed-node references for $N \in \{2, 6, 12, 20\}$ at the confinements where we compare with them ($\omega = 1, 0.5, 0.1$), and extends smoothly into the Wigner regime, where its energies are predictions.
2. A comparison of trained correlator and backflow implementations beyond their energy. At equal energy the message-passing correlator has lower local-energy variance and a higher fit to a selected operator dictionary; under the parametrisations used here it also concentrates its Fisher weight in fewer effective directions. The backflow sweep shows a rank-one collapse in the legacy conventional path, but capacity, input scaling and centre-of-mass handling prevent an architecture-only interpretation.

3. A hypothesis for when natural-gradient preconditioning earns its cost. Poor conditioning alone does not decide it, and that much is measured. The failures suggest that the metric must also be estimable from the batch.
4. Training with MCMC-free gradient updates by importance-sampled collocation, mapped to its limits: within about a tenth of a percent of DMC at strong confinement and within 0.35% on every seed across the whole confinement range at $N=6$ with an adaptive proposal; a proposal built from the measured shell geometry that raises the effective sample size fifteen- to thirtyfold over a fixed one; and a cascade that reaches a stable checkpoint at $\omega = 0.0035$, below which it no longer survives the step to a weaker trap.
5. Topology-resolved diagnostics of the Wigner crossover, which quantify, shell by shell, how far radial stiffening runs ahead of angular order, complementing classical and quantum studies of parabolic dots [2–4, 6, 7, 14].

Thesis structure. This introduction forms Part I. Part II sets out the physics and machine-learning background (Chapters 1 and 2). Part III gives the ansatz, the optimisation pipeline and the diagnostics (Chapters 3 to 5). Part IV reports the results: Chapter 6 the energies and the Wigner-molecule structure, and Chapter 7 the analysis of architecture, optimisation and sampling together with the MCMC-free update programme. Part V discusses and concludes. The appendices in Part VI develop the conditioning theory, the collocation–backflow obstruction that motivated the weak-form objective, the full diagnostic tables and the reproducibility record.

Part II

Theory

Men do not know how what is at variance agrees with itself. It is an attunement of opposite tensions, like that of the bow and the lyre.

HERACLITUS
Fragments

Chapter 1

The many-body problem

The system is N electrons in a two-dimensional harmonic trap, repelling one another through the Coulomb interaction. Its Hamiltonian is elementary, and three of its properties make the numerical solution difficult.

The electrons are identical fermions, so the wavefunction changes sign under exchange, and that sign structure is the expensive part of the problem. The interaction diverges where two electrons meet, so the local energy $\hat{H}\Psi/\Psi$ diverges with it unless the wavefunction is constructed to cancel the divergence. And the kinetic term is a second derivative, which exposes any method that evaluates the energy pointwise to the curvature of the function it is given.

1.1 The many-body problem

A quantum state is a normalised vector in a complex Hilbert space, and in the coordinate representation $\psi(x) = \langle x|\psi \rangle$ is the wavefunction. Which basis one expands in is not a formality: choosing it to match the physics (harmonic-oscillator orbitals for confined charges, plane waves for periodic systems) is what makes a representation compact, and it is why the orbitals of Chapter 3 are those of the trap itself.

For N particles the space is the N -fold tensor product $\mathcal{H}_1^{\otimes N}$, and a fermionic state expands exactly in Slater determinants built from N distinct spin-orbitals, $|\Psi\rangle = \sum_I C_I |\Phi_I\rangle$. That expansion is exact and, as §1.2 makes precise, unaffordable.

We work in atomic units ($\hbar = m_e = e = 1$) with the many-body Hamiltonian

$$H = -\frac{1}{2} \sum_{i=1}^N \nabla_{\mathbf{r}_i}^2 + \sum_{i=1}^N V_{\text{ext}}(\mathbf{r}_i) + \sum_{i<j} V_{\text{int}}(|\mathbf{r}_i - \mathbf{r}_j|), \quad H\Psi(R) = E\Psi(R), \quad (1.1)$$

where $R = (\mathbf{r}_1, \dots, \mathbf{r}_N)$ and Ψ is normalised. Identical particles restrict Ψ to a symmetry sector, anti-symmetric for the fermions studied here. For these quantum dots $V_{\text{ext}}(\mathbf{r}) = \frac{1}{2}\omega^2 r^2$ and $V_{\text{int}}(r) = 1/r$ in two dimensions. Everything that follows approximates the lowest eigenpair of Eq. (1.1).

1.2 Correlation, closed shells, and the cost of exactness

Three ideas from electronic structure are used throughout this thesis and computed nowhere in it. They are what the accuracy claims of later chapters are calibrated against.

The first is *correlation energy*. Hartree–Fock approximates the ground state by a single determinant whose orbitals are themselves optimised, each electron moving in the mean field of the others; the correlation energy $\Delta E = E_{\text{exact}} - E_{\text{HF}}$ is whatever that cannot reach. It is a small fraction of the total

and essentially all of the difficulty, which is why a method can look converged in total energy while representing the correlation badly—a distinction Chapter 7 makes quantitative.

The second is *closed-shell structure*. When the occupied orbitals fill complete shells the determinant is a spin singlet and is unique, with no degenerate manifold to choose from. Those particle numbers, $N \in \{2, 6, 12, 20\}$ in a two-dimensional parabolic trap, are the ones studied here.

The third is *exactness within a basis*, and its price. Expanding in all N -electron determinants of a given one-particle basis and diagonalising gives the exact ground state within that basis; what cannot be avoided is the $\binom{M}{N}$ growth of the space itself, which for the twenty-electron dot puts the determinant count far beyond reach. That combinatorial wall is what motivates variational methods. The occupation-number formalism of second quantisation, in which antisymmetry is carried by anticommuting creation and annihilation operators, not by the wavefunction, is the usual language for all three; our own ansatz is first-quantised throughout, so it appears here as vocabulary rather than machinery.

Our determinant is built from harmonic-oscillator orbitals, not self-consistent Hartree–Fock ones, and our external benchmark is diffusion Monte Carlo [15], which for these systems is both more accurate than an affordable configuration-interaction calculation and available at the particle numbers we care about.

1.3 Quantum dots

1.3.1 Model and units

We consider N spin- $\frac{1}{2}$ fermions in $d=2$ with harmonic confinement and Coulomb repulsion,

$$H = \sum_{i=1}^N \left(-\frac{1}{2} \nabla_{\mathbf{r}_i}^2 + \frac{1}{2} \omega^2 r_i^2 \right) + \sum_{i < j} \frac{1}{|\mathbf{r}_i - \mathbf{r}_j|}, \quad (\hbar = m_e = e = 1).$$

Spin does not appear in H , so the eigenstates can be classified by S and S_z and the spatial problem solved at fixed spin assignment. For more than two identical fermions a spin-adapted eigenstate of S^2 is in general a sum of spatial–spin products, not a single one; what we use computationally is the weaker and sufficient statement that with the numbers of up- and down-spin electrons held fixed, the antisymmetriser factorises and the spatial problem may be treated on its own.

1.3.2 One-particle structure and shell filling

We use two-dimensional harmonic-oscillator spin-orbitals. In the implementation these are the *Cartesian* 2D-HO eigenstates $\{\phi_{n_x n_y}(\mathbf{r}) \chi_\sigma\}$ with energies $E_{n_x n_y} = \omega(n_x + n_y + 1)$ and spin $\sigma \in \{\uparrow, \downarrow\}$. The principal index $k = n_x + n_y$ labels a shell of $(k+1)$ degenerate spatial states, each twofold spin-degenerate; for the closed shells studied here this Cartesian basis spans the same filled subspace as any complete basis of those shells, so the reference determinant is basis-independent. Filling from the bottom yields the *closed-shell* electron numbers

$$N_{\text{closed}} = (K+1)(K+2), \quad K = 0, 1, 2, \dots \quad (2, 6, 12, 20, 30, \dots).$$

All other N correspond to a *partially filled* top shell (open shell), with a degenerate noninteracting manifold.

1.3.3 Spin structure and Slater form

Write single-particle spin-orbitals as $\varepsilon_\alpha(x) = \phi_\alpha(\mathbf{r}) \chi_\sigma(s)$, with $x = (\mathbf{r}, s)$. Because $\chi_\uparrow$ and $\chi_\downarrow$ are orthogonal and H is spin-independent, the Slater determinant simplifies:

Closed shells. For even N with doubly occupied spatial orbitals $\{\phi_1, \dots, \phi_{N/2}\}$, fixing the spin labels makes the spin-orbital determinant block diagonal, with one spatial determinant for each spin:

$$\Psi_{\text{closed}}(X) = \frac{1}{(N/2)!} \det[\phi_a(\mathbf{r}_i)]_{i \in \uparrow, a=1 \dots N/2} \det[\phi_a(\mathbf{r}_j)]_{j \in \downarrow, a=1 \dots N/2},$$

which is the computational ansatz used here. This fixes $S_z = 0$, but it does not explicitly project the interacting trial state onto a total-spin eigenstate. The expression has the *form* of a restricted closed-shell

determinant, but its orbitals are the harmonic-oscillator eigenstates of the trap, not orbitals optimised self-consistently in the mean field of the other electrons. It is therefore the closed-shell *non-interacting* reference, not an RHF one, and no self-consistent-field calculation is performed anywhere in this thesis. The distinction matters for how the later results should be read: everything the correlator and the backflow achieve is measured against a determinant that has not been given the benefit of a mean field, so what they recover includes the orbital relaxation an HF calculation would have supplied as well as the correlation beyond it.

1.3.4 Kato cusp conditions in two dimensions

For Coulomb interactions the potential diverges as $1/r$ at coalescence. Finiteness of the local energy

$$E_L(R) = \frac{(H\Psi)(R)}{\Psi(R)}$$

requires that the kinetic singularity cancels the $1/r$ divergence. Kato's theorem, first proved in [16], gives the necessary short-distance behaviour of the (spherically averaged) wavefunction. Let r_{ij} be the interparticle distance, $\mu_{ij} = m_i m_j / (m_i + m_j)$ the reduced mass (in a.u.), Z_i the charge number (electron $Z = -1$, nucleus $Z = +Z_A$), and D the spatial dimension. Then

$$\left. \frac{\partial}{\partial r_{ij}} \ln \bar{\Psi} \right|_{r_{ij} \rightarrow 0} = \frac{2\mu_{ij} Z_i Z_j}{D-1} \quad (\text{s-wave / opposite-spin channel}).$$

For *identical fermions* the s-wave is excluded; the leading channel is *p*-wave (relative angular momentum $k = 1$), and the wavefunction carries a linear node at coalescence. The cusp condition then applies to the *regular part* left after that node is factored out, and this distinction is easy to lose. Writing $\bar{\Psi}(r) = r \bar{u}(r)$ with $\bar{u}$ regular and nonzero at contact, the logarithmic derivative of the full wavefunction diverges, $\partial_r \ln \bar{\Psi} = 1/r + \partial_r \ln \bar{u}$, so it is $\bar{u}$ and not $\bar{\Psi}$ that obeys a finite cusp condition:

$$\left. \frac{\partial}{\partial r_{ij}} \ln \bar{u} \right|_{r_{ij} \rightarrow 0} = \frac{2\mu_{ij} Z_i Z_j}{2k + D - 1} = \frac{2\mu_{ij} Z_i Z_j}{D + 1} \quad (\text{like-spin, } k = 1).$$

In a Slater–Jastrow form the antisymmetric factor supplies the linear zero and the Jastrow supplies $\bar{u}$, so this is exactly a condition on the Jastrow slope—which is how it is imposed in Chapter 3.

Specializing to 2D electrons (atomic units). With $m_e = 1$ and $\mu_{ee} = 1/2$:

$$\boxed{a_{ee}^{\uparrow\downarrow} = \left. \frac{\partial}{\partial r_{ij}} \ln \bar{\Psi} \right|_0 = 1, \quad a_{ee}^{\uparrow\uparrow} = \left. \frac{\partial}{\partial r_{ij}} \ln \bar{u} \right|_0 = \frac{1}{3} .}$$

The two derivatives are taken on different objects: the full spherically averaged wavefunction in the antiparallel case, where it is finite at contact, and its regular part in the parallel case, where the antisymmetric node has been removed. The like-spin value follows from the *p*-wave (linear-node) analysis carried out in full in Appendix B.2, the canonical derivation used throughout this thesis; it gives the familiar $\frac{1}{4}$ in three dimensions and $\frac{1}{3}$ here in two. The 2D like-spin cusp is therefore *not* simply half the opposite-spin value—a coincidence that holds only in 3D—and the implemented value is $1/(D+1) = \frac{1}{3}$. For electron–nucleus coalescence ($\mu_{eA} \approx 1$, $Z_i Z_j = -Z_A$):

$$\boxed{a_{eA} = \left. \frac{\partial}{\partial r_{iA}} \ln \bar{\Psi} \right|_0 = -2Z_A .}$$

Why this matters. These conditions ensure the kinetic energy's r^{-1} singularity cancels the Coulomb r^{-1} term so that E_L remains finite as particles coalesce. Enforcing the cusp explicitly improves conditioning.

1.4 Structural diagnostics and Wigner markers

A Wigner molecule is not defined by a single number, so the crossover has to be read from several markers at once. This section says what each one measures and why it responds to Wigner order; the estimators, thresholds and stability protocol are defined once, in the Analysis chapter (§5.1), and the measurements themselves are in Chapter 6.

Radial localisation. The pair distribution $g(r)$ and the radial probability $P(r)$ say how far apart the electrons sit and how tightly that distance is fixed. The informative quantity is not the typical radius, which simply grows as the trap weakens, but the *ratio* of the radial width to it. A distribution that becomes relatively narrower around a growing radius is a shell stiffening; one that merely dilates is a droplet expanding. This is the finite-system analogue of a Lindemann criterion.

Shell topology. In a parabolic trap the classical minimum energy configuration is a set of concentric rings with definite occupancies—(1, 5) at $N=6$, (3, 9) at $N=12$ [3, 4]. Detecting those occupancies frame by frame turns a qualitative picture into a distribution over topologies, which is what makes it possible to say that a state is *mostly* (3, 9) with real weight on (2, 10)—a rotating molecule fluctuating between near-degenerate shellings rather than a single frozen crystal.

Orientational order. Radial structure alone cannot distinguish a ring of electrons at fixed angular spacing from a ring free to slide. The bond-orientational order parameter Φ_m measures m -fold angular order on a detected ring after removing global rotation, and an angular Lindemann number measures how stiff those spacings are. Both are standard in two-dimensional Coulomb and Yukawa systems [14, 17], and together with the radial markers they separate angular from radial melting—the two need not happen at the same ω .

Classical scaling. Finally, the strong-coupling limit makes a parameter-free prediction: inter-particle distances should approach $r \propto \omega^{-2/3}$. Fitting the measured exponent and watching it approach $-2/3$ from above is an external check on the whole picture, since nothing in the ansatz knows about that number.

How to read them together. As ω falls, the width ratios and angular Lindemann numbers shrink, shell gaps sharpen, the shell-resolved $|\Phi_m|$ rise above their finite-ring random-angle baseline, and the radial exponent approaches its classical value. No one of these is decisive—a narrow $P(r)$ can be an artefact of poor sampling, and a large $|\Phi_m|$ can be a fluctuation on a short chain—but they fail in different ways, which is why Chapter 6 reports them together and Chapter 5 states the stability checks each one is subjected to. Quantum Wigner markers in parabolic dots are reviewed in Refs. [2, 5, 14, 17].

1.5 Quantum Monte Carlo benchmarks

Two stochastic methods supply the numbers this thesis is measured against, and the difference between what they guarantee is the whole of its external evidence. Variational Monte Carlo optimises a trial state under the variational principle. Diffusion Monte Carlo projects toward the ground state in imaginary time. Only the first is what we do; only the second is what we are scored against; and neither is exact for fermions.

1.5.1 Variational Monte Carlo

For any normalised trial state $\Psi_T(\mathbf{R}; \boldsymbol{\alpha})$ the Rayleigh quotient

$$E[\Psi_T] = \frac{\langle \Psi_T | \hat{H} | \Psi_T \rangle}{\langle \Psi_T | \Psi_T \rangle} \geq E_0$$

bounds the exact ground-state energy from above, so minimising it over $\boldsymbol{\alpha}$ can only improve the estimate. Writing the expectation as an average of the *local energy*

$$E_L(\mathbf{R}) = \frac{\hat{H}\Psi_T(\mathbf{R})}{\Psi_T(\mathbf{R})} \tag{1.2}$$

over configurations drawn from $P(\mathbf{R}) = |\Psi_T(\mathbf{R})|^2 / \int |\Psi_T|^2$ turns the integral into a sampling problem,

$$E[\Psi_T] \approx \frac{1}{N_{\text{MC}}} \sum_i E_L(\mathbf{R}_i),$$

with the configurations generated by Metropolis–Hastings [18] at acceptance

$$A(\mathbf{R} \rightarrow \mathbf{R}') = \min\left(1, \frac{|\Psi_T(\mathbf{R}')|^2}{|\Psi_T(\mathbf{R})|^2} \frac{T(\mathbf{R}' \rightarrow \mathbf{R})}{T(\mathbf{R} \rightarrow \mathbf{R}')}\right),$$

optionally drifted by the quantum force $\mathbf{F} = 2\nabla \ln |\Psi_T|$ toward regions of high density [19].

The zero-variance principle. One property of Eq. (1.2) does more work in this thesis than the energy itself. If Ψ_T were an exact eigenstate, $\hat{H}\Psi_T = E\Psi_T$ pointwise, and E_L would be *constant* over the whole of configuration space. Its variance

$$\text{Var}(E_L) = \langle E_L^2 \rangle - \langle E_L \rangle^2 \quad (1.3)$$

therefore vanishes for an exact eigenstate. It is often a useful measure of wavefunction quality, but need not decrease monotonically with the variational energy. It is independent of any external reference—which matters here for two reasons. It keeps saying something after the energy has saturated, when two ansätze agree to four figures and the energy can no longer tell them apart. And it is defined on the state rather than on any component of it, so it survives changes of parameterisation that quantities such as a component’s rank do not. Chapter 7 uses it in both roles.

1.5.2 Diffusion Monte Carlo, and the fixed-node constraint

DMC reduces the amplitude error of a trial state by imaginary-time projection. In imaginary time $\tau = it$ the Schrödinger equation becomes a diffusion equation,

$$-\frac{\partial \Psi(\mathbf{R}, \tau)}{\partial \tau} = (\hat{H} - E_T) \Psi(\mathbf{R}, \tau),$$

whose long-time limit is the ground state, provided the initial state overlaps it. The evolution is simulated by an ensemble of walkers that diffuse, drift along the quantum force, and are branched or killed with weight $w = \exp[-(E_L(\mathbf{R}) - E_T)\Delta\tau]$. In the standard importance-sampled formulation, the stationary walker density is the mixed distribution proportional to $\Psi_T\Phi_0$, where Φ_0 is the projected state; it is not generally $|\Phi_0|^2$.

That argument works because a diffusion equation propagates a *probability*. For bosons the ground state is nodeless and can be treated as one directly. For fermions it cannot: antisymmetry forces Ψ to change sign, and a signed density is not a probability. Splitting it into positive and negative populations is formally possible but their difference is swamped by their sum at a rate exponential in $N\tau$ —the *fermion sign problem* [20].

The standard escape is to impose the sign structure instead of sampling it. The *fixed-node* approximation confines the walkers to the nodal pockets of a trial state Ψ_T , killing any walker that crosses a node. Within each pocket the wavefunction has a definite sign, the projection is well posed, and the method returns the exact ground-state energy *of the problem with those nodes imposed as a boundary condition*. That energy is a rigorous upper bound on E_0 , and it is exact only if the nodes are. Otherwise it carries a systematic *nodal bias* that no amount of sampling removes. The bound is a statement about the fixed-node eigenvalue in the limit of zero time step and infinite walker population; a published DMC number is an estimate of it and carries time-step, population-control and finite-statistics errors of its own, which is part of why comparing a variational energy against one at the fifth decimal place requires the matched uncertainty analysis we did not undertake (§8.1).

Three consequences follow. A published DMC energy is not ground truth; it is the best energy attainable inside the nodal surface of the trial state used to produce it. A variational state with better nodes than that surface may therefore lie *below* a fixed-node DMC number without violating anything, since the variational principle bounds the energy from below by E_0 and not by another approximation (§8.1). And improving nodes becomes the thing to attempt, which is what the coordinate backflow of Chapter 3 is for.

1.5.3 Requirements on the trial state

Both methods take a trial state as given. VMC is only as good as the state it optimises, and fixed-node DMC only as good as the nodes it is handed, so the quality of the ansatz stops being an implementation detail and becomes the object of study. Three requirements bear on it at once.

Antisymmetry has to hold exactly and by construction, since the sign structure is what makes the problem hard in the first place. In practice that means a determinant, multiplied by a symmetric correlation factor of the kind Jastrow introduced for strongly interacting systems [21]. The nodes have to be placed well, since that is where the remaining systematic error lives once the amplitude is right. And the state has to be smooth enough that $E_L = H\Psi/\Psi$, containing as it does a Laplacian and a potential that diverges at coalescence, can be evaluated without the two singularities amplifying one another. The cusp conditions of §1.3.4 say that those two divergences can be made to cancel exactly, and hard-wiring that cancellation is the first design decision of Chapter 3.

The remaining design choice is the functional form of the smooth correlation factor. The next chapter takes it from a literature that grew up asking different questions.

Chapter 2

Neural quantum states and variational geometry

2.1 The variational learning objective

Machine learning is usually introduced as *empirical risk minimisation*: an unknown joint distribution $P(X, Y)$ generates data, a loss ℓ scores predictions, and one minimises the empirical average $\hat{\mathcal{R}}(\theta) = \frac{1}{n} \sum_i \ell(f_\theta(x_i), y_i)$ over a finite sample as a proxy for the population risk $\mathcal{R}(\theta) = \mathbb{E}_P[\ell(f_\theta(X), Y)]$. Much of the standard machinery concerns the gap between the two: overfitting, the bias–variance tradeoff, regularisation, held-out validation [22, 23].

The variational problem considered here differs from that setting in two respects. There is no fixed labelled dataset, and so no conventional train/test generalisation problem. What replaces it is not the absence of a finite-sample difficulty but a different one: the objective is a population expectation that must be estimated afresh at every step, and an optimiser can certainly overreact to a particular finite batch—which is exactly what the effective-sample-size and weight-tail diagnostics of Chapter 7 measure. The objective is an expectation,

$$E[\Psi_\theta] = \mathbb{E}_{\mathbf{R} \sim |\Psi_\theta|^2} [E_L(\mathbf{R})], \quad (2.1)$$

under a distribution the model itself defines, re-estimated from fresh samples at every step. The optimisation target is the physical state rather than predictive performance on unseen data.

What replaces generalisation as the difficulty is already visible in the form of Eq. (2.1). Its measure $|\Psi_\theta|^2$ moves: it changes with every update, so the gradient, the samples used to estimate it, and the metric used to precondition it are all functions of the current parameters. In supervised learning the data distribution sits still and none of this arises. Meanwhile $E_L = \hat{H}\Psi_\theta/\Psi_\theta$ contains a Laplacian and a potential that diverges, so the objective reads the network’s second derivatives, not its values, and reads them near a singularity. Smoothness stops being a stylistic preference and becomes a requirement of the loss.

One term from the usual taxonomy recurs below. The objectives here are *unsupervised*: no labelled targets exist, and the loss comes entirely from the physics the network is asked to satisfy. The exception is the second phase of residual pretraining, where a published reference energy is used as an explicit target, and that is why those runs are later described as trained without a Markov chain rather than as unsupervised solutions of the Schrödinger equation. Reinforcement learning enters obliquely, through the REINFORCE estimator borrowed for the collocation gradient.

2.2 Optimisation

Whatever the objective, minimising it over a nonlinear, high-dimensional hypothesis class admits no closed form, so the parameters $\theta \in \mathbb{R}^p$ are updated iteratively according to the local geometry of the loss.

2.2.1 First-order methods

Gradient descent, $\theta^{(t+1)} = \theta^{(t)} - \eta \nabla_{\theta} \hat{\mathcal{R}}(\theta^{(t)})$, moves along the direction of steepest local decrease; too small a learning rate η converges slowly and too large a one diverges. Evaluating the full gradient is usually unaffordable, so it is estimated from a mini-batch. For independent, uniformly drawn batches this estimate is unbiased, and its variance sets the noise scale of the trajectory [24]. Momentum accumulates a running average of past gradients, damping oscillation across narrow directions and building displacement along consistent ones. Adam [25] adds a per-parameter normalisation by a running second moment,

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t, \quad v_t = \beta_2 v_{t-1} + (1 - \beta_2) g_t^2, \quad \theta^{(t+1)} = \theta^{(t)} - \eta \frac{\hat{m}_t}{\sqrt{\hat{v}_t + \epsilon}}, \quad (2.2)$$

with g_t the mini-batch gradient and $\hat{m}_t, \hat{v}_t$ bias-corrected. That normalisation is a diagonal rescaling of parameter space, and the question of whether a non-diagonal one is worth its cost is what §2.2.2 and, empirically, §7.2 are about. Adam is the default optimiser in this work [26].

2.2.2 Second-order methods and natural gradient

First-order methods only use gradient information. In principle, convergence can be accelerated by incorporating curvature via the Hessian $H(\theta) = \nabla_{\theta}^2 \hat{\mathcal{R}}(\theta)$. Newton's method updates parameters as

$$\theta^{(t+1)} = \theta^{(t)} - H(\theta^{(t)})^{-1} \nabla_{\theta} \hat{\mathcal{R}}(\theta^{(t)}). \quad (2.3)$$

However, computing and inverting the Hessian is usually infeasible for large models.

An attractive alternative is the *natural gradient* [27, 28], which replaces the Euclidean metric with the geometry of probability distributions. The key object is the *Fisher information matrix* (FIM),

$$F(\theta) = \mathbb{E}[\nabla_{\theta} \log p_{\theta}(x) \nabla_{\theta} \log p_{\theta}(x)^{\top}], \quad (2.4)$$

where p_{θ} denotes the model distribution. The natural gradient update is

$$\theta^{(t+1)} = \theta^{(t)} - \eta F(\theta^{(t)})^{-1} \nabla_{\theta} \hat{\mathcal{R}}(\theta^{(t)}). \quad (2.5)$$

This preconditioning makes updates invariant to reparameterizations of θ and often improves conditioning of the optimisation. In practice, computing F^{-1} exactly is intractable, but approximations (e.g. stochastic reconfiguration in variational Monte Carlo, Kronecker-factored approximations in deep learning) make the method viable [9].

2.2.3 The variational tangent space: geometric tensor, tangent kernel, and effective dimension

For a variational quantum state the Fisher metric takes a particularly concrete form, and that form is the analytical lens used throughout the results of this thesis. Write the per-sample log-derivatives, the *score*, of an ansatz Ψ_{θ} as

$$O_{ki} = \frac{\partial \log |\Psi_{\theta}(\mathbf{x}_k)|}{\partial \theta_i}, \quad k = 1, \dots, B, \quad i = 1, \dots, P, \quad (2.6)$$

for a batch of B configurations drawn from $|\Psi_{\theta}|^2$ and P parameters θ (here θ collects *all* variational parameters). Its centred Gram matrices define two dual objects. Throughout, O is taken *column-centred*, $O_{ki} \mapsto O_{ki} - \frac{1}{B} \sum_{k'} O_{k'i}$, which is what makes the Gram matrix a covariance rather than a second moment and is how it is formed in the implementation (§4.4). The *quantum geometric tensor* (QGT) is then

$$S = \frac{1}{B} O^{\top} O \in \mathbb{R}^{P \times P}, \quad (2.7)$$

the Fisher–Rao metric of the state written in terms of the amplitude score. Every state studied here is real and positive up to the determinant sign, at zero magnetic field, so the amplitude score is the whole story; for a general complex wavefunction the quantum geometric tensor also carries phase sensitivity, and the identification with the classical Fisher matrix is then only of its real part. It is proportional to the Fisher matrix $F(\theta)$ of the previous subsection specialised to $p_{\theta} = |\Psi_{\theta}|^2$: because $\nabla_{\theta} \log |\Psi_{\theta}|^2 = 2 \nabla_{\theta} \log |\Psi_{\theta}|$, one has $F = 4S$, the factor of four being a convention absorbed into the learning rate. Stochastic

reconfiguration is natural-gradient descent under this metric, $\dot{\theta} = -S^{-1}\nabla_{\theta}E$ [29, 30]. Its Gram dual, the *neural tangent kernel* (NTK) [31],

$$K = OO^{\top} \in \mathbb{R}^{B \times B}, \quad (2.8)$$

shares the nonzero spectrum of S up to the overall factor B (its nonzero eigenvalues are $B\lambda_a$), and is cheaper to diagonalise when $B < P$.

The analysis is organised around two scalars read from that spectrum. The *condition number* $\kappa(S) = \lambda_{\max}/\lambda_{\min}$ measures how anisotropic the tangent metric is. How far a damped natural-gradient step departs from a Euclidean step also depends on the gradient direction, regularisation and spectral cutoff. Chapter 7 shows that κ alone does not decide when SR is useful, which motivates the estimability hypothesis developed there. The *effective dimension*

$$d_{\text{eff}}(S) = \frac{(\sum_a \lambda_a)^2}{\sum_a \lambda_a^2} \quad (2.9)$$

is the participation ratio of the spectrum: it counts how many independent tangent directions carry appreciable Fisher weight locally around the solution, regardless of the raw parameter count P . A small d_{eff} means the local tangent sensitivity is concentrated in a few collective directions; a large one means it is spread across many. Because these are properties of the tangent geometry, not of the raw parameter count, they permit architectures, optimisers and training paradigms to be compared on a common footing—provided the comparison is made under matched conventions. In the comparisons of §7.1.1 the measured d_{eff} proved stable across seeds and across the choice of probe measure; that is an empirical finding at those settings, not a general property.

The participation ratio d_{eff} is parameterisation-dependent. It is computed from the QGT eigenvalues in the ordinary Euclidean parameter coordinates, and those eigenvalues are *not* invariant under an arbitrary reparameterization: rescaling a single parameter $\theta_1 \mapsto c\theta_1$ leaves the physical state untouched but changes d_{eff} (for a diagonal metric it moves from 2 toward 1 as $c \rightarrow \infty$). d_{eff} is therefore *not* a coordinate-free intrinsic dimension of the wavefunction; it is a property of the (architecture, parameterisation, normalisation) triple. We use it accordingly—as a comparison between fixed architectures under matched conventions and a common probe measure—and read a smaller d_{eff} as “this architecture concentrates its Fisher weight into fewer active directions,” not as an absolute manifold dimension.

2.2.4 Tangent-space diagnostics

Convergence speed is not what matters about the optimiser here. A natural-gradient step is only as good as the metric it inverts, so the estimator of S becomes part of the method rather than an implementation detail, and Chapter 7 turns that into a three-regime result. Separately, $\kappa(S)$ and d_{eff} can be read off a trained state, which makes them diagnostics as much as convergence aids, and they are used that way from Chapter 6 onward.

2.3 Smooth networks for Laplacian objectives

A feed-forward network composes affine maps with elementwise nonlinearities,

$$h^{(0)} = x, \quad h^{(\ell)} = \sigma(W^{(\ell)}h^{(\ell-1)} + b^{(\ell)}), \quad f_{\theta}(x) = W^{(L)}h^{(L-1)} + b^{(L)},$$

and without the nonlinearity σ the composition collapses to a single affine map. For most applications the choice of σ is settled by convention or by a sweep. Here the objective settles it, because the local energy $E_L = \hat{H}\Psi_{\theta}/\Psi_{\theta}$ contains a Laplacian and training therefore backpropagates through second derivatives of the network.

The activation. $\text{ReLU}(x) = \max(0, x)$ has a second derivative that vanishes almost everywhere and is distributional at its kinks, so it is unsuitable for the pointwise autograd Laplacian used here. Less obviously, being C^{∞} does not by itself determine how useful an activation is for a Laplacian objective. What matters heuristically is also how its higher derivatives are distributed over the input range. When the second and third derivatives are sharply localised, curvature information can become concentrated in narrow regions of a deep composition. Tanh and the logistic sigmoid are perfectly smooth, yet their higher derivatives are sharply peaked and change sign. We use $\text{SiLU}(x) = x \sigma(x)$, whose higher derivatives are

distributed more broadly. Figure 2.1 compares several smooth alternatives, including GELU and Mish. This is a design heuristic rather than a theorem: the figure establishes the shapes of the derivatives, not a bound on the Laplacian variance of a deep composition, and we did not run an activation ablation.

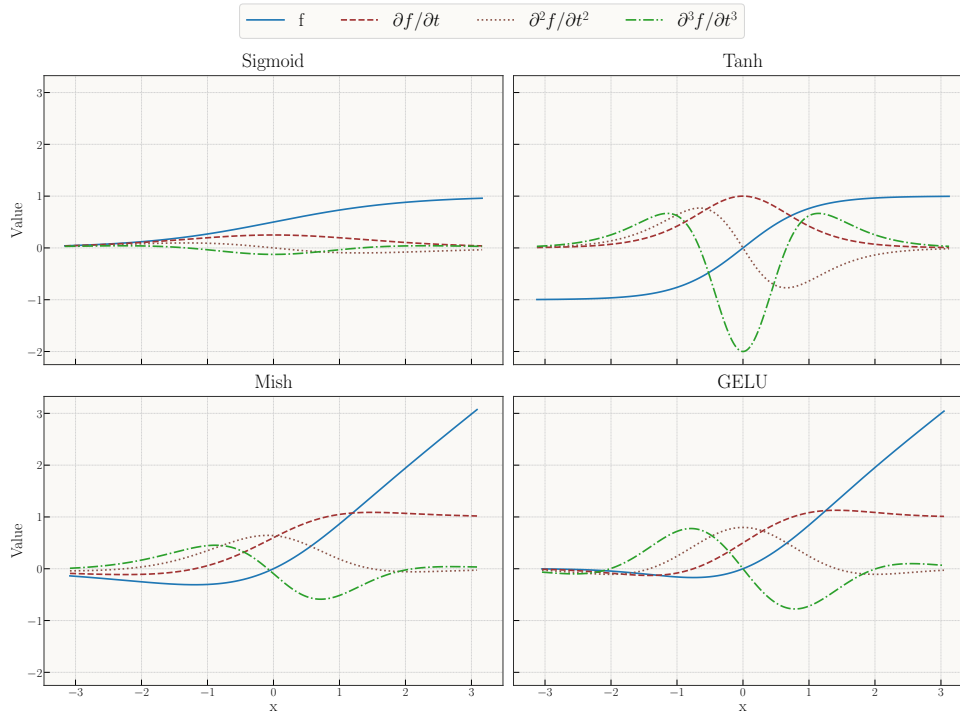


Figure 2.1: Common activation functions and their first three derivatives. All of GELU, Mish, tanh and the logistic sigmoid are C^∞ , so the distinction is not smoothness but the *reach* of the higher derivatives: for GELU and Mish the second and third derivatives are broad and decay slowly to zero, while for tanh and sigmoid they are sharply peaked near the origin and change sign, so curvature accumulates unevenly when such units are composed through depth. ReLU and SiLU are not shown; SiLU is the activation used in the reported models.

Initialisation. The standard schemes (Xavier/Glorot, scaling weight variance as $1/(n_{\text{in}} + n_{\text{out}})$, and Kaiming/He, scaling as $2/n_{\text{in}}$ for rectified units) are designed to keep forward activations and backward gradients from shrinking or exploding through depth [32, 33]. That is necessary but not sufficient when the loss contains second derivatives, because a scheme that stabilises $\partial f/\partial x$ says nothing directly about $\partial^2 f/\partial x^2$. We use Xavier-style initialisation with the smooth activations above, and—more consequentially than any initialisation choice—zero-initialise the final layer of the backflow, so that the ansatz begins exactly at the Slater nodes and the learned nodal deformation grows from nothing rather than from noise (§3.5.3).

None of this makes the Laplacian well conditioned; it only stops the network from making it worse. The conditioning itself comes from the coordinate scaling, the soft-core pair features and the analytic cusp of Chapter 3, and is analysed in Appendices A and B.

2.4 Architectures for many-body wavefunctions

The feed-forward network of the previous section maps a fixed-size input vector to an output. A many-body wavefunction is not a generic function of such a vector: identical particles impose a permutation symmetry, and correlation is intrinsically *relational*—what one particle should do depends on where the others are. A plain multilayer perceptron acting on the concatenated coordinates $(\mathbf{x}_1, \dots, \mathbf{x}_N)$ respects neither property; it would have to learn the symmetry from data, and its parameter count would grow with N . This section introduces the two architecture classes that build the required structure in, permutation-

invariant pooling and message passing, and situates the specific message-passing network used in this thesis as one instance of the latter.

2.4.1 Permutation symmetry and equivariance

For identical particles the modulus $|\Psi|$ is invariant under exchange (the sign is carried by the antisymmetric Slater reference of Chapter 3). The two learnable objects of the ansatz therefore have distinct symmetry requirements. A scalar *correlator* f must be permutation-*invariant*,

$$f(P_\sigma R) = f(R),$$

while a per-particle *backflow* displacement field $\mathbf{g} = (\mathbf{g}_1, \dots, \mathbf{g}_N)$ must be permutation-*equivariant*,

$$\mathbf{g}(P_\sigma R) = P_\sigma \mathbf{g}(R),$$

for any permutation σ (here P_σ relabels the particles). Enforcing these by construction is both more data-efficient than learning them and yields a parameter count independent of N , because the same shared maps are applied to every particle and pair.

2.4.2 Permutation-invariant pooling: DeepSets

The simplest way to build an invariant function is to embed each particle independently and pool the embeddings symmetrically. This is the DeepSets construction [34]: any suitably regular permutation-invariant function admits the form

$$f(R) = \rho\left(\sum_i \phi(\mathbf{x}_i)\right),$$

with a per-particle embedding ϕ , a symmetric pool $\sum_i$, and a readout ρ . A two-body extension encodes pairs, $\phi(\mathbf{x}_i, \mathbf{x}_j)$, and pools over them, capturing pairwise correlation. What characterises this class is that each particle, or pair, is embedded *independently* and only then pooled: no information is exchanged between particles before the pool. We refer to such networks as *separable*, and they serve as the non-message-passing baseline for the correlator.

2.4.3 Message passing and graph networks

To let a particle’s representation depend on the others’ in an iterated, higher-order way, one places the particles on a graph—nodes are particles, edges are pairs—and passes learned messages along the edges [35, 36]. In a generic round ℓ , each edge gathers information from its endpoint nodes and each node aggregates its incoming edge messages,

$$\mathbf{m}_{ij}^{(\ell)} = \phi_e(\mathbf{h}_i^{(\ell)}, \mathbf{h}_j^{(\ell)}, \mathbf{e}_{ij}^{(\ell)}), \quad \mathbf{h}_i^{(\ell+1)} = \phi_v(\mathbf{h}_i^{(\ell)}, \text{Agg}_{j \neq i} \mathbf{m}_{ij}^{(\ell)}),$$

with shared maps ϕ_e, ϕ_v and a symmetric aggregation Agg , so that permutation equivariance holds by construction; a final readout produces an invariant scalar (correlator) or an equivariant field (backflow). Both DeepSets and message passing use pairwise information, but only message passing lets that information *propagate*: after several rounds a particle’s representation reflects not just its neighbours but their neighbours’ responses. This iterated propagation is what we call the *relational channel*, and asking what it buys, in the correlator and in the backflow, is one of the central questions of this thesis. Message passing has been used in exactly this spirit for neural quantum states of extended fermion systems [12, 13].

2.4.4 The CTNN: one message-passing instance

The message-passing correlator and backflow studied here are realised by a particular network we refer to as the *CTNN*: a copresheaf-style message-passing network that maintains explicit *node and edge* features and transports information between them through learned maps, rather than recomputing and discarding one-shot messages. The correlator iterates this exchange over a down/up “V-cycle”; the backflow applies a single such copresheaf round. Its concrete update equations, hidden dimensions, and parameter counts are specified in the Methods chapter (§3.5.2, Eqs. (3.13)–(3.15)). Two things follow. First, the CTNN is *one* concrete realisation of the message-passing class defined above, and the comparison run in this thesis is between two specific inductive biases—the CTNN against a DeepSets-type separable baseline—not between the classes they belong to. Conclusions are stated for message passing *as instantiated by the*

CTNN, relative to the separable baseline studied here; whether they generalise to other message-passing neural quantum states is a plausible extrapolation and not something these experiments test. Wherever we contrast a “message-passing” network with a “separable” one, the separable member is a DeepSets-type network and the message-passing member is the CTNN. Second, we write “copresheaf-style” deliberately: the name reflects the node/edge transport structure of the updates, not a claim to a formal sheaf-theoretic construction.

2.4.5 Relevance to the ansatz

Both learnable objects of the ansatz—the Jastrow correlator and the coordinate backflow—can be built in either form: as a separable DeepSets-type network, or as the message-passing CTNN. Chapter 7 compares these implementations through the geometry of the tangent kernel introduced in §2.2.3. Because the archived models also differ in capacity, features and coordinate handling, the comparison is descriptive rather than an isolated test of the relational channel.

2.5 Approximation theory and its limits

The usual justification for using a neural network as a variational ansatz is that it can represent anything. That argument needs disposing of early, since this thesis is about what happens after it stops being relevant.

The universal approximation theorems say that a single hidden layer with a non-polynomial activation is dense in $C(K)$ on compacta: for any continuous f and any $\varepsilon > 0$ there exist a width and a parameter vector achieving $\sup_{x \in K} |f(x) - f_\theta(x)| < \varepsilon$ [37–39]. They establish one thing, best put negatively: there is no *fundamental* representability obstruction. They say nothing about the particular 20k–236k networks used here, nor about whether gradient-based training will find such a representation, at what sample cost, or how the answer depends on the parameterisation. The existence proofs are non-constructive, the widths they guarantee are astronomical, and the same function can be represented by two networks whose optimisation behaviour is entirely different—which is, in miniature, the comparison Chapter 7 makes.

One further result does more than assert possibility. In high dimension a generic continuous target requires a network size growing exponentially in d , so approximation alone cannot explain why these methods work at all in Nd dimensions. The escapes are all statements about *structure*, not capacity: targets of finite Barron norm admit dimension-independent rates [40]; compositional targets admit efficient representation by deep networks when the composition matches the architecture; and imposing a symmetry removes the redundant copies the network would otherwise have to learn, reducing the burden on the representation rather than the dimension of the domain. The last of these is the one we exploit directly, and the middle one is why the message-passing architectures of §2.4 deserve attention: their usefulness comes from a compositional form matching the relational structure of the physics, not from added capacity.

The rest of the thesis therefore assumes representability and measures the other things: trainability, conditioning, and the geometry of the parameterisation.

2.6 Residual objectives and their difficulties

In scientific applications the governing equations already encode most of what is known about a system, and a model fitted only to data throws that information away. *Physics-informed neural networks* (PINNs) [41] put it back by penalising the residual of the equation itself: for a differential operator $\mathcal{N}$ and a network u_θ , the loss is a sampled sum of $|\mathcal{N}[u_\theta](\xi_j)|^2$ over collocation points $\{\xi_j\}$, together with boundary and data terms. Training a network against a differential operator predates the name by two decades [42]. Raissi, Perdikaris and Karniadakis consolidated and popularised the modern PINN formulation in the automatic-differentiation era [41], in which derivatives such as ∇u_θ and Δu_θ can be evaluated directly and used to train on the residual.

Four difficulties follow from differentiating through the operator, and each recurs by name in the results of this thesis [43].

Approximation in Sobolev norms. Universal approximation controls function error in L^2 or L^∞ , but a strong residual depends on *derivatives*. Making it small requires approximation in $W^{\ell,p}$ with ℓ the order of the equation, which forces smooth activations for $\ell \geq 2$ and architecture choices that control higher derivatives. This is why the activation here is chosen for its second derivative, not its values.

Residual-to-error stability. A small residual implies a small solution error only where the forward problem is stable. This is why a small residual is never taken here as evidence of a good state without an independent energy evaluation.

Sampling. Collocation minimises a sampled surrogate of a continuum loss, and the empirical residual can be small while the population residual is not. In this setting the question becomes whether an importance-weighted estimator has a finite variance at all, which is what bounds the MCMC-free update programme of Chapter 7.

Conditioning. Residual losses couple terms of disparate scale and produce ill-conditioned gradient dynamics; in the linearised view, convergence is governed by the spectrum of a problem-dependent kernel. That spectrum is the subject of the entire tangent-kernel analysis.

The last of these determined the design of the collocation programme. Near a generic simple node, E_L has a pole, and differentiating a strong residual through it produces a heavy-tailed gradient. The tested strong-form implementations stalled at the accuracy floor documented in Appendix C. The weak form instead keeps the local energy as a detached reward, which removes that derivative and proved substantially more stable.

2.7 Constraints carried into the method

The two traditions of this chapter and the last have been describing the same objects in different words. The quantum geometric tensor of variational Monte Carlo is the Fisher metric of information geometry; stochastic reconfiguration is natural-gradient descent under it; the neural tangent kernel is that metric's Gram dual; and the sampling measure a physicist chooses for a Metropolis chain is what a statistician calls a proposal distribution. Neither community needs the other's vocabulary in order to do its own work, which is why the identifications are easy to miss. They matter here because the architecture determines the spectrum of that metric, the optimiser inverts it, and the sampling measure determines how well it can be estimated.

Four properties of this problem bear on that metric.

The objective contains a second-order operator. The local energy carries a Laplacian, so training differentiates through second derivatives of the network. This is what dictates smooth activations, controlled initialisation, and a determinant that is never asked to do anything violent (§2.3).

The interaction is singular, and the state carries nodes. The Coulomb term diverges at coalescence, and the trial function vanishes on a surface whose location is itself being learned. Both are places where $E_L = \hat{H}\Psi/\Psi$ can blow up, and the cusp conditions of §1.3.4 say that the first divergence can be cancelled analytically if the short-range form is built in, not learned.

The measure is part of the method. The objective is an expectation under $|\Psi_\theta|^2$, so the sampling distribution enters the gradient, the metric, and the estimate of both. Whether that metric can be estimated at all turns out to decide whether preconditioning helps or hurts (§2.2.3).

Correlation is relational. What one electron should do depends on where the others are, and the two architecture classes of §2.4 differ precisely in how far that dependence is allowed to propagate.

The first two are met by construction in Chapter 3, which hard-wires antisymmetry and the cusp and conditions the geometry so that the learned part can remain small and smooth. The third is treated in Chapter 4, where the sampling measure is a design variable rather than a fixed background. The fourth is not settled here: both architectures are constructed, and the comparison between them is reported in Chapter 7.

Part III

Methods

*No great thing is created suddenly, any more than a
bunch of grapes or a fig.*

EPICTETUS
Discourses

Chapter 3

The trial wavefunction

The trial state is constructed to satisfy fermionic antisymmetry and the electron–electron cusp analytically, and to keep the learned terms smooth under the Laplacian. Antisymmetry requires a determinant. The Coulomb divergence admits an analytic cancellation, so it is imposed, not learned. Second derivatives enter the loss, so the features fed to the network are bounded and smooth and the coordinates are scaled to keep those derivatives $\mathcal{O}(1)$ across three decades of confinement. The form of the learned part is not fixed by these constraints; both candidate architectures are constructed here and compared in Chapter 7. Unless stated otherwise we work in atomic units.

3.1 Preliminaries

Hamiltonian and units. We study N spin- $\frac{1}{2}$ fermions in two spatial dimensions confined by a harmonic trap and interacting via Coulomb repulsion,

$$H = \sum_{i=1}^N \left(-\frac{1}{2} \nabla_{\mathbf{r}_i}^2 + \frac{1}{2} \omega^2 r_i^2 \right) + \sum_{1 \leq i < j \leq N} \frac{1}{|\mathbf{r}_i - \mathbf{r}_j|}, \quad (\hbar = m_e = e = 1). \quad (3.1)$$

Spin does not appear in H , so eigenstates may be classified by L_z , S and S_z . The calculations below fix the populations in the two spin sectors, and hence S_z , without an explicit projection onto total spin S .

Coordinates and scaling. We collect positions as $R = (\mathbf{r}_1, \dots, \mathbf{r}_N) \in \mathbb{R}^{2N}$ and use trap scaling $\tilde{\mathbf{r}}_i = \sqrt{\omega} \mathbf{r}_i$, $\tilde{R} = \sqrt{\omega} R$. This aligns typical length scales to $\mathcal{O}(1)$, improves conditioning of second derivatives, and makes the noninteracting ground state ω -independent. When useful, we also use pair distances $r_{ij} = |\mathbf{r}_i - \mathbf{r}_j|$ and their scaled versions $\tilde{r}_{ij} = |\tilde{\mathbf{r}}_i - \tilde{\mathbf{r}}_j|$.

Sectors and focus. We primarily report *closed-shell* cases (unique spin singlet at the noninteracting level). Open shells introduce a degenerate manifold and additional choices for the reference; we keep this to brief remarks only where needed.

Spin-orbitals and basis. One-particle spatial orbitals are the Cartesian two-dimensional harmonic-oscillator eigenstates $\{\phi_{n_x n_y}(\mathbf{r})\}$ with $E_{n_x n_y} = \omega(n_x + n_y + 1)$, $n_x, n_y \in \mathbb{N}_0$ (the basis used in the implementation; for the closed shells studied it spans the same filled shells as any complete basis of those shells). Spin-orbitals are $\varepsilon_\alpha(x) = \phi_{n_x n_y}(\mathbf{r}) \chi_\sigma(s)$, $\sigma \in \{\uparrow, \downarrow\}$.

3.2 Overview of the ansatz

Our trial state combines exact antisymmetry, an analytic short-range cusp, and a compact neural residual, while optionally evaluating the Slater determinant on backflowed coordinates:

$$\Psi_{\theta,\beta}(R) = \text{SD}_\omega(R') \exp\left(\sum_{i<j} u_{\sigma_i\sigma_j}(r_{ij}) + W_\theta(R)\right), \quad (3.2)$$

$$R' = R + \Delta_\beta(R). \quad (3.3)$$

Here SD_ω evaluates the harmonic-oscillator determinant at physical coordinates, u hard-wires the Kato cusp, W_θ learns smooth medium- and long-range structure, and Δ_β adjusts the nodes. The modules receive R and perform any trap scaling internally; the correlator is evaluated at the original coordinates, not at R' . Per-particle displacements $\Delta_{\beta,i}$ are written $\Delta\mathbf{x}_i$ in the results chapters.

Design principles. (i) *Separate guaranteed physics from learned details:* antisymmetry and the cusp are hard-wired; the network learns the rest. (ii) *Condition the geometry:* use trap scaling where stated and soft-core pair features so that first/second derivatives are bounded. (iii) *Keep the learned object small and smooth:* once the cusp is explicit, a compact NQS suffices across (N, ω) .

3.3 Slater reference

For closed shells we use a restricted product of spin-block determinants of the lowest $N/2$ harmonic oscillator orbitals (trap units):

$$\text{SD}(\tilde{R}) = \det[\phi_a(\tilde{\mathbf{r}}_i)]_{i \in \uparrow, a \leq N/2} \det[\phi_a(\tilde{\mathbf{r}}_j)]_{j \in \downarrow, a \leq N/2}. \quad (3.4)$$

The correlator reweights amplitudes *within* nodal pockets; without backflow, the nodes are those of (3.4).

3.4 Neural correlator W_θ

As with the backflow, the correlator comes in two forms that are compared in §7.1: a *separable* network that encodes particle and pair features independently before pooling, and a copresheaf *message-passing* (CTNN) network. Both include the same analytic cusp, but their learned pair features are not identical; the comparison is therefore between the implemented model families, not a one-variable ablation.

3.4.1 Separable correlator (DeepSets baseline)

The separable NQS aggregates *particlewise* and *pairwise* embeddings with permutation-invariant pooling and appends two global statistics, all in $\tilde{R}$. A core safety requirement is that pair channels satisfy $ds_k/d\tilde{r} \rightarrow 0$ as $\tilde{r} \rightarrow 0$ so that W_θ never injects spurious $1/\tilde{r}$ terms into $\nabla \log |\Psi|$ or $\Delta \log |\Psi|$.

Particle branch (DeepSets).

This part of the network is typically referred to as Deep Sets [34]. The permutation invariance of quantum many-body mechanics is naturally captured by this architecture. A neural network ϕ creates a latent embedding for each particle individually. All embeddings are pooled together to form a global representation, which is used later in the readout. Formally, we have

$$\mathbf{h}_i^\phi = \phi(\tilde{\mathbf{x}}_i) \in \mathbb{R}^{d_L}, \quad \bar{\mathbf{h}}^\phi = N^{-1} \sum_i \mathbf{h}_i^\phi \in \mathbb{R}^{d_L}.$$

Pair branch (soft-core, even channels).

Although the particle branch could theoretically produce pair correlations, in practice it struggles to represent sharp near-field structure. We therefore add a dedicated pair branch that builds features from interparticle distances. This network processes all pairs $(i, j), i \neq j$ individually and pools the resulting latent embeddings. The Euclidean norm is nondifferentiable at exact coalescence and its Hessian is singular there, which destabilises Laplacian evaluations. We therefore build *safe* pair features whose first and second radial derivatives remain bounded as $\tilde{r} \rightarrow 0$. With $\tilde{r}^{\text{soft}} = \sqrt{\tilde{r}^2 + \varepsilon^2}$ ($\varepsilon \in [0.15, 0.30]$ in trap units), we build six “safe” features

$$\begin{aligned} s_1 &= \log(1 + (\tilde{r}^{\text{soft}}/\varepsilon)^2), \quad s_2 = \frac{\tilde{r}^2}{\tilde{r}^2 + \varepsilon^2}, \quad s_3 = (\tilde{r}^{\text{soft}}/\varepsilon)^2 \exp[-(\tilde{r}^{\text{soft}}/\varepsilon)^2], \\ \text{rbf}_g &= \exp(-g s_1), \quad g \in \{0.25, 1, 4\}, \quad \mathbf{u}_{ij} = [s_1, s_2, s_3, \text{rbf}_{0.25}, \text{rbf}_1, \text{rbf}_4] \in \mathbb{R}^6. \end{aligned} \quad (3.5)$$

An MLP ψ maps $\mathbf{u}_{ij} \mapsto \mathbf{h}_{ij}^\psi \in \mathbb{R}^{d_L}$. A short-range gate

$$\chi(\tilde{r}) = \frac{\tilde{r}^2}{\tilde{r}^2 + r_g^2}, \quad r_g \approx 0.3, \quad \chi(0) = \chi'(0) = 0,$$

suppresses learned responses *at* coalescence; downstream we use $\chi \mathbf{h}_{ij}^\psi$.

Equations (3.5) describe the intended dimensionless construction. In the archived separable implementation, however, the distance is first converted to trap units and the feature and gate scales are then multiplied by a_{ho} . The effective feature scale therefore changes with ω . The reported sweep retains this implementation and is interpreted with that limitation in Chapter 7.

3.4.2 Analytic short-range cusp (shared)

The analytic cusp below is shared by both correlator variants (and is independent of the backflow). Because the safe features are necessary in the pair branch, the NQS cannot represent the Kato cusp condition exactly. We therefore add an analytic cusp term to the exponent in (3.2), written in *physical* coordinates so that its slope at contact is manifestly correct:

$$u_{\sigma\sigma'}(r) = \gamma_{\sigma\sigma'} r e^{-r/a_{\text{ho}}}, \quad a_{\text{ho}} = 1/\sqrt{\omega}, \quad \gamma_{\uparrow\downarrow} = \frac{1}{d-1}, \quad \gamma_{\uparrow\uparrow} = \frac{1}{d+1}. \quad (3.6)$$

Because the linear prefactor is the physical separation r , the slope at contact is $\partial_r u|_0 = \gamma_{\sigma\sigma'}$, exactly the Kato condition in general d ; for $d = 2$ this gives $\gamma_{\uparrow\downarrow} = 1$ and $\gamma_{\uparrow\uparrow} = \frac{1}{3}$, the standard two-dimensional quantum-dot values, derived—including the parallel-spin case, where antisymmetry already forces a linear node—in Appendix B.2. Only the exponential taper carries the trap length a_{ho} , suppressing long-range interference so the NQS need not unlearn cusp structure at medium and long range; in trap units ($\tilde{r} = \sqrt{\omega} r$) the same term reads $u = (\gamma_{\sigma\sigma'}/\sqrt{\omega}) \tilde{r} e^{-\tilde{r}}$.

3.4.3 Separable readout

With contributions from the particle and pair branches, we form a global representation for the final readout. In addition to the pooled embeddings $\bar{\mathbf{h}}^\phi$ and $\bar{\mathbf{h}}^\psi$, we also include two global statistics that help the network gauge overall system size and density. We average pairs by default or apply degree-normalised radial attention

$$w_{ij} = \exp[-(\tilde{r}_{ij}/r_c)^p], \quad \hat{w}_{ij} = w_{ij} / \sum_{k \neq i} w_{ik}, \quad (r_c, p) = (0.4, 6),$$

then form $g_i = \sum_{j \neq i} \hat{w}_{ij} \mathbf{h}_{ij}^\psi$ and $\bar{\mathbf{h}}^\psi = N^{-1} \sum_i g_i$. Two system-level scalars in trap units,

$$\mathbf{g}(\tilde{R}) = \left[\frac{1}{Nd} \sum_i \|\tilde{\mathbf{x}}_i\|^2, \frac{2}{N(N-1)} \sum_{i < j} s_1(\tilde{r}_{ij}) \right] \in \mathbb{R}^2,$$

are concatenated with $\bar{\mathbf{h}}^\phi$ and $\bar{\mathbf{h}}^\psi$ and mapped to a scalar by a tiny MLP ρ :

$$W_\theta(\tilde{R}) = \rho(\bar{\mathbf{h}}^\phi \parallel \bar{\mathbf{h}}^\psi \parallel \mathbf{g}(\tilde{R})). \quad (3.7)$$

3.4.4 Message-passing correlator (CTNN)

The message-passing correlator keeps the analytic cusp but uses its own node and edge embeddings. Its defining change is the copresheaf transport of §2.4.4: at each round the endpoint nodes are transported into an edge update and the updated edges are transported back and aggregated onto the nodes,

$$\mathbf{h}_{ij} += F_e(\mathbf{h}_{ij} \parallel \rho_{v \rightarrow e} \mathbf{h}_i \parallel \rho_{v \rightarrow e} \mathbf{h}_j), \quad \mathbf{h}_i += F_v\left(\mathbf{h}_i \parallel \text{Agg}_{j \neq i} w_{ij}^{\text{spin}} \rho_{e \rightarrow v} \mathbf{h}_{ij}\right), \quad (3.8)$$

with residual updates and per-round transport maps. Unlike the single-round backflow, the correlator stacks these rounds in a multiscale “V-cycle”: $n_\downarrow$ down-pass rounds, a linear coarsening to a bottleneck width, then $n_\uparrow$ up-pass rounds with skip connections back to the down-pass features (defaults $n_\downarrow = n_\uparrow = 2$). A permutation-invariant readout over the final node and edge statistics returns the scalar $W_\theta(R)$. The Slater reference, cusp and training protocol are shared with the separable model, but the CTNN uses relative vectors, their norm and square, together with a separate raw-distance readout, rather than the six features and gate above. The comparison in §7.1 therefore measures the two implemented correlator families rather than isolating message routing alone.

3.5 Backflow

Backflow deforms the physical coordinates used inside the Slater reference so that the nodal surface can adapt to interparticle correlations:

$$R' = R + \Delta_\beta(R), \quad \Psi_{\theta,\beta}(R) = \text{SD}_\omega(R') \exp\left(\sum_{i < j} u_{\sigma_i \sigma_j}(r_{ij}) + W_\theta(R)\right). \quad (3.9)$$

The correlator W_θ and cusp u are evaluated on R ; only the determinant sees R' . This choice cleanly separates (i) amplitude reweighting within nodal pockets (handled by $u + W_\theta$) from (ii) nodal *geometry* (handled by backflow).

Design goals. (i) *Stability near coalescence*: messages are built from mollified radii and short-range gates so that $\partial \Delta_\beta / \partial \tilde{r}$ stays bounded; (ii) *Permutation symmetry*: permutation equivariance holds by construction (shared maps and symmetric aggregation); (iii) *Close to identity at initialisation*: a zero-initialised last layer and a positive learnable scale keep $\Delta_\beta \approx 0$ at the start. Rotational symmetry, by contrast, is only *approximate*: the displacement head reads out a d -vector from node features that include the raw coordinates, so the map is not rotation-equivariant by construction, and it is encouraged instead by the geometry-based edge features and by rotationally-augmented sampling. A global translation is not a symmetry of the trap in any case; the CTNN’s zero-mean projection is a regulariser that removes a spurious centre-of-mass drift, not an imposed physical symmetry.

We study two archived backflow architectures. Both map a configuration to a bounded, per-particle displacement field and use the same caller, which adds that field directly to R . They do not use the same preprocessing or centre-of-mass constraint. The conventional network is the baseline in Chapter 7; the copresheaf CTNN is the production ansatz. Their comparison is therefore descriptive of these implementations, not a controlled architecture ablation.

3.5.1 Conventional backflow (single-round messages)

The baseline uses a single round of pairwise messages. For each ordered pair (i, j) with $i \neq j$,

$$\mathbf{m}_{ij} = \phi_\beta(\mathbf{x}_i, \mathbf{x}_j, \mathbf{r}_{ij}, r_{ij}^{\text{soft}}, (r_{ij}^{\text{soft}})^2) \cdot w_{ij}^{\text{spin}}, \quad r_{ij}^{\text{soft}} = \sqrt{r_{ij}^2 + \varepsilon_{\text{bf}}^2}, \quad (3.10)$$

with ϕ_β a small MLP and $w_{ij}^{\text{spin}} \in \{0, 1\}$ a spin mask (same-spin only or all pairs). Messages are aggregated to a node state and a node network returns the displacement,

$$\mathbf{m}_i = \text{Agg}_{j \neq i} \mathbf{m}_{ij}, \quad \delta \mathbf{x}_i = \psi_\beta(\mathbf{x}_i \parallel \mathbf{m}_i), \quad \text{Agg} \in \{\text{sum}, \text{mean}, \text{max}\}. \quad (3.11)$$

There is no persistent edge state: each message is recomputed from the raw geometry and discarded after aggregation.

3.5.2 CTNN backflow (copesheaf)

The message-passing architecture instead maintains explicit *node and edge* features and transports information between them through learned linear maps—the copresheaf-style step of §2.4.4. Node features are embedded from the position and a spin scalar, edge features from the relative geometry,

$$\mathbf{h}_i^{(0)} = W_v [\tilde{\mathbf{x}}_i, s_i], \quad \mathbf{h}_{ij}^{(0)} = \text{MLP}_e(\mathbf{r}_{ij}, |\mathbf{r}_{ij}|, |\mathbf{r}_{ij}|^2). \quad (3.12)$$

A node-to-edge transport $\rho_{v \rightarrow e}$ feeds the endpoint nodes into an edge update; an edge-to-node transport $\rho_{e \rightarrow v}$ returns the updated edges to the nodes, which are refined by a residual node update; a linear head reads out the displacement:

$$\mathbf{h}'_{ij} = \text{MLP}'_e(\mathbf{h}_{ij}^{(0)} \parallel \rho_{v \rightarrow e} \mathbf{h}_i^{(0)} \parallel \rho_{v \rightarrow e} \mathbf{h}_j^{(0)}), \quad (3.13)$$

$$\mathbf{m}_i = \text{Agg}_{j \neq i} w_{ij}^{\text{spin}} \rho_{e \rightarrow v}(\mathbf{h}'_{ij}), \quad (3.14)$$

$$\mathbf{h}_i = \mathbf{h}_i^{(0)} + \text{MLP}_v(\mathbf{h}_i^{(0)} \parallel \mathbf{m}_i), \quad \delta \tilde{\mathbf{x}}_i = W_\Delta \mathbf{h}_i. \quad (3.15)$$

The learned transport maps $\rho_{v \rightarrow e}, \rho_{e \rightarrow v}$ are the channel whose ablation collapses the backflow at the Wigner crystal (Chapter 7). The backflow uses a single copresheaf round; the message-passing *correlator* of the architecture comparison (CTNNJastrowVCycle) iterates the same style of update over a down/up “V-cycle”.

3.5.3 Constraints, symmetries and scaling

Both variants bound their output and start at the Slater nodes, but their coordinate conventions differ:

$$\Delta_\beta^{\text{conv}}(R) = s_\beta \tanh[\delta X_{\text{conv}}(R)], \quad \Delta_\beta^{\text{CTNN}}(R) = s_\beta P_{\text{COM}} \tanh[\delta X_{\text{CTNN}}(\sqrt{\omega}R)], \quad (3.16)$$

where P_{COM} subtracts the particlewise mean. The common caller adds both outputs directly to the physical coordinates R . Their learned scales therefore do not have a guaranteed common displacement unit. The final layer is zero-initialised and $s_\beta = \text{softplus}(s_\beta^{\text{raw}})$ begins in $[0.02, 0.10]$.

Near coalescence derivatives of the pair norm can make $\nabla \cdot \Delta_\beta$ spiky. The conventional path mollifies r_{ij} as in (3.10); the CTNN also permits an optional short-range gate in its internal trap units,

$$\chi_{\text{bf}}(\tilde{r}_{ij}) = \frac{\tilde{r}_{ij}^2}{\tilde{r}_{ij}^2 + r_g^2}, \quad \chi_{\text{bf}}(0) = \chi'_{\text{bf}}(0) = 0, \quad r_g \approx 0.3, \quad (3.17)$$

or use degree-normalised radial attention. The numerical mollifier $\varepsilon_{\text{bf}} = 10^{-6}$ is fixed rather than tuned; the optional hard gate was disabled in the reported architecture diagnostics unless stated otherwise. Its dormant extra trap-scaling factor was removed after the reported runs; because the gate was disabled, this code correction does not change the results.

Permutation symmetry is automatic. In the CTNN path a spurious centre-of-mass drift is removed by

$$\Delta_\beta \leftarrow \Delta_\beta - \frac{1}{N} \sum_{i=1}^N \Delta_{\beta,i}, \quad (3.18)$$

taken over particles within each configuration. The conventional path returns its raw field; the caller does not project it. This difference changes both the shifted determinant coordinates and the ceiling of the rank diagnostic in §7.1.2. Together with the input-scale difference in Eq. (3.16), it prevents the archived sweep from isolating backflow architecture.

Because only the determinant is evaluated at R' , backflow moves the *nodes* while the correlator exponent stays on R , and no Jacobian term arises. This Feynman–Cohen construction [44] targets fixed-node error directly; its modern quantum-Monte-Carlo form is due to Holzmann and Ceperley [45]. Compute and memory scale as $O(BN^2(d+H))$ for batch size B and hidden width H , which is what limits N in practice.

3.6 Architecture sizes and hyperparameters

Table 3.1 collects the representative settings. The two comparisons in Chapter 7 stand on different footings in this respect.

The correlator study includes a capacity-matched pair and a separable-network ladder from 20k to 164k parameters against a fixed 80k message-passing model. Capacity is therefore tested directly, but the learned feature sets remain unmatched as noted above.

The *backflow* comparison is not. At equal width the copresheaf network is intrinsically the heavier object—it carries a persistent edge state where the conventional network carries only a per-particle aggregate—so at the shared production setting (message width 128, node MLP 128×3) it holds 182k parameters against 51k, a factor of 3.6. No matched backflow pair was trained. Its preprocessing and centre-of-mass constraint also differ, so the sweep reports the behaviour of two production implementations rather than an architecture-only comparison.

Table 3.1: The three sets of models trained in this thesis. They exist for different purposes and are never compared with one another. Widths are hidden dimensions; “rounds” are message-passing steps ($n_\downarrow + n_\uparrow$ for the V-cycle). The main correlator pair differs in size, so §7.1.1 also reports the matched 89.4k separable network and the full 20k–164k ladder. The backflow variants are not matched in capacity or preprocessing. Settings follow the accompanying code (`scripts/run_depth_analysis.py`, `scripts/source_state_check.py`).

Programme	Correlator	Backflow	Total
Physics production (§6.1)	separable; width 128, 2 layers; 54k	conventional; msg 128, node 128×3 ; 51k	105k
	(as above)	message-passing; same widths; 182k	236k
Architecture study (§7.1)	separable; pair-MLP 64×4 , out 32; 47.7k	as production; not matched	—
	message-passing; node/edge 32, bottleneck 16, 2+2 rounds; 79.8k		—
Collocation (§7.3)	message-passing V-cycle; 25k	none	25k
	(as above)	copresheaf; $h_{bf}=128$; +24k	49k

Readout for every correlator: hidden 64, 2–3 layers, SiLU. Optimiser: Adam at $\text{lr} = 5 \times 10^{-4}$ (backflow) and 5×10^{-5} (Jastrow); CG-SR with Tikhonov damping λ , a parameter-step cap and CG restarts. Collocation batch $n_{\text{coll}} \sim 3 \times 10^3$ with oversampling $m = 8$ –16; the collocation families are described further in §4.5.3.

Implementation correspondence. For replication: the separable correlator is PINN and the message-passing one CTNNJastrowVCycle; the conventional backflow is BackflowNet, whose ϕ_β and ψ_β are its message and node MLPs, and the copresheaf backflow is CTNNBackflowNet, whose embeddings, transport maps and update MLPs realise (3.13)–(3.15). Aggregation is selectable (`sum|mean|max`); the bounded output of (3.16) is `out_bound="tanh"` with $s_\beta = \text{softplus}(\text{bf_scale_raw})$. The comparison tiers are produced by `scripts/run_depth_analysis.py` and the production widths by `scripts/source_state_check.py`. Schedule and solver options carry the names `alpha_start`, `alpha_end`, `alpha_decay_frac`, `max_param_step`, `damping`, `micro_batch` and `grad_clip`.

Chapter 4

Optimisation

4.1 Primary objective and two-stage training

Given a parameterised wavefunction Ψ_θ , the variational ground-state energy is

$$E(\theta) = \frac{\langle \Psi_\theta | H | \Psi_\theta \rangle}{\langle \Psi_\theta | \Psi_\theta \rangle} = \mathbb{E}_{\tilde{R} \sim \pi_\theta} [E_L(\tilde{R})], \quad \pi_\theta(\tilde{R}) = \frac{|\Psi_\theta(\tilde{R})|^2}{\int |\Psi_\theta|^2}. \quad (4.1)$$

The local energy is evaluated in *physical* coordinates, consistently with the Hamiltonian (3.1),

$$E_L(R) = -\frac{1}{2} \sum_{i=1}^N \left[\Delta_{\mathbf{r}_i} \ln |\Psi_\theta| + \|\nabla_{\mathbf{r}_i} \ln |\Psi_\theta|\|^2 \right] + \frac{1}{2} \omega^2 \sum_{i=1}^N r_i^2 + \sum_{i < j} \frac{1}{r_{ij}}, \quad (4.2)$$

and the zero-variance principle holds: $\text{Var}_{\pi_\theta}[E_L] = 0$ iff Ψ_θ is an eigenstate. The network *features* are computed in trap units $\tilde{\mathbf{r}} = \sqrt{\omega} \mathbf{r}$ for conditioning, but the local energy itself is formed in physical units, so no ω factors are absorbed into the kinetic, trap, or Coulomb terms. (In trap units the same operator reads $E_L = \omega \left[-\frac{1}{2} \sum_i (\Delta_{\tilde{\mathbf{r}}_i} \ln \Psi_\theta + \|\nabla_{\tilde{\mathbf{r}}_i} \ln \Psi_\theta\|^2) + \frac{1}{2} \sum_i \tilde{r}_i^2 \right] + \sum_{i < j} \sqrt{\omega}/\tilde{r}_{ij}$; the kinetic and trap terms scale by ω and the Coulomb term by $\sqrt{\omega}$.)

Our optimisation follows two stages that mirror the implementation:

1. **Residual-based pretraining (stage I).** We fit the residual of E_L to a moving scalar target E_{eff} under the current collocation proposal q_t , $\mathcal{L}_{\text{res}}(\theta) = \mathbb{E}_{R \sim q_t} [(E_L(R) - E_{\text{eff}})^2]$. We begin with a *self-target* (stabiliser) $E_{\text{eff}} = \mu$ where $\mu = \mathbb{E}[E_L]$ over the current batch. This minimises the *sample* variance of E_L under the current collocation measure—which is not the same object as $\text{Var}_{|\Psi|^2}(E_L)$ unless the two measures coincide—and avoids imposing an external energy target at the start. We then anneal toward a trusted reference E_{DMC} via

$$E_{\text{eff}}(\alpha) = \alpha E_{\text{DMC}} + (1 - \alpha) \mu, \quad \alpha \in [0, 1], \quad (4.3)$$

with a smooth cosine schedule $\alpha = \alpha(t)$ that ramps from α_{start} to α_{end} over a prescribed fraction of epochs. Setting the **objective** to "residual" uses $E_{\text{eff}} = \mu$; "energy" uses $E_{\text{eff}} = E_{\text{DMC}}$; and "energy_var" uses the annealed blend above.

2. **Stochastic Reconfiguration (stage II).** After residual pretraining, we refine with SR (natural-gradient VMC). Writing $\bar{O}(R) = \partial_\theta \log |\Psi_\theta(R)| - \mathbb{E}_\pi[\partial_\theta \log |\Psi_\theta|]$ for the centred score,

$$S = \mathbb{E}_\pi[\bar{O}^\top \bar{O}], \quad g = 2 \mathbb{E}_\pi[(E_L - \mathbb{E}_\pi[E_L]) \bar{O}], \quad (4.4)$$

we solve the damped normal equations

$$(S + \lambda I) \Delta \theta = -g, \quad (4.5)$$

using (preconditioned) conjugate gradients with restarts, a cap on the final parameter step, and batchwise centering/whitening of O .

This pipeline uses residual pretraining to shape W_θ and, where present, the backflow before applying the geometry-aware SR update.

4.2 Configuration–space sampling

Guiding principle (permutation invariance). Within each spin sector, relabelling particles does not change the state. The sampler can therefore target geometric motifs—clusters, shells, dimers and long tails—rather than fixed particle identities. We also permute labels within spin sectors and rotate configurations randomly to improve coverage without favouring an orientation.

4.2.1 Stratified capped–simplex mixture

We draw collocation batches $X \in \mathbb{R}^{B \times N \times d}$ from a five–component mixture,

(centre, tails, mixed, shells, dimers),

with nonnegative weights $w \in \Delta^4$ projected to a *capped simplex* $\{w : \sum_k w_k = 1, 0 \leq w_k \leq 0.3\}$ to enforce coverage. Each component proposes x as follows (all in physical units; the code internally scales by $a_{\text{ho}} = 1/\sqrt{\omega}$):

- **Centre:** narrow isotropic Gaussian around the trap centre (samples near the density core).
- **Tails:** wide Gaussian (probes low–density outskirts and large– r behaviour).
- **Mixed:** hybrid batches mixing narrow and moderate spreads per configuration (tests robustness to inhomogeneous spreads).
- **Shells:** points placed on K hyperspherical shells with trap–scaled radii $\{r_k\}$ and occupancy $\{q_k\}$, with small radial/tangential jitter (captures ring/shell order at weak traps).
- **Dimers:** we induce a few close pairs per configuration by sampling short interparticle offsets with random directions and log–uniform distances (enforces near–coalescence coverage without singularities).

We then apply a random in–plane rotation (for $d = 2$), and a random permutation of particle indices; both preserve the target expectations by symmetry.

Adaptive mixture and shell occupancy. Per epoch we compute simple difficulty scores and update w and q by exponentiated–gradient (EG) steps with momentum, temperature, and an exploration term; w is then projected back to the capped simplex. Mixture difficulty uses the residual $(E_L - E_{\text{eff}})^2$ per component; shell difficulty uses a proxy

$$\underbrace{V_i}_{\text{harm.+Coulomb per particle}} + \underbrace{\gamma_{g^2} \|\nabla_{\mathbf{x}_i} \log |\Psi|\|^2}_{\text{stiff gradients}} + \underbrace{\gamma_{\text{cusp}} r_{\text{min}}^{-1}}_{\text{coalescence pressure}},$$

aggregated by nearest shell index. This shifts probability mass toward underfit regions *without* collapsing coverage thanks to the caps.

Hard–example injection. We maintain a small buffer of previously challenging configurations (based on residuals) and stochastically inject them, with mild multiplicative and additive jitter, into the next batch. This provides a curriculum–like pressure while avoiding overfitting to outliers.

Robustified updates. We remove the most extreme local-energy values by quantile (default 3%) before forming an update. This reduces variance but changes the finite-sample objective. We also micro-batch the evaluation and clip parameter gradients.

4.3 Residual-based training (stage I)

For a batch $\{\tilde{R}_b\}_{b=1}^B$ we compute $E_L(\tilde{R}_b)$ and the batch mean $\mu = \frac{1}{B} \sum_b E_L(\tilde{R}_b)$. The residual loss is

$$\mathcal{L}_{\text{res}}(\theta; \alpha) = \frac{1}{B} \sum_{b=1}^B (E_L(R_b) - E_{\text{eff}}(\alpha))^2, \quad E_{\text{eff}}(\alpha) = \alpha E_{\text{DMC}} + (1 - \alpha) \mu. \quad (4.6)$$

For draws from q_t , this is an expectation under q_t . Written as a squared Schrödinger residual, it carries the weight $q_t(R)/|\Psi_\theta(R)|^2$; it is not the constant-weight residual analysed schematically in Appendix A.

- **Self-target warmup** ($\alpha = 0$). Minimising $\mathbb{E}[(E_L - \mu)^2]$ is variance minimisation *under the sampling measure in use*; μ is treated as a detached constant, so no gradient flows through the batch mean. It teaches the network to produce smooth, low-variance amplitudes inside nodal pockets and reduces sensitivity to early sampler noise.
- **Anneal to reference** ($\alpha \uparrow$). As capacity grows, we blend in the external target, gently steering the mean energy while retaining the variance pressure. The cosine ramp avoids abrupt objective switches.
- **Reference-target residual** ($\alpha = 1$). Stage I may end with $E_{\text{eff}} = E_{\text{DMC}}$. Despite its name in the code, this is not variational energy minimisation: it is a squared residual against a fixed scalar, and it can in principle drive $\langle E_L \rangle$ up toward a target the ansatz could beat. The variational optimisation is stage II.

In practice we compute $\nabla_\theta \mathcal{L}_{\text{res}}$ by autodiff, use micro-batches, apply optional gradient clipping, and log per-component usage and shell radii in physical units.

Local-energy derivatives in practice. We support three Laplacian backends:

1. **Exact:** chunked second partials (reference-accurate, heavier).
2. **HVP-Hutchinson:** $\Delta \log |\Psi| = \mathbb{E}_v[v^\top H_{\log |\Psi|} v]$ with Rademacher v (fast and stable; fallback to FD if a probe is non-finite).
3. **FD-Hutchinson:** centred finite differences of directional gradients (robust in corner cases; slightly noisier).

We average a small number of probes (2–16 in ablations), and trim quantiles before forming losses.

4.4 Stochastic Reconfiguration (stage II)

Stochastic reconfiguration [29, 46] preconditions the energy gradient with the (inverse) quantum geometric tensor, and is the imaginary-time / natural-gradient step for a variational state [27, 30]. It uses the covariance form of the energy gradient

$$\partial_{\theta_k} E = 2 \text{Cov}_\pi(E_L, O_k) = 2 \left(\langle E_L O_k \rangle - \langle E_L \rangle \langle O_k \rangle \right), \quad (4.7)$$

and approximates a natural-gradient step by solving $(S + \lambda I) \Delta \theta = -g$ with the quantum geometric tensor S of §2.2.3. Implementation details:

- **Centering/whitening.** We centre O per batch and whiten by the diagonal of S before CG to improve conditioning.
- **Step cap.** The final update is scaled to satisfy a maximum Euclidean parameter-step norm.
- **Damping.** A small λ stabilises low-variance directions.

The damping and step cap limit unstable finite-sample updates. They do not guarantee well-behaved statistics when node-moving scores develop heavy tails; Appendix C examines that limitation.

Optimiser variants compared. Three first-order updates appear in the results, and their comparison is itself a finding (§7.2):

- **Adam** [25]: a diagonal adaptive step, used as the robust default, with separate learning rates for the Jastrow and backflow blocks.
- **Plain SR**: the natural-gradient step above with a fixed diagonal damping λ .
- **CG-SR**: the same step solved matrix-free by preconditioned conjugate gradients, with Tikhonov damping and a parameter-step cap. It is most useful where S is anisotropic but still statistically resolved. Whether that condition holds, and what follows when it does not, is §7.2.

Solving the SR step without forming S is what makes it affordable at $\mathcal{O}(10^5)$ parameters. The same obstacle is addressed from the other side by the minSR and kernel-formulation reductions, which exploit the fact that the number of samples, rather than the number of parameters, sets the true rank of the problem [47, 48]. Whether the extra cost buys anything over a diagonal preconditioner is measured in §7.2.

4.5 Sampling paradigms and the weak-form collocation objective

The alternative pipeline described in this section draws its batches from an analytic proposal rather than a Markov chain—the SR route of §4.4 samples $|\Psi_\theta|^2$ by Metropolis—so the choice of *sampling measure* must be stated explicitly: the results treat it as a variable in its own right. Two paradigms appear in this thesis. In *variational Monte Carlo* (VMC) the batch is drawn from $|\Psi_\theta|^2$ itself, usually by a Metropolis chain, so the estimators of the QGT and of the gradient share the very metric they are meant to precondition. In *collocation* the batch is drawn from a fixed analytic proposal $q(\mathbf{R})$ and reweighted to $|\Psi_\theta|^2$ by importance weights $w_i = |\Psi_\theta(\mathbf{R}_i)|^2/q(\mathbf{R}_i)$, so that the gradient training requires no Markov chain. A Metropolis probe is used only outside the gradient loop—for periodic checkpoint selection and for the final energy certification.

For the collocation objective we use a *weak form*: the local energy E_L enters only as a reward in a score-function (REINFORCE) gradient [49],

$$\nabla_\theta \langle E \rangle = 2 \mathbb{E}_{\mathbf{R} \sim |\Psi_\theta|^2} [(E_L(\mathbf{R}) - b) \nabla_\theta \log |\Psi_\theta(\mathbf{R})|], \quad (4.8)$$

with a variance-reducing baseline b (a running mean of E_L), and the Laplacian inside E_L is never differentiated through. This is deliberate. Differentiating the Laplacian through a backflow-shifted determinant is the source of the nodal instability analysed in Appendix C; the weak form keeps the full local-energy signal as a reward while sidestepping that pathology. Two robustness controls stabilise the reward: local energies are clipped to a robust median/MAD location–scale band ($\sim 5\sigma$), and the most extreme rewards are quantile-trimmed before the gradient is formed.

4.5.1 Analytic proposal and importance resampling

The collocation points are drawn from an analytic mixture of isotropic Gaussians centred at the trap minimum and reweighted to $|\Psi_\theta|^2$ [19],

$$q(\mathbf{R}) = \frac{1}{K} \sum_{k=1}^K \mathcal{N}(\mathbf{R}; 0, \sigma_{f,k}^2/\omega I_{Nd}), \quad \sigma_f \in \{0.8, 1.3, 2.0\}, \quad w_i = \frac{|\Psi_\theta(\mathbf{R}_i)|^2}{q(\mathbf{R}_i)}, \quad (4.9)$$

where the $1/\omega$ scaling ties the proposal width to the oscillator length $\ell = 1/\sqrt{\omega}$ and the three components cover core, bulk, and tails. A batch of n_{coll} points is obtained by multinomial resampling on $p_i = w_i / \sum_j w_j$ from $m n_{\text{coll}}$ candidates ($m \approx 8\text{--}16$). Reweighting quality is tracked by the effective sample size $\text{ESS} = (\sum_i w_i)^2 / \sum_i w_i^2$ and, as a heavier-tailed diagnostic, the Pareto shape parameter $\hat{k}$ of the weights [50]. The second is not a second opinion on the first. Importance weights are heavy-tailed, and $\hat{k}$ estimates the tail index of the generalised Pareto distribution fitted to the largest of them. The thresholds are statements about the moments of the *fitted* tail. For $k < 1/2$ the weights behave as though

the variance were finite and the reweighted estimator obeys a central limit theorem. Between 1/2 and 1 the variance behaves as infinite though the mean does not, and convergence is slow enough that $\hat{k} \approx 0.7$ is the practical limit of trustworthiness. Above 1 the fitted tail has no finite mean.

That last statement needs care, because in this setting the underlying ratio does have a finite mean by construction: $w = |\Psi_\theta|^2/q$ with q a proper density gives $\mathbb{E}_q[w] = \int |\Psi_\theta|^2 < \infty$ whenever Ψ_θ is square-integrable, so asymptotically $\hat{k} < 1$. A measured $\hat{k} > 1$ is therefore a *finite-sample* diagnostic: it says the sample is deep in a pre-asymptotic regime where the realised estimator behaves as if no finite mean existed, not that the expectation fails to exist [50]. We use it in that sense throughout. An ESS can look survivable while $\hat{k}$ says the estimator has no variance to speak of, which is why we track both. Both degrade when q ceases to overlap $|\Psi_\theta|^2$, and that degradation is the failure mode which bounds the paradigm at large N and weak ω (Chapter 7, §7.3.4). Three controls manage the tails: weight tempering ($w \rightarrow w^\alpha$, $\alpha < 1$), log-weight clipping at an upper quantile, and ESS-adaptive oversampling that raises m when the achieved ESS falls below a floor. The first two are deliberate bias–variance trades: they change the distribution the reward is averaged over, so the reweighted energy readout is no longer an estimate of $\langle E_L \rangle_{|\Psi_\theta|^2}$. That is why the final certification is always a separate evaluation under $|\Psi_\theta|^2$ (§7.2). An instability-rollback reverts to the last good checkpoint and decays the learning rate if the energy jumps beyond a threshold.

4.5.2 Physics-informed proposal for the Wigner regime

The origin-centred proposal (4.9) places its mass where a Wigner molecule has almost none: at strong correlation the density collapses onto a shell at the classical ring radius r_0 , and $|\Psi_\theta|^2$ has essentially no support near the origin, so the ESS collapses. We therefore replace q by a proposal built from the Wigner shell itself—a radial Gaussian centred at r_0 (set by balancing confinement against Coulomb repulsion) times an angular distribution over the ring, so that proposed configurations resemble electrons spread around the ring instead of clustered at the centre. Where the ring itself develops angular structure the uniform angular factor is replaced by a Dirichlet model of the angular gaps. Difficult confinements are reached by *cascade warm-starting*: parameters converged at one ω initialise the next, following $\omega : 1.0 \rightarrow 0.5 \rightarrow 0.1 \rightarrow 0.01 \rightarrow 0.001$. What this construction achieves, and where it stops, is §7.4.

4.5.3 Ansatz families used in the collocation programme

Three ansatz families are trained under this objective in Chapter 7. The Jastrow-only and backflow variants use the Slater–Jastrow core $\Psi(\mathbf{R}) = \det[\phi_i(\mathbf{r}_j + \Delta\mathbf{r}_j)] e^{J(\mathbf{R})}$; the third replaces the determinant with a neural Pfaffian. They are smaller than the production networks of §3.6 to reduce the cost of the collocation campaign. Comparisons with the production models are therefore not capacity matched.

Jastrow-only. The correlator alone at $\sim 25\text{k}$ parameters, in its message-passing form (CTNNJastrow-VCycle), acting on node features (single-particle orbital values) and edge features (soft-core pair distances). It is used both standalone and as the initialiser for the backflow runs.

Backflow + Jastrow. Adds the copresheaf displacement network (CTNNBackflowNet), $\sim 49\text{k}$ parameters in total at $h_{\text{bf}}=128$. Its edge tensor scales as $\mathcal{O}(N^2 h_{\text{bf}})$, which is the memory bottleneck at large N : at $N=20$ the hidden width must be cut to $h_{\text{bf}} \in \{48, 64\}$.

Neural Pfaffian. Replaces the determinant with $\text{Pf}[A_{ij}]$, a more general antisymmetric form admitting richer nodal topology [51].

4.6 Units, scaling, and batch construction

The wavefunction evaluator receives physical coordinates. The sampler expresses Gaussian widths and shell radii through $a_{\text{ho}} = 1/\sqrt{\omega}$ so that its components cover comparable dimensionless regimes over $\omega \in [0.01, 1]$. The CTNN feature paths also scale their coordinates internally. The archived separable correlator and conventional backflow do not follow the same convention, as documented above; results comparing those implementations are therefore not treated as scale-controlled architecture tests.

Proposal candidates are drawn independently, so the batch carries no serial Markov-chain autocorrelation. The retained batch is not i.i.d. in the strict sense, because multinomial resampling can repeat candidates and the hard-example buffer reinjects earlier ones. The design removes a mixing time, not all statistical dependence.

4.7 The two workflows

Both routes use the same ansatz form and end at an energy evaluated under $|\Psi_\theta|^2$. They differ in the measure that defines the loss in between.

SR-VMC (§4.3–§4.4)

stratified mixture $\rightarrow$ residual pretraining
 $\downarrow$
 $|\Psi_\theta|^2$ sampling (Metropolis)
 $\downarrow$
 SR step $(S + \lambda I)\Delta\theta = -g$
 $\downarrow$
 energy under $|\Psi_\theta|^2$

Collocation (§4.5)

pretrained checkpoint
 $\downarrow$
 analytic proposal q , importance reweighting
 $\downarrow$
 weak-form REINFORCE gradient (Adam)
 $\downarrow$
 Metropolis probe for checkpoint choice,
 then energy under $|\Psi_\theta|^2$

The right-hand route places no Markov chain in any gradient; the probe that selects its checkpoints sits outside the update loop entirely. The left-hand route, which carries the ground-state energies of §6.1, keeps the early dynamics variance-dominated and sampler-aware and then transitions to a geometry-aware natural-gradient refinement—a division that matters most at weak traps, where the shell and dimer coverage of the stratified mixture is what makes the early stage converge at all.

4.8 Statistical uncertainties

Every energy and observable we report carries a statistical error, and two distinct sources contribute. Within a single evaluation, successive MCMC samples are autocorrelated, so a naive standard error over B samples understates the sampling uncertainty; this is what blocking addresses. Across training runs, the optimisation is itself stochastic and lands on different states from different seeds; blocking says nothing about that, and it is assessed separately by repeating runs at independent seeds wherever the claim depends on it (§7.3.4, §7.1.1). We therefore estimate the error on a scalar mean $\bar{A} = \frac{1}{B} \sum_i A(\mathbf{R}_i)$ by *blocking*: successive samples are averaged into blocks of growing size and the plateau of the block standard error as the block size exceeds the autocorrelation time is taken as the uncertainty [52, 53]. For the final ground-state energy this is the dominant reported error; a parenthetical digit, e.g. 155.8738(7), denotes one standard error on the last quoted figure.

For quantities that are *nonlinear* functions of sample statistics (participation ratios, effective ranks, PCA cosines, and the coupling diagnostics defined in §5.2.3 and reported in Chapter 7) a blocked standard error is not available in closed form, and we use a *block bootstrap* instead: blocks (of the plateau size above) are resampled with replacement, the statistic is recomputed on each resample, and its spread across resamples is the reported error (§5.2.3). Where a number is quoted as a seed average, we additionally report the spread across independent random seeds, which captures optimisation variability and not only sampling variability; the two are distinguished in the captions. Reference energies drawn from the literature carry the uncertainties quoted by their sources.

Chapter 5

Analysis

This chapter describes *what* we measure on fresh $|\Psi_\theta|^2$ samples, *how* we compute the diagnostics, and *why* each quantity is informative. Results and numbers are presented in Chapter 6.

5.1 Structural diagnostics and Wigner markers

Unless stated otherwise, curves are accumulated in *trap units* $\mathbf{y} = \sqrt{\omega} \mathbf{x}$ to compare shapes across ω , while all headline distances are reported in *Bohr* (physical units $\mathbf{x}$). Production chains are thinned; we split samples into halves for stability checks (acceptance window, JSD between halves, and reproducibility under restarts). Using trap units to compare shapes across confinements is standard in parabolic dots; see the overview in [5].

5.1.1 Pair distribution and radial statistics

We estimate an *effective* pair distribution—a shell-area-corrected pair-distance density—in trap units. For a finite trapped dot there is no homogeneous-system normalisation, because the one-body density varies strongly across the cloud, so A_{eff} below is a finite-dot convention rather than a thermodynamic volume and $g(r)$ does not tend to unity at large r . Peak positions and relative profiles, which is what the analysis uses, are insensitive to that choice; absolute amplitudes are not.

$$g(r_\alpha) = \frac{2A_{\text{eff}}}{N(N-1)} \frac{\langle \sum_{i < j} \mathbf{1}\{r_{ij} \in [r_\alpha^-, r_\alpha^+]\} \rangle}{2\pi r_\alpha \Delta r_\alpha}, \quad r_{ij} = \|\mathbf{y}_i - \mathbf{y}_j\|_2, \quad (5.1)$$

with $A_{\text{eff}} = \pi R_{0.95}^2$ and $R_{0.95}$ the 95% quantile of single-particle radii in trap units. This $g(r)$ -based structural analysis follows established practice in mesoscopic 2D electron systems and quantum dots [2, 5, 14]. The pair-distance probability is therefore denoted $P_{\text{pair}}(r_{ij}) \propto r_{ij}^{d-1} g(r_{ij})$. It is used for pair-structure diagnostics, distinct from the one-body radial probability $P_1(r) \propto r \rho_1(r)$, from which we report the headline mode r_{mode} , mean $\bar{r}$, standard deviation σ_r , and FWHM (all in Bohr). A compact one-body width-to-scale marker is

$$\gamma = \sigma_r / r_{\text{mode}}, \quad (5.2)$$

which tracks the width of P_1 relative to its modal radius. It is interpreted alongside shell separation and pair/angular diagnostics, rather than as a universal monotone Wigner marker [2, 14].

5.1.2 Automatic shell detection and topologies

For each frame we sort the single-particle radii $s_1 \leq \dots \leq s_N$ (trap units), compute gaps $g_k = s_{k+1} - s_k$, and flag the frame as *two-shell* if the largest gap exceeds a robust threshold:

$$\max_k g_k \geq \tau \text{median}(g_k), \quad \tau \in [2.5, 3.5] \text{ (default 3.0)}. \quad (5.3)$$

The cut radius is the midpoint of the maximal gap; the inner multiplicity n_{in} defines the topology $(n_{\text{in}}, N - n_{\text{in}})$.

More than two shells. At $N=20$ and weak confinement a single cut is not enough, and the detector is applied recursively. Having placed one cut, we re-apply the same robust-gap test *within* each resulting group, accepting a further cut only if that group contains at least three particles and its own maximal gap again exceeds $\tau \text{median}(g_k)$ computed over the whole frame—so the threshold is global and a shell is never split by a gap that is small on the scale of the configuration. Recursion stops when no group qualifies, which for the systems here never yields more than three shells. A frame is labelled by the resulting occupancy vector $(n_1, \dots, n_S)$ ordered outward, so that the two-shell case above is the $S=2$ member of the same family; the $N=20$ deep-Wigner topology (1, 7, 12) is an $S=3$ assignment with cuts after the first and eighth particle. Topology fractions and their robustness under τ are reported exactly as in the two-shell case. Shelling and discrete ring occupancies are well known in parabolic Coulomb clusters and quantum dots [3–5]; our detector provides a reproducible, data-driven assignment in that spirit. We report: (i) the fraction of two-shell frames, (ii) the histogram over n_{in} among two-shell frames, and (iii) for selected topologies (e.g. (1, 5) or (3, 9)) the fraction among *all* frames.

Shell-resolved reconstruction of $g(r)$. On two-shell frames we decompose pairs into II, IO, and OO and accumulate histograms $g_{\text{II}}, g_{\text{IO}}, g_{\text{OO}}$. With empirical weights $w_{\text{II}}, w_{\text{IO}}, w_{\text{OO}}$ (combinatorial counts) we reconstruct $\tilde{g}(r) = w_{\text{II}}g_{\text{II}} + w_{\text{IO}}g_{\text{IO}} + w_{\text{OO}}g_{\text{OO}}$ and report $\cos \angle(g, \tilde{g})$. Values ≈ 1 indicate that the observed $g(r)$ shape is explained by shell geometry, not sampling artefacts, in line with shell-based descriptions in [3–5].

5.1.3 Orientational order and “ring stiffness”

On each detected ring (inner or outer) we take the particle angles $\{\phi_k\}_{k=1}^n$ in the lab frame. No rotation is removed at this stage: the modulus $|\Phi_m|$ below is already invariant under a global rotation, and the registration of Eq. (5.7) is applied only where the *phase* is needed, which is the cross-shell comparison. Bond-orientational order is

$$\Phi_m = \frac{1}{n} \sum_{k=1}^n e^{im\phi_k}, \quad m = n, \quad (5.4)$$

so that $|\Phi_m| \in [0, 1]$ measures n -fold orientational order (e.g. $m=5$ for (1, 5), $m=9$ for the outer ring of (3, 9)), analogous to standard bond-orientational order parameters used in 2D Coulomb/Yukawa crystals [17]. As a complementary, dimensionless *angular Lindemann* number we use

$$\lambda_\phi = \frac{\text{std}(\Delta\phi)}{\mathbb{E}[\Delta\phi]}, \quad \Delta\phi = \text{nearest-neighbour angular spacing on the ring.} \quad (5.5)$$

Smaller λ_ϕ indicates a stiffer (more Wigner-like) ring. This is the quantity abbreviated “Lind” in Chapter 6; where a shell index is attached, as in λ_{in} and λ_{out} , it is the same estimator evaluated on that shell alone.

Lab-frame and cross-shell order. For a frame with shells $s = 1, \dots, S$ and occupancies n_s , the lab-frame statistic

$$C_{\text{global}} = \left| \left\langle \frac{1}{S'} \sum_{s: n_s \geq 3} \Phi_{n_s} \right\rangle_{\text{frames}} \right|, \quad (5.6)$$

is the modulus of the frame-averaged complex order parameter. A freely rotating molecule therefore has $C_{\text{global}} \rightarrow 0$ even when it is internally ordered: this statistic measures pinning to the laboratory frame.

Cross-shell locking requires a branch-invariant phase comparison. For shells of occupancies m and n , let $z_m = \Phi_m/|\Phi_m|$, $z_n = \Phi_n/|\Phi_n|$ and $g = \text{gcd}(m, n)$. Then

$$C_{m:n} = \left| \left\langle z_m^{n/g} (z_n^*)^{m/g} \right\rangle_{\text{frames}} \right| \quad (5.7)$$

is unchanged by a rigid rotation and contains no choice of phase branch. The earlier $N=20$ cross-shell values were produced by an analysis script that was not retained and used a phase-registration convention that cannot now be confirmed. They are therefore not used quantitatively; the shellwise moduli $|\Phi_{n_s}|$ and angular-spacing statistics are unaffected.

The random-angle null. None of these is interpretable without a baseline. For n particles at independent uniform angles, $n\Phi_n$ is a two-dimensional random walk of n unit steps, so $\langle |\Phi_n|^2 \rangle = 1/n$ exactly, and the mean modulus follows from Klyuyver’s integral,

$$\langle |\Phi_n| \rangle_{\text{random}} = \frac{1}{n} \int_0^\infty \frac{1 - J_0(t)^n}{t^2} dt, \quad (5.8)$$

which we evaluate numerically. The familiar Rayleigh form $\sqrt{\pi}/(2\sqrt{n}) \approx 0.886/\sqrt{n}$ is its large- n limit and undershoots it at the ring sizes that occur here, by 2.6% at $n = 3$, 1.3% at $n = 5$ and 0.4% at $n = 19$ —the same size as the effects being tested at strong confinement, which is why the exact value is used throughout. Because shells are assigned by radial rank alone and no angular cut is applied, it is also exactly the null of our pipeline: redrawing every angle uniformly while keeping each frame’s radii reproduces it within its bootstrap error in all fifteen (N, ω) cells (Appendix F.2). A five-particle ring with no angular order whatever returns $|\Phi_5| \approx 0.40$. We therefore report $|\Phi_n|$ together with its ratio to this null wherever the ordering claim depends on it (Appendix F.2). This is a Lindemann-type orientational metric tailored to finite rings, motivated by the use of Lindemann criteria to diagnose intershell rotational (angular) melting in mesoscopic Coulomb clusters and dots [4, 14].

5.1.4 Scaling check for $N = 2$

To connect with the classical strong-coupling limit we fit

$$r_{\text{mode}}(\omega) = C \omega^\alpha \quad (5.9)$$

by linear regression in $(\log \omega, \log r_{\text{mode}})$ and compare the slope with the classical reference $\alpha_{\text{cl}} = -2/3$ (two charges in a harmonic trap at strong coupling; see, e.g., the reviews in [5]). Approach toward α_{cl} as $\omega \downarrow$ is one of our Wigner crossover indicators, consistent with the crossover phenomenology reported in [2, 14].

5.1.5 Reporting and stability checks

For all scalars and histograms we report standard errors calculated over thinned blocks or frames. We verify: (i) acceptance in the target window, (ii) Jensen–Shannon divergence and ℓ_2 distance between halves of production (stationarity), (iii) robustness of topology fractions and $|\Phi_m|$ under $\tau \in [2.5, 3.5]$ and mild pre-smoothing of radii, and (iv) reproducibility of $g(r)$ and $P(r)$ under independent sampler restarts. All diagnostics operate directly on saved bundles $\{\mathbf{X}_{\text{Bohr}}, r, g(r)\}$.

5.2 Representation analysis

We probe what the networks learn by evaluating fixed, trained models on fresh $|\Psi_\theta|^2$ samples. Unless stated otherwise, features are computed in *trap units* $\mathbf{y} = \sqrt{\omega} \mathbf{x}$ to match the correlator’s internal scaling. All estimates use thinned MCMC frames, split into two equal blocks A and B ; PCA fits on A and reports on B (and vice versa) to avoid circularity. We use `float64` throughout.

Feature taps. Let $z_\theta \in \mathbb{R}^D$ denote the concatenated correlator representation

$$z_\theta = [\bar{\phi} \mid \bar{\psi} \mid \mathbf{g}],$$

where $\bar{\phi}$ are per-particle summaries, $\bar{\psi}$ are pairwise summaries, and $\mathbf{g}$ are global/extras. The readout ρ_θ is a two-layer MLP, not a linear map, so

$$f_{\text{base}}(z) = \rho_\theta(z), \quad f = f_{\text{base}} + f_{\text{cusp}}, \quad (5.10)$$

and the ablations below re-evaluate ρ_θ on modified inputs instead of manipulating a weight vector. Back-flow returns a displacement field $\Delta x \in \mathbb{R}^{N \times d}$ that enters only the Slater determinant; the correlator and cusp are evaluated on the unshifted $\tilde{R}$ (§3.5). The coupling diagnostics below additionally probe the correlator on the shifted coordinates as a counterfactual sensitivity.

5.2.1 Representation geometry

Preprocessing and PCA. Given a sample matrix $Z \in \mathbb{R}^{K \times D}$ (rows are frames), we *mean-centre* the columns, $Z_c = Z - \mu$, and take the SVD of Z_c to obtain orthonormal loadings $U = [u_1, \dots, u_D]$; scores are $S = Z_c U$. Centring but not rescaling is deliberate: the readout ρ_θ was trained on the raw features, so every ablation below is performed in raw coordinates and the mean is restored before ρ_θ is applied. Where a per-column standardisation is used—for the branch-power decomposition of §5.2.2, which compares loadings across branches carrying different natural scales—it is stated explicitly. PCA is fitted on block A and reported on the held-out block B unless noted.

Entropy effective rank. Let $\{\lambda_i\}_{i=1}^D$ be the eigenvalues of $\text{Cov}(Z_c)$ (on the held-out block). With $p_i = \lambda_i / \sum_j \lambda_j$, the *entropy effective rank* is

$$r_{\text{eff}}(Z) = \exp\left(-\sum_{i=1}^D p_i \log p_i\right),$$

which equals 1 for perfectly one-dimensional representations and increases with spectral spread. Two spectral counts appear in this thesis and they are not the same number. $r_{\text{eff}} = \exp(H_1)$ above is the exponential of the Shannon entropy of the spectrum; the *participation ratio* $d_{\text{eff}} = (\sum_a \lambda_a)^2 / \sum_a \lambda_a^2$ of §2.2.3 is the exponential of the Rényi-2 entropy. Since Rényi entropies decrease in their order, $d_{\text{eff}} \leq r_{\text{eff}}$ always, with equality only for a flat spectrum. Appendix E reports r_{eff} for feature covariances and Chapter 7 reports d_{eff} for the tangent metric and for the displacement field, so the same object can carry two slightly different counts in the two chapters; each table names which is used.

5.2.2 Probes and ablations

Principal-component ablations of the head. To test how many latent axes the readout truly uses, we project the centred features onto the top- k PCs and restore the mean before re-evaluating the trained readout *without refitting*,

$$\Pi_k = \sum_{i=1}^k u_i u_i^\top, \quad z^{(k)} = \mu + \Pi_k(z - \mu), \quad \hat{f}^{(k)} = \rho_\theta(z^{(k)}). \quad (5.11)$$

Restoring μ matters: ρ_θ is a nonlinear map trained on uncentred inputs, so feeding it a centred or standardised vector would evaluate it off its training domain and the resulting error would measure that displacement rather than the loss of PC content. As a null, the same projection is repeated onto k *random* orthonormal directions and the two error curves are compared; a top- k curve that is not below the random- k curve would mean the leading PCs carry no privileged information for the readout. We report a *normalised* error on block B ,

$$\text{rel-MAE}(k) = \frac{\mathbb{E}_B |\hat{f}^{(k)} - f_{\text{base}}|}{\text{std}_B(f_{\text{base}})},$$

together with the learning curve in k .

PC1 block-power decomposition. To attribute the leading axis to branches, we partition u_1 conformably with $z = [\bar{\phi} \mid \bar{\psi} \mid \mathbf{g}]$ and report

$$\text{power}(\bar{\phi}) = \frac{\|u_{1,\bar{\phi}}\|_2^2}{\|u_1\|_2^2}, \quad \text{power}(\bar{\psi}) = \frac{\|u_{1,\bar{\psi}}\|_2^2}{\|u_1\|_2^2}, \quad \text{power}(\mathbf{g}) = \frac{\|u_{1,\mathbf{g}}\|_2^2}{\|u_1\|_2^2}.$$

These are computed on standardised features so that branches carrying different natural scales are comparable. One confound remains and should be kept in mind when reading them: a branch holding d_b of the D coordinates would already carry about d_b/D of the squared loading mass under an isotropic random direction, so a large share partly reflects a wide branch. The shares below are therefore read as *relative* shifts across ω at fixed architecture—where the branch widths are constant and the confound cancels—and not as absolute statements that one branch dominates.

Linear probes to coarse physics. We regress interpretable summaries per configuration on the first k PC scores $S^{(k)}$ using ridge regression (tiny penalty $\alpha = 10^{-6}$):

$$y \approx S^{(k)}\beta, \quad y \in \{\bar{r}, \text{Var}(r), \text{Pr}(r < r_0), \text{shell contrast}\}.$$

We report R^2 on the held-out block; r_0 is a fixed near-contact cutoff in trap units.

Near-field gradient concentration. For each frame we compute the minimum pair distance $r_{\min}$ (trap units) and the gradient of the residual head with respect to particle coordinates, $\nabla f_{\text{base}}(\mathbf{y})$. The quantity reported is an *enrichment factor*: the mean gradient power among the closest-contact frames divided by the mean over all frames,

$$\mathcal{E}(q) = \frac{\mathbb{E}[\|\nabla f_{\text{base}}\|_2^2 \mid r_{\min} \leq Q_q]}{\mathbb{E}[\|\nabla f_{\text{base}}\|_2^2]}, \quad (5.12)$$

with Q_q the q -quantile of $r_{\min}$. This is a ratio of a *conditional* mean to an unconditional one and is therefore unbounded above, unlike an indicator-weighted mass fraction, which could not exceed unity. $\mathcal{E} \approx 1$ means gradient power is spread in proportion to frame count, so the analytic cusp is handling most short-range structure; $\mathcal{E} \gg 1$ means residual stiffness is concentrated near contact.

5.2.3 Backflow–correlator coupling

Backflow geometry and energy-locality shares.

Field spectrum. We vectorize the backflow displacement per frame, $\delta = \text{vec}(\Delta x) \in \mathbb{R}^{Nd}$, standardize across frames, and compute the entropy effective rank $r_{\text{eff}}(\Delta x)$ of Sec. 5.2.1. Chapter 7 reports the participation ratio of the same spectrum instead, which is the smaller of the two counts; the ceiling $2N - d$ set by the zero-mean projection (3.18) is common to both.

Where backflow acts. On the same thinned frames we evaluate local energies with and without backflow, $E_{\text{loc}}^{\text{BF}}$ and $E_{\text{loc}}^{\text{noBF}}$, using the *same* coordinates. We form the signed correction $\Delta E_{\text{loc}} = E_{\text{loc}}^{\text{BF}} - E_{\text{loc}}^{\text{noBF}}$ and report near-field shares at quantile q of $r_{\min}$:

$$\text{NF-}\Delta E(q) = \frac{\mathbb{E}[|\Delta E_{\text{loc}}| \mathbf{1}\{r_{\min} \leq Q_q\}]}{\mathbb{E}[|\Delta E_{\text{loc}}|]}.$$

(Using absolute values isolates magnitude; signed averages are also available.)

Alignment of correlator representations. We collect two standardized feature matrices $\tilde{Z}^{\text{noBF}}$ and $\tilde{Z}^{\text{BF}}$ on identical frames (each standardized by its own μ, σ), compute their PC1 loadings u_1^{noBF} and u_1^{BF} , and report the cosine similarity

$$\cos(u_1^{\text{noBF}}, u_1^{\text{BF}}) = \frac{\langle u_1^{\text{noBF}}, u_1^{\text{BF}} \rangle}{\|u_1^{\text{noBF}}\|_2 \|u_1^{\text{BF}}\|_2}.$$

Cosines near 1 indicate that backflow acts as a structured reweighting along the same dominant axis, rather than rotating the correlator’s manifold.

Centred kernel alignment. The PC1 cosine compares one direction. To compare the two feature *geometries* as a whole we use linear centred kernel alignment [54]. With $\tilde{Z}^{\text{noBF}}, \tilde{Z}^{\text{BF}} \in \mathbb{R}^{K \times D}$ column-centred on the same K frames,

$$\text{CKA}(\tilde{Z}^{\text{noBF}}, \tilde{Z}^{\text{BF}}) = \frac{\|(\tilde{Z}^{\text{noBF}})^\top \tilde{Z}^{\text{BF}}\|_F^2}{\|(\tilde{Z}^{\text{noBF}})^\top \tilde{Z}^{\text{noBF}}\|_F \|(\tilde{Z}^{\text{BF}})^\top \tilde{Z}^{\text{BF}}\|_F} \in [0, 1], \quad (5.13)$$

which is invariant to orthogonal transformations and to isotropic scaling of either representation, and equals 1 only when the two Gram matrices are proportional. It is the quantity reported in Table E.2. Two cautions attach to it. CKA compares the *geometry of the feature cloud*, so a high value says the correlator’s representation is unchanged by the presence of backflow—not that the backflow occupies the same subspace, and not that the two components are functionally independent. And CKA is known to saturate near 1 whenever both representations are dominated by a few shared high-variance directions, which is exactly the regime here (§5.2.1); values above 0.99 should therefore be read as “no detectable reorganisation” rather than as a quantitative distance.

Uncertainty and stability. For scalars we report standard errors over blocks. For PCA-based quantities and cosines we use a block bootstrap (resampling blocks with replacement) to obtain 68%/95% CIs. Stability checks include: (i) swapping train/held-out blocks for PCA, (ii) varying the near-field quantiles $q \in \{1, 5, 10\}\%$, (iii) re-running analyses on independent sampler restarts.

Reproducibility package. All diagnostics consume the saved bundles $\{\mathbf{X}_{\text{Bohr}}, r, g(r)\}$ and the model snapshot, and emit JSON/CSV tables with r_{eff} , $\text{rel-MAE}(k)$ curves, block-power shares, R^2 probes, near-field shares, $r_{\text{eff}}(\Delta x)$, and noBF/BF cosines. As in Sec. 5.1, we log JSD/ℓ_2 splits to confirm stationarity.

5.3 Tangent-kernel diagnostics

The representation diagnostics above describe the ansatz’s *features*; a second family, developed with the geometry of §2.2.3, describes its *tangent space*, and it is these that carry the architecture, optimiser, and paradigm comparisons of Chapter 7. From a batch we assemble the centred score matrix $O_{ki} = \partial_{\theta_i} \log |\Psi_{\theta}(\mathbf{x}_k)|$ and form the quantum geometric tensor $S = O^{\top} O / B$ from the *centred* O (or, when $B < P$, the tangent kernel $K = O O^{\top}$, whose nonzero spectrum equals that of S up to the overall factor B). From the spectrum $\{\lambda_a\}$ we report the effective dimension $d_{\text{eff}} = (\sum_a \lambda_a)^2 / \sum_a \lambda_a^2$ and the condition number $\kappa(S) = \lambda_{\text{max}} / \lambda_{\text{min}}$. Cross-architecture comparisons of d_{eff} are made on a *common probe set* (all models scored on the same configurations, not on each model’s own $|\Psi|^2$) so that the comparison reflects the ansatz and not its sampling measure, and we report only participation ratios that have converged in the batch size.

A selected operator dictionary. Counting tangent directions says how many; naming them says which. We therefore project the leading tangent modes onto a small, fixed dictionary of one- and two-body operators chosen before the fits were run, each written as a scalar function of a configuration whose parameter-space image is obtained by the same score construction as O :

$$\begin{aligned} \mathcal{D}_1 &= \sum_i \tilde{r}_i^2 && \text{breathing / monopole, proportional to } \partial H / \partial \omega; \\ \mathcal{D}_2 &= \sum_i \tilde{r}_i^2 \cos 2\theta_i && \text{quadrupole, the lowest shape distortion;} \\ \mathcal{D}_3 &= \sum_{i < j} e^{-\tilde{r}_{ij}^2 / 2\sigma_h^2} && \text{correlation hole, with } \sigma_h \text{ the fitted hole width;} \\ \mathcal{D}_4 &= \sum_{i < j} \tilde{r}_{ij} && \text{pair-linear, the leading long-range pair response.} \end{aligned} \tag{5.14}$$

Writing $D \in \mathbb{R}^{B \times 4}$ for the centred matrix of these four functions on the probe batch and $v_a \in \mathbb{R}^B$ for the a th tangent-kernel eigenvector, the *naming* score of a mode is the coefficient of determination of its least-squares regression on the dictionary,

$$R_a^2 = 1 - \frac{\|(I - P_D)v_a\|^2}{\|v_a\|^2}, \quad P_D = D D^+, \tag{5.15}$$

and the *dictionary-orthogonal tangent mass* is the eigenvalue-weighted share of the leading A modes lying outside that span,

$$M_{\perp} = \frac{\sum_{a \leq A} \lambda_a \|(I - P_D)v_a\|^2}{\sum_{a \leq A} \lambda_a}, \quad A = 10. \tag{5.16}$$

Finally, *correlation-hole capture* is R^2 of Eq. (5.15) computed against the single operator $\mathcal{D}_3$, i.e. the share of the leading mode explained by the hole channel alone.

Here D^+ is computed by a tolerance-controlled singular-value decomposition. The dictionary is a *chosen, non-exhaustive* probe, and nothing here establishes that it spans the physically meaningful directions. A large $M_{\perp}$ means the tangent space is poorly accounted for by these four operators; it does not mean the remaining directions are unphysical. The comparison supports a relative statement between two architectures scored against the same dictionary on the same probe batch, and that is how Chapter 7 uses it. Because no random, dimension-matched dictionary or held-out fit was run, the numerical R^2 values should not be read as an absolute test of physical interpretability.

Reporting the condition number. $\kappa(S) = \lambda_{\max}/\lambda_{\min}$ is quoted with two caveats that its definition forces. Our batches have $B < P$, so S is rank-deficient by construction and $\lambda_{\min}$ is the smallest *sample-supported* eigenvalue, not an intrinsic one; the value of κ therefore depends on B and is a property of the empirical metric the optimiser actually sees, not of the variational manifold. That is the relevant object for the question we ask of it, whether preconditioning by S^{-1} helps, but it means κ may not be compared across batch sizes, and we hold B fixed wherever we compare it.

To ask whether two trained models are the *same* state rather than merely equal in energy, we use a symmetric, importance-sampled squared overlap,

$$\mathcal{S}^2(A, B) = \langle \Psi_B / \Psi_A \rangle_{|\Psi_A|^2} \langle \Psi_A / \Psi_B \rangle_{|\Psi_B|^2},$$

each factor evaluated by a log-sum-exp mean over samples of the corresponding state. It equals 1 for identical states, and on the analytic two-electron ground state the trained ansatz reproduces $\mathcal{S}^2 = 1.000000$, which anchors the tooling.

One convention must be stated, because the ratio Ψ_B/Ψ_A is signed for a fermionic state. We evaluate the ratios from $\log |\Psi|$, so the estimator is strictly a *modulus* overlap, $\langle |\Psi_B/\Psi_A| \rangle \langle |\Psi_A/\Psi_B| \rangle$. It is an upper bound on signed agreement and does not certify nodal agreement, because independently trained backflows can move configurations across different nodes. We therefore use it only as an amplitude-overlap diagnostic. Energy and $\text{Var}(E_L)$ are properties of the state and are unaffected by how correlation is divided between the correlator and the backflow; the energetic price of ablating a component is not, since two parameterisations that divide correlation differently report different ablation costs.

Part IV

Results

The argument of Part IV

The two results chapters run a loop between solver and physics.

We first show that the trained states agree closely with the external energy references we compare them against—an agreement rather than an independent validation, and one that stops at $\omega = 0.1$ for $N \geq 6$, below which the energies are predictions. That agreement is the precondition for using the states as instruments in the physics analysis that follows. Read for the physics they contain, they show a crossover from compact dot to finite Wigner molecule that is *staged*: the cloud expands and its shells stiffen radially well before the electrons acquire angular order within them.

We then turn from the state to the solver. The implemented model families have different variational geometries even where their total energies agree. Under the parametrisations used here, the message-passing correlator concentrates its tangent weight on fewer directions that are better described by the selected operator dictionary. The message-passing backflow also avoids the rank collapse seen in the smaller conventional implementation. Architecture, optimiser and sampling measure then meet in a single object, the geometry of the wavefunction’s log-derivatives, and the evidence suggests that natural-gradient preconditioning earns its cost only while that geometry can still be estimated from a batch.

Finally, a generic collocation proposal loses the Wigner state. The structural measurements of Chapter 6 supply the remedy: a proposal built from the observed shell geometry carries a cascade with MCMC-free updates to a stable state at $\omega = 0.0035$, where transfer to the next confinement fails. The component decomposition is consistent with a disturbed Slater–Jastrow balance but does not isolate it.

Three differently sized sets of models appear below—the production states that carry the physics, the capacity ladder of the architecture study, and the smaller collocation networks. They are kept distinct, and any cross-set comparison is identified as unmatched. Table 3.1 gives all three. The four research questions of §I are answered here:

RQ4 — Physics	§6.2
RQ1 — Architecture	§7.1
RQ2 — Optimiser	§7.2
RQ3 — MCMC-free gradient updates	§7.3–§7.4

*The least initial deviation from the truth is multiplied
later a thousandfold.*

ARISTOTLE
On the Heavens

Chapter 6

Energies and Wigner-molecule structure

This chapter is about the *state*: whether it is accurate enough to trust, and what physics it turns out to contain. Why the solver behaves as it does is the subject of Chapter 7.

The quantum dot is an unforgiving test. The Coulomb interaction is singular wherever two electrons meet, and the kinetic energy carries second derivatives, which magnify any error in the wavefunction’s curvature. The confinement ω then sweeps the physics across three decades of correlation strength: from a compact, kinetic-energy-dominated droplet at $\omega = 1$ to a dilute Wigner molecule at $\omega = 10^{-3}$, where the electrons settle several oscillator lengths apart and their mutual repulsion rather than the trap determines the structure of the state. We work at $N \in \{2, 6, 12, 20\}$ electrons and $\omega \in \{1.0, 0.5, 0.1, 0.01, 0.001\}$.

Section 6.1 benchmarks the energies against published references: fixed-node diffusion Monte Carlo for $N \in \{6, 12, 20\}$ [55, 56] and compiled DMC values, with analytic anchors where available, for $N=2$ [55, 57]. We compare at $\omega = 1, 0.5$ and 0.1 for every N , and at $\omega = 0.01$ for $N=2$; below that, for $N \geq 6$, we have no reference to compare with, and the boundary is marked wherever it occurs. Section 6.2 then uses the trained wavefunction as an instrument and follows the Wigner crossover from two electrons to twenty. Its result is that the crossover is staged: the cloud expands and its shells stiffen radially well before the electrons acquire angular order inside them. How the network internally represents these states is a question about the solver, not about the physics, and is deferred to Chapter 7 and Appendix E.

6.1 Benchmark energies

Training is residual-based pretraining to a well-conditioned initialiser followed by a Stochastic Reconfiguration tail, run for 10^3 – 3×10^3 iterations on batches of $\sim 3 \times 10^3$ samples, that refines the last digits. Wherever this thesis says “SR tail” without qualification, SR is applied to variational Monte Carlo under $|\Psi|^2$ sampling; the same optimiser reappears in Chapter 7 on the collocation objective, where the measure differs and so does its role.

Two production ansätze carry this chapter. They share the separable correlator of §3.4.1 at ≈ 54 k parameters and use different backflow implementations: a *conventional* single round of pairwise messages aggregated per particle (≈ 51 k, total ≈ 105 k), or a *message-passing* field that maintains node and edge features and transports information between them (≈ 182 k, total ≈ 236 k). They are named *conv. BF* and *MP BF*. Besides capacity, the archived paths differ in coordinate preprocessing and centre-of-mass projection (§3.5.3), so their comparison is not an architecture-only ablation. The counts are independent of N , since interactions enter through permutation-equivariant features, not per-particle weights, and they are not the widths of the architecture study of §7.1 (Table 3.1).

6.1.1 Energies

Across every externally benchmarked system, the trained states agree with the available fixed-node reference energies to between 10^{-5} and 5×10^{-4} in relative terms (Tables 6.1 and

6.2, Fig. 6.1). All tabulated deviations are below 0.05%; the available runs do not resolve a trend with particle number or confinement. Everything after this section depends on it. It is less than a validation of the wavefunction, for reasons set out below, but it is the precondition for treating the state as an instrument in §6.2, although it does not validate every property of the wavefunction.

Where the agreement holds matters as much as its size. A method that matches at $\omega = 1$ and drifts at $\omega = 0.1$ differs from one that holds across the benchmarked range, since the physics changes character between those points while the architecture does not. Below $\omega = 0.1$ we have no reference of any kind for $N \geq 6$, and the values reported there are variational predictions rather than validated ones; that boundary is marked wherever it occurs.

Where a reference exists it is also the target that residual pretraining anneals toward (§4.1), so the agreement is not a fully independent validation. The SR tail that follows strengthens it considerably, since it minimises the variational energy itself and no longer sees the target; but the final states still descend from reference-guided training. Nor does closeness to a reference certify the state. The variational principle guarantees that the expectation does not fall below the exact E_0 , not that it lies near it, and a flexible ansatz can reproduce an approximate reference without describing every property of the exact state correctly. §8.1 takes this up properly.

Table 6.1 begins with the easiest case, and its purpose is diagnostic, not competitive: two electrons are few enough that the architecture can be tested nearly in isolation. It reports two-electron energies from *residual-only* training, with and without the backflow. For $N=2$, backflow is *not* required to reach reference accuracy; it slightly reduces the variance and smooths convergence, and nothing more. The closed-shell two-electron singlet has no nontrivial same-spin fermionic node for backflow to repair. This matters because it establishes a baseline against which the backflow’s later contribution, at larger N and weaker traps, can be read as a genuine effect rather than as an artefact of having added parameters.

Table 6.1: Ground-state energies (Hartree) for two electrons in a harmonic trap using *residual-only* training—that is, *without* the SR tail. To be read against Table 6.2, which reports the same systems after SR refinement: the comparison isolates what pretraining alone achieves, and the residual bias at $\omega = 0.01$ visible here is the one the SR tail removes. The reference column is the same as in Table 6.2; see its notes for provenance. Parentheses denote 1σ on the last digits.

ω	correlator + BF	correlator only	Reference
1.00	2.99998(3)	2.999940(5)	3.00000(1)
0.50	1.65975(2)	1.65979(3)	1.65977(1)
0.10	0.440795(3)	0.440833(1)	0.44079(1)
0.01	0.0740724(6)	0.0741679(6)	0.073839(2)

Table 6.2 compares diffusion Monte Carlo (DMC) references with our *final* conv.-BF energies after residual pretraining and an SR tail. From the ultra-shallow two-electron case ($\omega=0.01$) to the many-electron systems, absolute deviations are small; for $N=2$ they lie within DMC uncertainties, and for $N \in \{6, 12\}$ the relative deviations span 0.008–0.05% (see also Fig. 6.1).

Table 6.2: Variational energies (Hartree) from residual pretraining plus an SR tail, for the conventional-backflow and message-passing-backflow ansätze. Reference values for $N \geq 6$ are Høgberget’s fixed-node diffusion Monte Carlo energies [55], as tabulated in Ref. [56]; provenance and the $N=2$ attribution are given in the notes below. Parentheses denote the 1σ blocked statistical error on the last digit(s) for the single optimised state reported; they do not include the variation between independent training runs, which was not measured for these production states, and at $N=20$ in particular each entry is one training run. For scale: the backflow comparison of §7.1.2 trains the same two backflow architectures with two seeds per cell, and there the seeds differ by up to 0.09%, comparable to the deviations in this table. “Rel. dev.” is $100(E - E_{\text{ref}})/E_{\text{ref}}$, labelled deviation and not error because the $N \geq 6$ references are themselves fixed-node upper bounds, so a negative entry is not necessarily wrong (§8.1). **These are production energies and the two ansätze are not capacity matched** ($\approx 105\text{k}$ against $\approx 236\text{k}$); the controlled architecture comparison is §7.1, and this table is not an architecture ablation. Rows whose reference column reads *prediction*—all $\omega = 0.001$, and $\omega = 0.01$ for $N \geq 6$ —have no reference value to compare with. They are variational predictions, not validated energies, and in the deep-Wigner rows they are the lowest states found at fixed spin projection (§6.2).

N	ω	Reference	conv. BF	rel. dev.	MP BF	rel. dev.
2	1.00	3.00000	2.99999(1)	−0.0003	3.00000(2)	+0.0000
	0.50	1.65977(1)	1.65976(2)	−0.0006	1.65976(1)	−0.0006
	0.10	0.44079(1)	0.44079(1)	+0.0000	0.440796(4)	+0.0014
	0.01	0.073839(2)	0.073838(1)	−0.0014	0.0738382(5)	−0.0011
	0.001	<i>prediction</i>	0.0137948(8)	—	0.013778(1)	—
6	1.00	20.15932(8) ^H	20.1610(6)	+0.0083	20.1585(3)	−0.0041
	0.50	11.78484(6) ^H	11.7895(4)	+0.0395	11.7847(2)	−0.0012
	0.10	3.55385(5) ^H	3.5549(1)	+0.0295	3.55388(5)	+0.0008
	0.01	<i>prediction</i>	0.69125(2)	—	0.69036(1)	—
	0.001	<i>prediction</i>	0.140913(1)	—	0.140832(1)	—
12	1.00	65.7001(1) ^H	65.717(1)	+0.0257	65.69556(5)	−0.0069
	0.50	39.1596(1) ^H	39.1786(8)	+0.0485	39.1604(3)	+0.0020
	0.10	12.26984(8) ^H	12.2731(2)	+0.0266	12.2743(2)	+0.0363
	0.01	<i>prediction</i>	2.48620(5)	—	2.47363(4)	—
	0.001	<i>prediction</i>	0.515823(3)	—	0.515365(4)	—
20	1.00	155.8822(1) ^H	—	—	155.8738(7)	−0.0054
	0.50	93.8752(1) ^H	—	—	93.8789(5)	+0.0039
	0.10	29.9779(1) ^H	—	—	29.9888(2)	+0.0364
	0.01	<i>prediction</i>	—	—	6.14645(5)	—
	0.001	<i>prediction</i>	—	—	1.293033(6)	—

Reference-energy provenance: values marked ^H are Høgberget’s fixed-node diffusion Monte Carlo energies [55], as tabulated in the published compilation of Nordhagen *et al.* [56]. The same values are reproduced by Haas [58], whose compilation was used during development. The $N=2$ values come from the reference column of the same compilation, which draws on both Høgberget’s DMC and Taut’s analytic solutions [57]; since Taut’s construction is quasi-exactly solvable only at particular frequencies, we do not assign individual $N=2$ rows between the two, beyond noting that 3.00000 at $\omega = 1$ is the exact analytic value. The foundational DMC study of these dots is Pederiva *et al.* [15] (spanning $N \leq 13$; see also its erratum [59]), cited here for the method, not for these particular numbers. Rows marked *prediction* have no reference value to compare with.

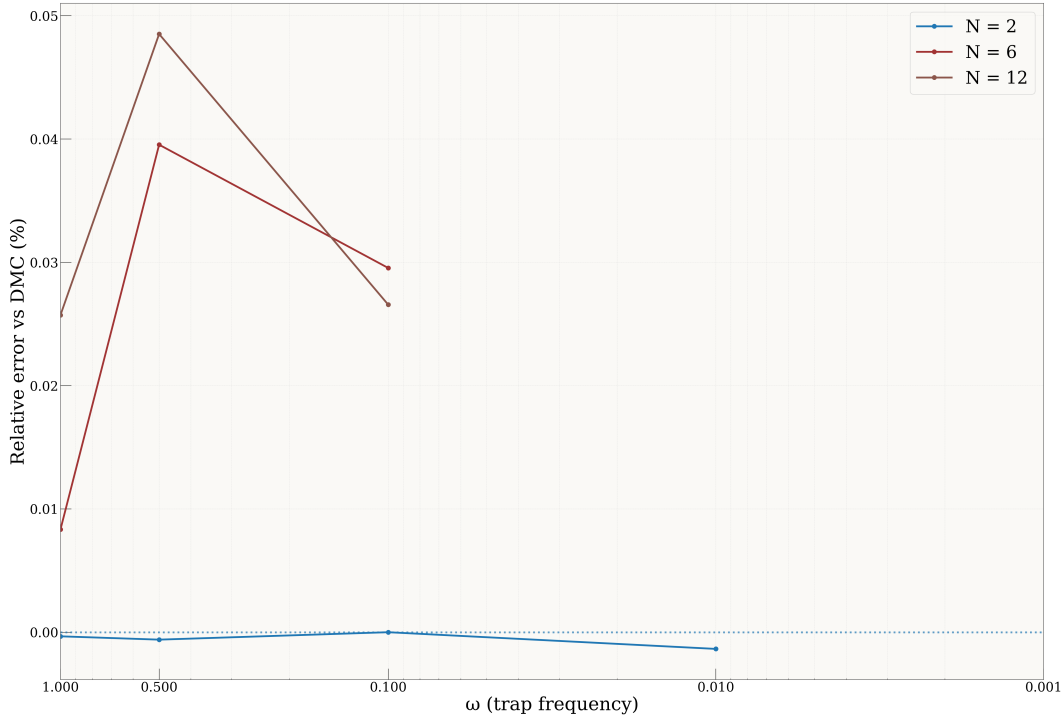


Figure 6.1: Relative deviation of the conv.-BF energies from the external references (Table 6.2) at every (N, ω) where one exists: fixed-node DMC [55] for $N=6$ and 12, and the compiled DMC and analytic [57] values for $N=2$. Deviations are at the 10^{-3} % level for $N=2$, comparable to the reference’s own precision, and 0.008–0.05 % for $N=6$ and 12, with no trend in ω .

The same benchmark, without a Markov chain. Everything above was trained with a Markov chain: residual pretraining, then an SR tail that draws its samples from $|\Psi|^2$ by Metropolis. A second route removes the chain from the gradient entirely, replacing it with an analytic proposal and importance weights. It reaches comparable energies over part of this range. Its performance, its failure boundary and the mechanism behind both are the subject of §7.3, and the energies are tabulated there (Table 7.5) and not here, because they are not interpretable until that mechanism is in hand.

Where the tables stop. One subtlety matters here, because the MP-BF ansatz sits *below* the quoted DMC reference in four cells, and at $N=12$ and $N=20$, $\omega = 1$, by more than ten combined standard errors (for example 65.6956 vs. 65.7001 Ha at $N=12$). This does not violate the variational principle, which bounds the energy from below only by the *exact* ground state, $E_{\text{VMC}} \geq E_0$; the DMC benchmarks are *fixed-node* calculations and carry their own nodal bias. A better nodal surface in the network would produce exactly this pattern. It is not the only thing that would, but one alternative can be set aside. A difference in the convention for the Hamiltonian or its units is ruled out: the same ansatz reproduces the exact two-electron value 3.00000 at $\omega = 1$, and a convention mismatch would shift every cell, not four. What remains is a time-step or population extrapolation in the reference, an underestimated statistical error on either side, or an error somewhere in the chain of compilations by which the reference reached us (see the notes to Table 6.2), and nothing here distinguishes these from better nodes. The clean test is a fixed-node DMC calculation that uses the network’s own nodes as its trial surface, which we have not run. The sub-reference values are therefore recorded as deviations and not interpreted.

Wherever an external check exists, the pattern is the same. For $N=2$ the ansatz reaches reference accuracy at every ω at which it is compared with one, and for $N \in \{6, 12\}$ the relative deviation stays within 0.008–0.05% across the benchmarked range ($\omega \geq 0.1$; Fig. 6.1). Below $\omega = 0.1$ there is no DMC reference for $N \geq 6$, and we make no claim of absolute accuracy there; what we can say is that the energies join smoothly onto the benchmarked regime and, in the deep-Wigner limit, follow a power law whose fitted exponent, 0.689 for $N=6$ (§7.4), sits close to the classical Wigner value $2/3$. The efficiency is as notable as the accuracy: the SR tail converges in 10^3 – 3×10^3 iterations of $\sim 3 \times 10^3$ samples each, a modest optimisation budget for a fully many-body wavefunction. We attribute this stability to a small set of conditioning choices— trap-unit scaling, soft-core pair features with $ds/dr \rightarrow 0$ at coalescence, and

an explicit analytic cusp—each of which keeps the gradients and the Laplacian numerically tame from the shallowest trap to the tightest.

6.2 Wigner–molecule crossover

Up to this point the wavefunction has been the thing under examination. Here it becomes the instrument.

The result of this section is that the crossover is staged. As the trap weakens the cloud first expands and its shells stiffen *radially*—the width-to-radius ratio falls, discrete occupancies emerge, the energy partition approaches classical force balance—while the electrons remain very nearly disordered in angle within those shells. Angular structure then develops in two steps of its own: a small, gradually growing correlation that is measurable well before the Wigner regime, and a large bond-orientational ordering confined to the final decade of confinement. Against the random-angle baseline the shell bond order runs at 0.99–1.01 of its null value from $\omega = 1$ down to $\omega = 0.1$, reaches 1.08 at $\omega = 0.01$ for $N=6$ (and stays within 1.02 at $N=12$ and $N=20$), and only at $\omega = 10^{-3}$ jumps to 1.16–1.63 on every ring at every particle number. Radial localisation and angular crystallisation are not one transition, and the diagnostics below are arranged to separate them.

Which states these are. Every state in this section is the lowest we find with $N_\uparrow = N_\downarrow$ held fixed, starting from the closed-shell singlet of the non-interacting problem. That is not a formality here. Fixing the spin projection does not fix the total spin, since an $S = 1$ state also has an $S_z = 0$ component, and we never compare sectors of different S . In parabolic dots the interaction is known to drive spin transitions as it strengthens—for instance $S = 0 \rightarrow S = 1$ at $N = 6$ in the strongly correlated regime [2]—so a change of ground-state spin is plausible precisely at the weakest traps, where the structural conclusions below are strongest. The deep-Wigner states are therefore the lowest states found within a fixed spin-projection sector, not established global ground states, and every statement below is about them. There is a reason to expect the *charge* geometry to be less sensitive than the energy ordering—deep in the Wigner regime exchange is weak and states of different S become nearly degenerate over the same classical arrangement—but that is an expectation we have not tested, not a result.

The subsections take that in order. §6.2.2 shows the cloud expanding and acquiring visible radial structure; §6.2.3 shows interactions taking over and the trap–interaction partition approaching its classical form; §6.2.4 shows the shells stiffening radially; and §6.2.5 shows angular order finally appearing, tested against a null in which the angles are random. §6.2.6 collects what the diagnostics agree on and what they do not establish.

What we are looking for cannot be seen in the observable one would naturally plot. In a rotationally symmetric trap, the electrons of a Wigner molecule localise in *relative* coordinates (concentric rings, polygon-like angular correlations) while the whole pattern is free to rotate. Average over that rotation, as any lab-frame density does, and the rings smear into something close to uniform. The order is there; the density hides it. Every diagnostic below exists to look past that averaging.

We combine one-body densities, label-invariant shell topology, rotation-registered angular order, shell radii and gaps, pair statistics, and Lindemann-type disorder, across $N = 2, 6, 12, 20$ and $\omega \in \{1, 0.5, 0.1, 0.01, 0.001\}$ [2, 5]. No one of them is decisive on its own. They fail in different ways, which is why they are reported together.

6.2.1 Observables and diagnostics

For each (N, ω) we draw $\mathbf{X} \sim |\Psi|^2$ and report (i) the pair distribution $g(r)$, (ii) the single-particle radial density $P(r) \propto r \rho(r)$, and summary statistics: r_{mode} , $\langle r \rangle$, σ_r , FWHM, quantiles q_{10}, q_{50}, q_{90} , and a localisation ratio $\gamma = \sigma_r / r_{\text{mode}}$ (smaller $\gamma \Rightarrow$ stiffer localisation). For multi-electron systems we additionally monitor shell-resolved bond-orientational order $|\Phi_m|$ and a dimensionless Lindemann-type ratio for positional fluctuations on each ring [17].

Density parameter r_s . Following Egger *et al.* [2], we estimate the Wigner–Seitz parameter from the first maximum r^* of the spin-summed pair correlation, $r_s \equiv r^* / a_B^*$. The Fermi–liquid $\rightarrow$ Wigner–molecule crossover occurs near $r_s \simeq 4$ in parabolic dots; our weak-trap cases lie well beyond this threshold [2, 14].

6.2.2 One-body densities: visual crossover

Figures 6.2 and 6.3 compare one-body densities at strong confinement ($\omega = 1$) and weak confinement ($\omega = 10^{-3}$) for $N = 2, 6, 12, 20$. At $\omega = 1$ the density remains compact and the dominant change with N is progressive shell filling under the harmonic trap. At $\omega = 10^{-3}$ it broadens by two orders of magnitude and develops pronounced *radial* structure at every N .

What that structure can and cannot establish follows from the symmetry argument above. The lab-frame density is an average over all orientations of the molecule, so it retains radial information and destroys angular information by construction. These figures are therefore evidence of *radial shell localisation* and are silent on whether the electrons are angularly ordered within those shells—a ring of six electrons at fixed 60° spacing and six electrons sliding freely around the same ring produce the same picture. The angular question is taken up in §6.2.5, where it needs a co-rotating frame and a null model to answer.

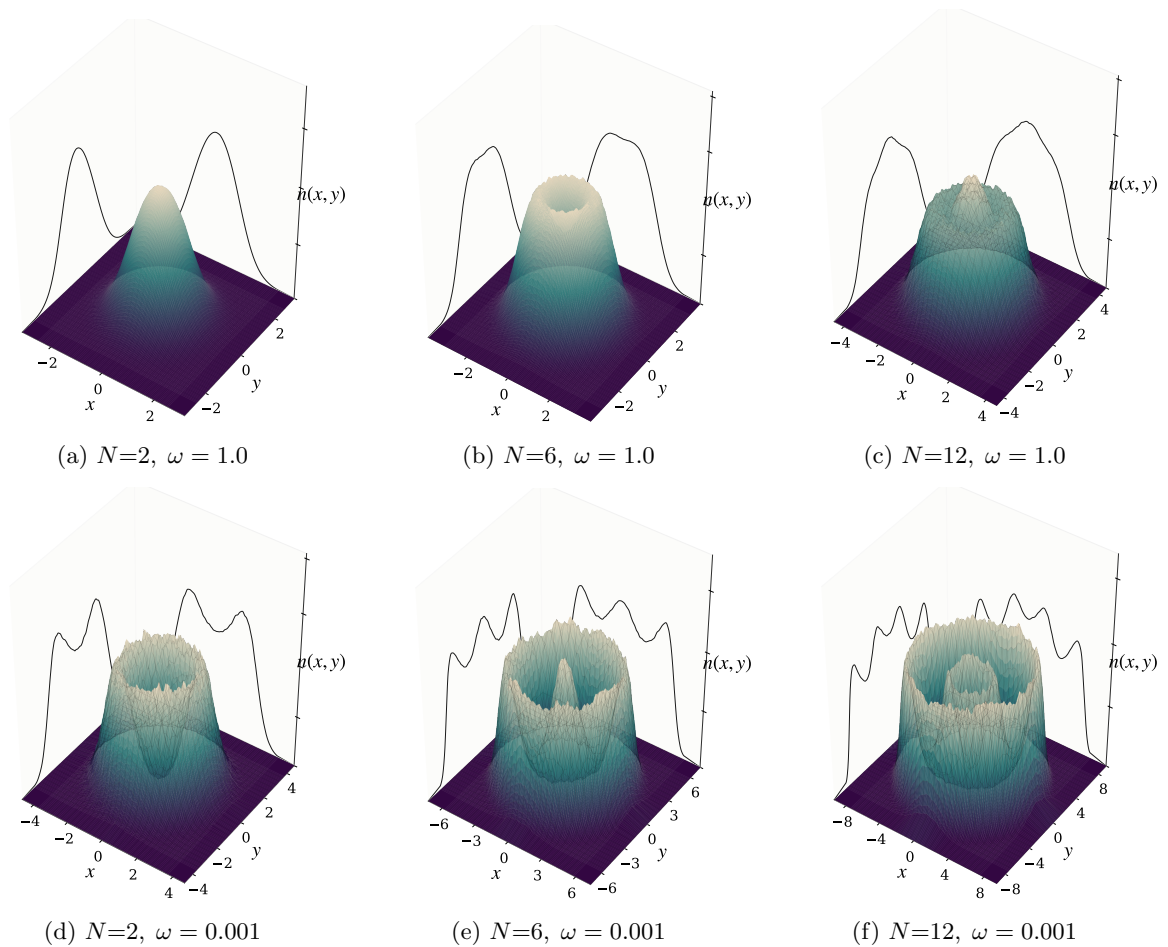


Figure 6.2: One-body densities for $N = 2, 6, 12$ at $\omega = 1.0$ (top) and $\omega = 0.001$ (bottom). The vertical axis is the one-body density $n(x, y)$ over the two Cartesian components of a single particle's position, in trap units; the curves on the back walls are its marginals. As confinement weakens the densities broaden and develop pronounced *radial* shell structure. Being lab-frame averages over all orientations of the molecule, they cannot show angular order: that requires the registered diagnostics of §6.2.5.

6.2.3 Energy-based crossover diagnostics

The energy partition supports two conclusions.

Interactions take over. The interaction-to-kinetic ratio $\Gamma = \langle V_{\text{int}} \rangle / \langle T \rangle$ rises across $\omega \in [1, 10^{-3}]$ from 0.9 to 8.6 at $N=2$, from 2.4 to 24.5 at $N=6$ and from 3.6 to 41.1 at $N=12$ (Fig. 6.4a). The dot stops being a kinetic-energy-dominated droplet and becomes a Coulomb-dominated one.

Classical force balance emerges. At $\omega = 10^{-3}$ the trap–interaction partition $2\langle V_{\text{trap}} \rangle / \langle V_{\text{int}} \rangle$ reaches 1.23, 1.07 and 1.05 at $N = 2, 6, 12$, so the twelve-electron dot satisfies the classical condition $V_{\text{int}} \approx 2V_{\text{trap}}$

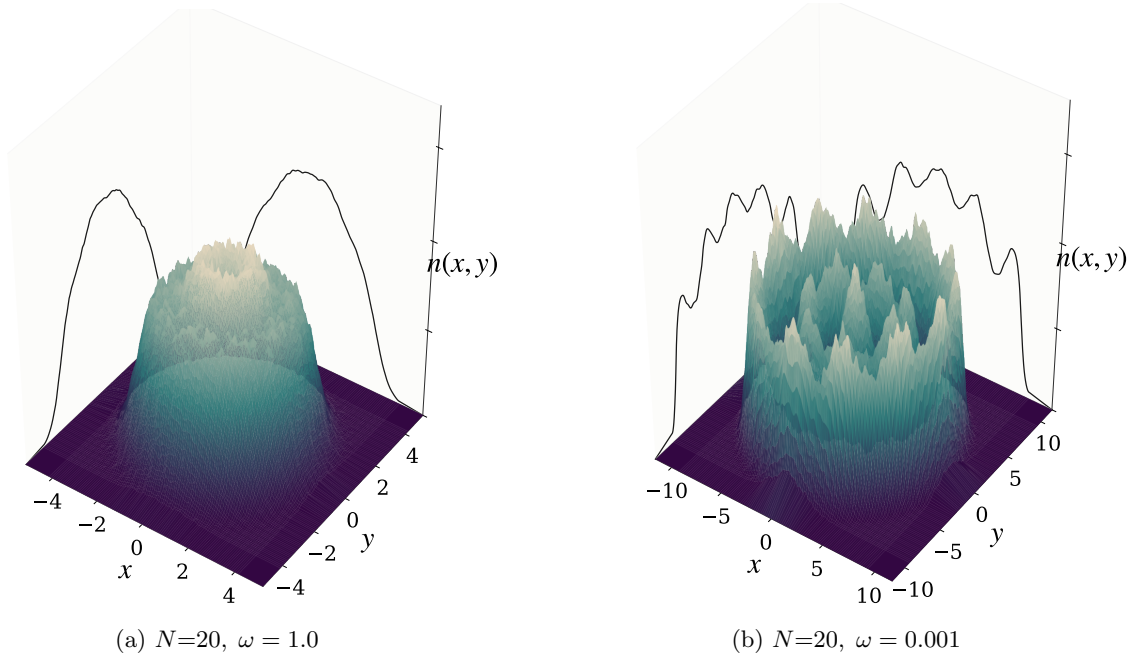
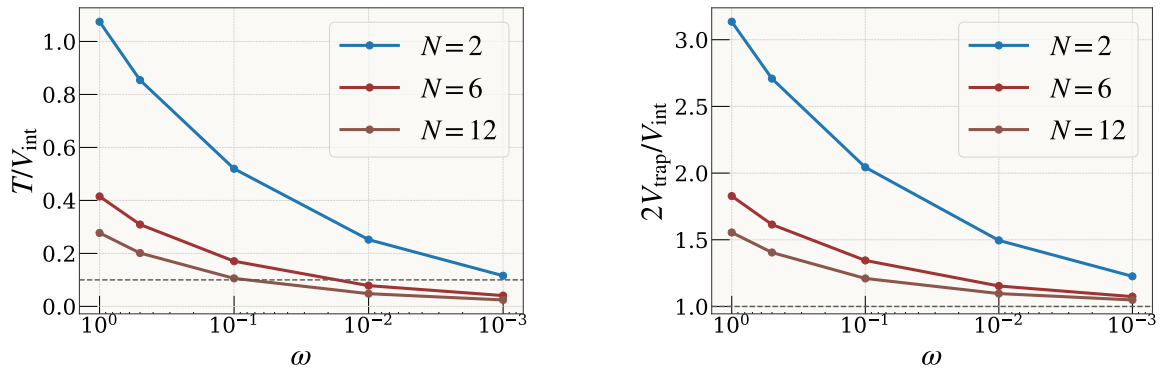


Figure 6.3: One-body densities for $N=20$ at strong and weak confinement; the vertical axis is $n(x, y)$ as in Fig. 6.2. The weak-trap case shows strong radial broadening and clear multi-shell organisation. As above, this is radial evidence only.



(a) T/V_{int} vs. ω for $N=2, 6, 12$. The horizontal line marks $T/V_{\text{int}}=0.1$.

(b) $2V_{\text{trap}}/V_{\text{int}}$ vs. ω for $N=2, 6, 12$. The horizontal line marks the classical virial ratio $2V_{\text{trap}} = V_{\text{int}}$.

Figure 6.4: Energy-based Wigner diagnostics as functions of trap frequency. As ω decreases the dots become interaction dominated ($T/V_{\text{int}} \ll 1$) and the trap–interaction partition approaches the classical virial form, most clearly for $N=12$.

to within 5% (Fig. 6.4b). This ratio approaches one over the sampled grid. By the virial relation it is linked to T/V_{int} , so it is a consistency view rather than an independent diagnostic.

Both are consistent with real-space expansion. The rms radius extracted from $V_{\text{trap}} = \frac{1}{2}\omega^2 \sum_i \langle r_i^2 \rangle$ grows from $1.1\text{--}2.0 a_B^*$ at $\omega = 1$ to $70\text{--}170 a_B^*$ at $\omega = 10^{-3}$, faster than the non-interacting $\omega^{-1/2}$ law, which is interaction-driven swelling. The fitted exponents for the individual energy components and for the radius, which support these statements without adding to them, are collected in Appendix F.1.

6.2.4 Radial localisation and pair correlations

Over the weakest three confinements, the typical one-body radius follows $r_{\text{mode}} \sim \omega^\alpha$ with $\alpha \simeq -0.626$ at $N=2$ —close to the classical two-body value $-2/3$, which nothing in the ansatz knows about. At $N=2$ the width-to-scale ratio $\gamma = \sigma_r/r_{\text{mode}}$ falls as well, from 0.49 to 0.29, so the pair stiffens rather than merely dilating. At $N=6$ the same global ratio barely moves (Table 6.3), and radial stiffening there has to be read from quantities that resolve the shell, not from this one.

Table 6.3: Radial localisation across the confinement range. r_{mode} is the mode of the single-particle radial density in Bohr; $\gamma = \sigma_r/r_{\text{mode}}$ is the width-to-scale ratio. Full diagnostics—mean, width, FWHM, quantiles, ring-angle spread and pair Lindemann ratio—are in Appendix F. The $N=6$ values at $\omega \leq 0.01$ come from an earlier run than the shell analysis and Fig. 6.6 (Appendix H).

ω	$N=2$		$N=6$	
	r_{mode}	γ	r_{mode}	γ
1.00	1.44	0.491	2.09	0.442
0.50	2.30	0.437	3.07	0.445
0.10	3.60	0.480	8.17	0.429
0.01	14.19	0.396	33.15	0.420
0.001	64.29	0.287	113.55	0.421

The exponent is not the sharpest form of this check. Because r_{mode} is the mode of the *single-particle* radial density, it mixes the central electron of a (1, 5) topology with the ring and is not the ring radius. The shell-resolved outer radius is, and comparing it with the classical Wigner ring $1.334\omega^{-2/3}$ tests a prefactor as well as a power. That ratio falls monotonically—1.247, 1.188, 1.098, 1.032, 1.017 from $\omega = 1$ to 10^{-3} —so at the weakest trap the quantum ring sits 1.7% outside the classical one, and the fitted exponent on that quantity is -0.637 over the full grid and -0.650 over the weakest three confinements (Appendix F.4). Nothing in the ansatz, the objective or the sampler knows the number 1.334.

The two sizes stiffen differently, and the difference is physical. At $\omega = 10^{-3}$ the $N=2$ pair sits at separation $r_{12} \approx 122$ with Lindemann ratio $\gamma_{r_{12}} \approx 0.20$ and a relative angle sharply peaked at π : a dilute, anti-aligned dimer. At $N=6$ the global γ plateaus near 0.42, and the evidence for stiffening comes from the shell-resolved and pair quantities instead: the outer ring’s own relative radial width falls from 0.32 at $\omega = 1$ to 0.15 at 10^{-3} (Table F.4), and the ring’s angular fluctuations and pair Lindemann ratio fall monotonically. A single-particle width is the weakest of the radial diagnostics: it mixes the central electron with the ring, and it is the one quantity here that does not resolve the shell structure it is being asked about.

6.2.5 Shell topology, angular order, and the rotating Wigner molecule

The densities plotted above look like smooth rings, yet the electrons are localised. Both are true, and the reason is symmetry. Because the Hamiltonian is rotationally invariant, so is its ground state, and any lab-frame average necessarily smears an angularly ordered configuration into a featureless ring. The order is real; it simply lives in relative coordinates, and looking for it in the lab frame is looking in the one place it is guaranteed not to be. Everything below follows from taking that seriously.

Concretely, a Wigner molecule’s internal angular order must be measured through rotation-invariant relative quantities. For a shell s with n_s particles we compute the bond-orientational modulus $|\Phi_{n_s}|$ and a corresponding Lindemann-type ratio quantifying angular stiffness. Lab-frame densities remain ring-like (the “rotating Wigner molecule” scenario), while the co-rotating frame reveals crystal-like polygon order [2, 5].

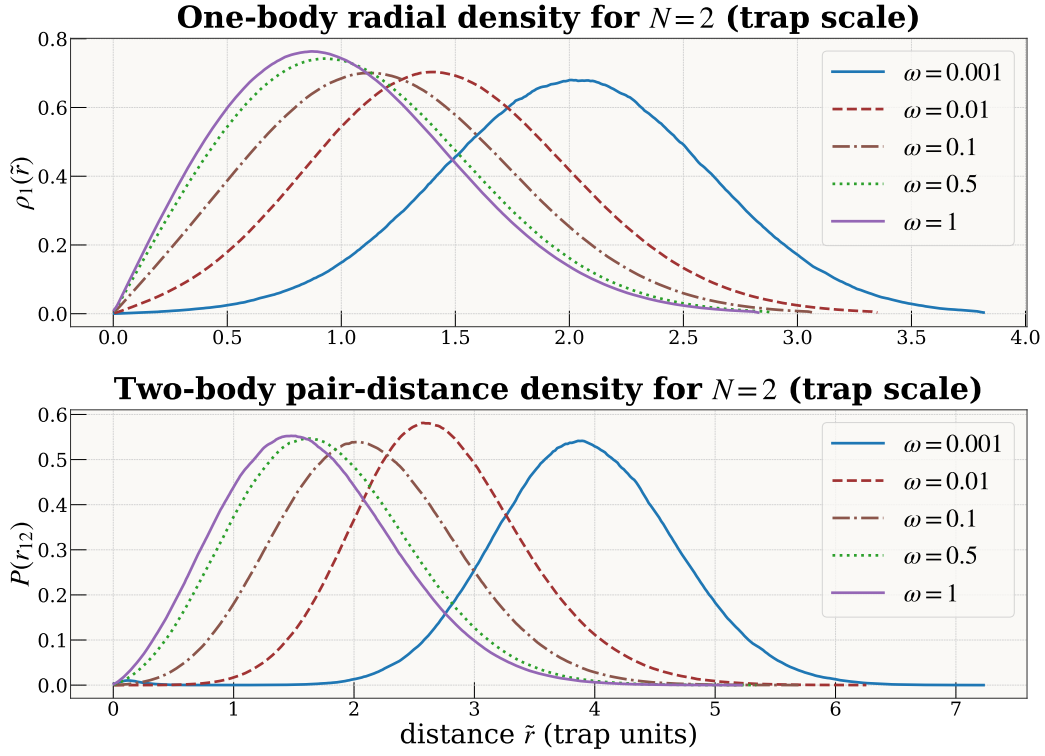


Figure 6.5: One-body radial density $\rho_1(\tilde{r})$ (top) and pair-distance density $P(r_{12})$ (bottom) for $N=2$ across ω , in trap units $\tilde{r} = \sqrt{\omega} r$.

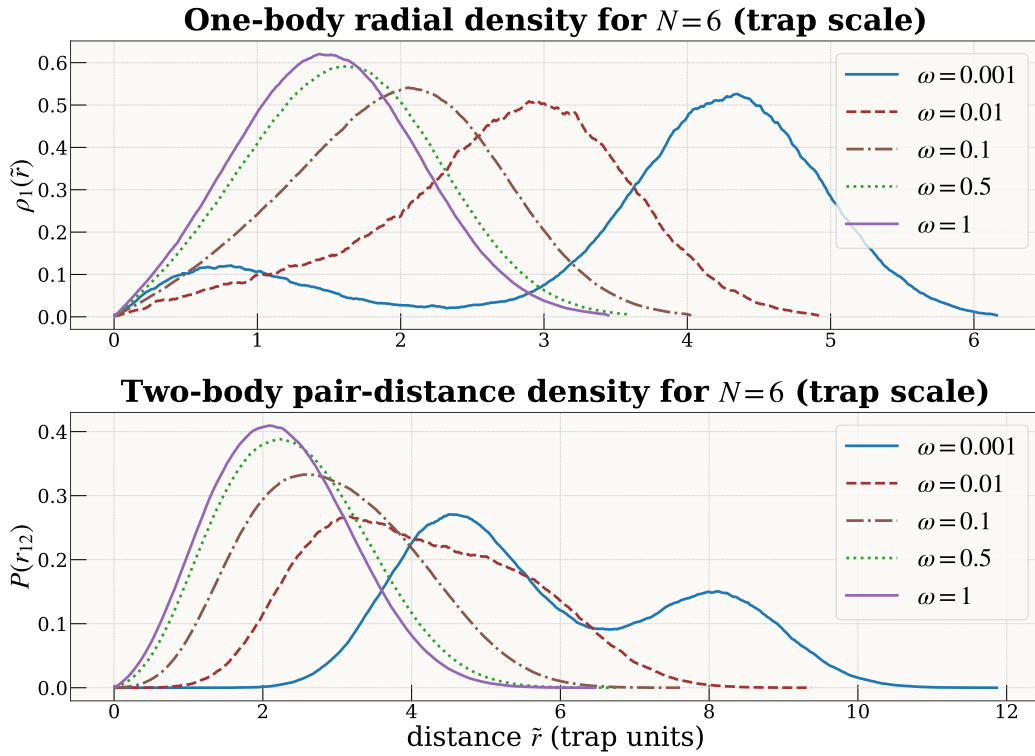


Figure 6.6: As Fig. 6.5, for $N=6$. At $\omega = 10^{-3}$ the one-body density separates into the central electron and the ring, and the pair distribution splits into centre–ring and nearest-neighbour distances (first peak) and cross-ring distances (second).

$N=6$: single ring with sector mixing. At $\omega = 0.01$ the system fluctuates between a $(1, 5)$ sector (fraction ≈ 0.69) and a $(0, 6)$ single-ring sector (fraction ≈ 0.31), with bond order $|\Phi_5| = 0.453 \pm 0.214$ and $|\Phi_6| = 0.466 \pm 0.206$ respectively. Two conventions apply here and below. The quoted $\pm$ is the frame-to-frame standard deviation of $|\Phi_m|$, not a standard error on its mean, which with 3×10^4 frames is smaller by two orders of magnitude. And these are *sector-conditioned* values, computed on the frames assigned to that sector in an earlier analysis pass than the modal-topology values of Appendix F.2; the two estimators are set side by side in Table F.3. Both must be read against the random-angle null, the exact mean of $|\Phi_n|$ for independent uniform angles (Eq. (F.1)), which is 0.402 for $n=5$ and 0.366 for $n=6$: the ratios here are 1.13 and 1.27, so the measured bond order at $\omega = 0.01$ exceeds the random-angle value by a modest margin rather than sitting on it. At $\omega = 10^{-3}$ the ring stiffens further: $(1, 5)$ fraction ≈ 0.80 , $(0, 6) \approx 0.20$, with $|\Phi_5| = 0.597 \pm 0.224$ (Lind ≈ 0.225) and $|\Phi_6| = 0.663 \pm 0.183$ (Lind ≈ 0.205)—now far above the random-angle null, by factors of 1.49 and 1.81. This is the confinement at which n -fold angular order becomes *large* rather than merely resolvable (Appendix F.2). The distinction matters because with 10^4 – 10^6 frames a 2% excess over the null is already resolved at tens of standard errors; detectability is therefore not what separates the regimes, magnitude is. Classically $(1, 5)$ is the ground-state topology and $(0, 6)$ a proximate competitor; the observed quantum mixture mirrors this hierarchy [3, 4]. In both weak-trap cases the global $g(r)$ is reproduced to numerical precision by the appropriate inner–inner / inner–outer / outer–outer (II/IO/OO) mixtures.

$N=12$: two shells and commensurability. At $\omega = 0.01$ two-shell formation is already frequent (fraction 0.761 at radial-gap threshold $\tau=3.0$), distributed across sectors $(1, 11) = 32.6\%$, $(2, 10) = 25.7\%$, $(3, 9) = 13.7\%$. At $\omega = 10^{-3}$ shelling becomes ubiquitous ($f_{2\text{shell}} \approx 0.99$ – 0.92 for $\tau = 2.0$ – 3.0) and the *commensurate* $(3, 9)$ split emerges as modal (43.3% of quality-cut frames), with strong bond order on both rings: $|\Phi_3| = 0.687 \pm 0.229$ (Lind_{in} = 0.223), $|\Phi_9| = 0.526 \pm 0.194$ (Lind_{out} = 0.224). Recombining the inner–inner, inner–outer and outer–outer histograms with the combinatorial weights of the ideal $(3, 9)$ topology reproduces the global $g(r)$ almost exactly. Because the component and total histograms use the same frames, this checks the bookkeeping rather than the shell model itself. We therefore use a held-out generative test (Table 6.4). Shell radii and widths are fitted on half the frames; each shell is modelled as a regular n_s -gon carrying a per-frame global rotation and a fitted angular jitter; and the model is scored on the other half. It reproduces the pair distribution to cosine similarity 0.987–0.999 across the whole grid, against 0.936–0.996 for an angle-randomised null built from the same fitted radial distributions. The difference between them, $\Delta \cos$, adds information about angular order beyond shell radii, and it grows monotonically as the trap weakens at every particle number—at $N=12$ from 0.006 at $\omega = 1$ to 0.026 at $\omega = 10^{-3}$, at $N=6$ from 0.012 to 0.058, at $N=20$ from 0.003 to 0.010. Shell radii alone account for most of the pair structure at strong confinement; angular structure accounts for a growing remainder as the molecule forms. How firmly is a separate question, because the smallest values are a few thousandths, and it was tested directly (Appendix F.3). Every $\Delta \cos$ carries a block-bootstrap interval that excludes zero—at $N=20$, $\omega = 1$ it is $[0.0027, 0.0030]$ —and the growth with $1/\omega$ is resolved at every particle number. It survives changes of histogram binning and of the train/test split essentially unchanged. What does move it is the angular-jitter model, by up to a third, and the shell threshold where that changes the partition; the size of $\Delta \cos$ is model-dependent, its sign and its trend are not. The candidate topology and radial partition were selected in the preceding full-data analysis, so the split tests conditional reconstruction rather than fully held-out model selection.

That growth has a different shape from the bond order. $\Delta \cos$ rises by a factor of 1.1–2.1 per decade at every particle number, monotonically and without a threshold, whereas the bond order stays flat at its null and then jumps. The two are not in conflict: $\Delta \cos$ responds to *any* angular correlation that improves the pair-distance density, while $|\Phi_n|$ responds specifically to n -fold order on a registered ring. Weak angular correlations build gradually from strong confinement onward, and crystalline n -fold order emerges from them only in the final decade—three stages, not a single threshold, with the middle one measured by a quantity whose sign and trend are robust but whose size depends on the shell model.

$N=20$: three-shell molecule and a rapid radial reorganisation. At $\omega = 10^{-3}$ the dominant shell topology is $(1, 7, 12)$ —a clear three-shell Wigner molecule with widely separated rings at radii $r_0 \approx 38$, $r_1 \approx 132$, $r_2 \approx 247$ (gaps ≈ 94 and ≈ 115). The lab-frame pinning metric remains small ($C_{\text{global}} \approx 0.06$), as expected for a freely rotating molecule. That number also answers a question the ansatz raises. Its displacement head is not rotation-equivariant by construction (§3.5.3), so a trained state could in principle carry an orientation fixed to the lab frame and read as angular order; such a state would show a large C_{global} , and this one does not. The order the per-shell moduli measure is internal to the molecule. The

Table 6.4: Conditional held-out shell reconstruction. Radii, widths and angular jitter are fitted on half the frames and scored on the other half. $\cos_{\text{poly}}$ and $\cos_{\text{uni}}$ compare the polygon and angle-randomised models with the empirical pair-distance density; $\Delta \cos$ is their difference. Intervals are 95% block-bootstrap intervals. The topology and radial partition were selected on the full data; sensitivity checks are in Appendix F.3.

N	ω	occ.	$\cos_{\text{poly}}$	$\cos_{\text{uni}}$	$\Delta \cos$	95% interval
6	1.0	1, 5	0.9988	0.9866	0.0122	[0.0122, 0.0136]
6	0.5	1, 5	0.9986	0.9836	0.0150	[0.0147, 0.0160]
6	0.1	1, 5	0.9974	0.9741	0.0234	[0.0219, 0.0236]
6	0.01	1, 5	0.9922	0.9536	0.0386	[0.0371, 0.0395]
6	0.001	1, 5	0.9942	0.9358	0.0585	[0.0556, 0.0592]
12	1.0	1, 11	0.9993	0.9937	0.0056	[0.0054, 0.0059]
12	0.5	1, 11	0.9990	0.9927	0.0063	[0.0060, 0.0066]
12	0.1	1, 11	0.9980	0.9896	0.0085	[0.0083, 0.0088]
12	0.01	1, 11	0.9929	0.9805	0.0124	[0.0124, 0.0132]
12	0.001	3, 9	0.9868	0.9604	0.0264	[0.0258, 0.0271]
20	1.0	1, 19	0.9992	0.9964	0.0028	[0.0027, 0.0030]
20	0.5	1, 19	0.9990	0.9958	0.0032	[0.0030, 0.0033]
20	0.1	1, 19	0.9982	0.9942	0.0040	[0.0038, 0.0040]
20	0.01	1, 19	0.9947	0.9900	0.0047	[0.0046, 0.0049]
20	0.001	1, 7, 12	0.9915	0.9820	0.0095	[0.0091, 0.0095]

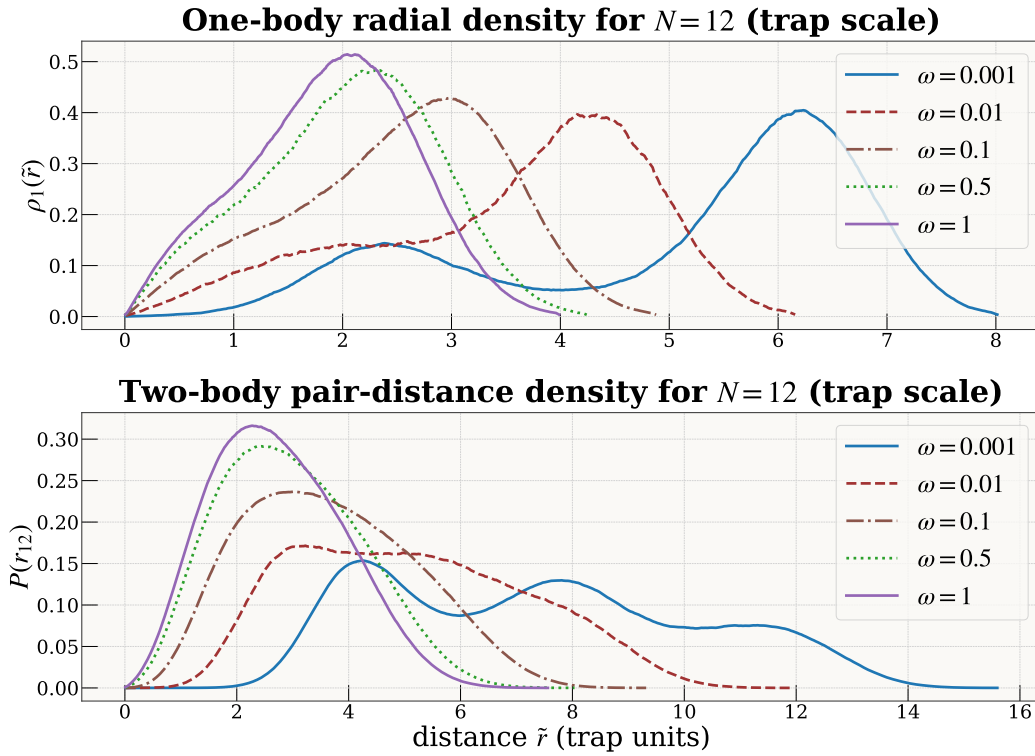


Figure 6.7: As Fig. 6.5, for $N=12$.

cross-shell quantity $\tilde{C}_{\text{global}}$ is omitted: it depends on a registration convention whose original analysis code was not retained (§5.1.3). The per-shell moduli and angular-spacing diagnostics below are rotation-invariant and carry no such dependence. Angular-spacing disorder on the two occupied outer shells is low: $\lambda_7 \approx 0.35$ and $\lambda_{12} \approx 0.27$.

Already at $\omega = 10^{-2}$ the radial gap detector returns a (1, 19) partition rather than a three-shell one: the shell radii contract by an order of magnitude ($r_0 \approx 10$, $r_1 \approx 45$), the outer-shell angular disorder rises to ≈ 0.57 . The wording matters here. At $\omega = 10^{-3}$ the (1, 7, 12) label describes a Wigner *topology*, because it is accompanied by separated shells, shellwise angular order and low Lindemann disorder. At $\omega = 10^{-2}$ and above, with angular order at its random-angle null and disorder high, (1, 19) is what the *radial classifier returns* on a state that is not angularly ordered at all—a decomposition of the radial density, not a crystalline shell structure. We use the occupancy notation in both cases because the detector does, but only the weakest-trap case earns the word topology. At $\omega = 1$ the classifier returns the same (1, 19) partition, with compact radii ($r_0 \approx 0.55$, $r_1 \approx 2.3$) and high angular disorder ($L_{\text{out}} \approx 0.69$). What the data show, then, is a rapid radial reorganisation somewhere between $\omega = 10^{-3}$ and 10^{-2} , not a collapse of one crystalline topology into another: the state at 10^{-2} has no demonstrated angular order to lose. How rapid, and where in that decade it happens, the grid cannot say. It samples one point per decade, and the radial classifier reports a partition at each point rather than tracking a structure between them.

6.2.6 The crossover across diagnostics

Six diagnostics, applied at four particle numbers and five confinements, together show the crossover. As ω falls from 1 to 10^{-3} the densities expand and develop rings, and the shell topology resolves into the discrete finite- N patterns that classical Coulomb clusters predict—(1, 5) at $N=6$, (3, 9) at $N=12$, (1, 7, 12) at $N=20$ [3, 4]. The shells separate and generally stiffen, although several inner-shell width estimates are nonmonotonic. Shellwise bond order stays within 1% of its random-angle null down to $\omega = 0.1$ and reaches 1.16–1.63 times it only at $\omega = 10^{-3}$ (Appendix F.2). Radial stiffening precedes n -fold angular ordering instead of accompanying it. Lab-frame order stays near zero throughout, as expected for a freely rotating molecule. The energy partitioning approaches the classical virial form, with Γ reaching ~ 40 for $N=12$ at the weakest trap. None of these was built into the ansatz. At $N=12$, (3, 9) is modal while (1, 11) and (2, 10) remain appreciably populated. This sector mixing is consistent with the rotating-Wigner-molecule picture [5, 14]. At $N=20$ the three-shell structure at $\omega = 10^{-3}$ gives way, by $\omega = 10^{-2}$, to a radial partition of a state without angular order; the grid is too coarse to place that reorganisation more finely than within the decade. Neither the (1, 7, 12) configuration nor the broad separation of radial and angular ordering is new [6, 7, 14]. The specific contribution here is the combination of sector-conditioned bond-order and angular-spacing statistics, finite- n null calibration, topology-resolved measurements across the selected grid, and a held-out polygon reconstruction of $g(r)$ against an angle-randomised baseline.

The diagnostics do not carry equal weight, and they do not all support the same claims. Table 6.5 sets out which one carries which statement and how firmly. The strongest evidence is at the deepest Wigner points; the qualitative staging is better established than where, within the coarse ω grid, each stage sets in.

Two qualifications bound all of it. The states are the lowest we find within a fixed spin-projection sector, and we have not scanned total spin, so these are not proven global-spin ground states. And the combinatorial reconstruction of $g(r)$ establishes decomposability rather than sampling correctness; the held-out polygon fit of Appendix F.3 is the version that could have failed, and sampling correctness itself rests on the independent restart and chain diagnostics.

The two angular diagnostics stage the crossover slightly differently, and the difference is informative. Radial localisation (expansion, shell separation, falling shell-resolved width, classical force balance) develops progressively from $\omega = 1$ down. Bond-orientational order is flat at its null through $\omega = 0.1$, departs at $\omega = 0.01$ clearly only at $N=6$, and becomes large at 10^{-3} everywhere. The held-out polygon model, sensitive to any angular correlation rather than to n -fold order specifically, instead grows monotonically throughout, roughly doubling per decade. Read at face value, weak angular correlations build gradually across the whole range while *crystalline* n -fold order appears only in the last decade: a three-stage picture rather than a two-stage one. The two-stage separation of radial and n -fold angular order is the firmer statement; the gradual middle stage is resolved in sign and trend, but its size depends on the shell model used to measure it.

These structural facts then enter the method. The shell radius approaching its classical $\omega^{-2/3}$ scaling—

Table 6.5: Which diagnostic supports which claim about the crossover. “Firm” means a large effect in a state-level or shell-resolved quantity; “weak” marks evidence that is suggestive but small or carried by a quantity known to mix structures.

Claim	Carried by	Strength
The cloud expands faster than the non-interacting $\omega^{-1/2}$	rms radius from $\langle V_{\text{trap}} \rangle$; r_{mode} exponent	firm, all N
Interactions take over	$\Gamma = \langle V_{\text{int}} \rangle / \langle T \rangle$ rises tenfold	firm, $N \leq 12$
Classical force balance emerges	$2\langle V_{\text{trap}} \rangle / \langle V_{\text{int}} \rangle \rightarrow 1.23, 1.07, 1.05$	firm at $N=12$
Shells stiffen radially	shell-resolved widths ($N=6$ ring: $0.32 \rightarrow 0.15$) and pair Lindemann ratios; global γ only at $N=2$	firm; global γ weak at $N=6$
The ring approaches its classical radius	outer radius over $1.334\omega^{-2/3}$: $1.247 \rightarrow 1.017$	firm, $N=6$
Discrete Wigner topologies form	radial classifier with shellwise angular order: $(1, 5), (3, 9), (1, 7, 12)$	firm at 10^{-3} ; above it, radial partitions only
Weak angular correlation builds gradually	held-out polygon model against angle-randomised null, $\Delta \cos$ (block bootstrap)	firm in sign and trend; size depends on the jitter model
n -fold angular order appears late	$ \Phi_n $ against its exact null, which is also the pipeline’s null	firm at 10^{-3} ; absent above 0.1 (resolved departures $\lesssim 1\%$)
Where each stage sets in	one grid point per decade	not established

within 1.7% of the classical prefactor at $\omega = 10^{-3}$, Appendix F.4—and the angular order sharpening in the last decade are exactly the quantities a sampler must respect to reach the Wigner regime without a Markov chain. An origin-centred proposal aims where the wavefunction no longer has any mass; a proposal built from the measured shell radius and angular order tracks it, and carries MCMC-free gradient training far into the correlated regime (§7.4). What the trained model reveals about the physics is what later lets us train it more cheaply.

That is the state seen from outside. What the network had to hold internally in order to produce it is a question about the solver, and Chapter 7 takes it up.

Chapter 7

Architecture, optimisation and sampling

The previous chapter showed that the ansatz works: it reaches reference quality wherever it was compared with a reference, and the states it produces carry the staged Wigner-molecule structure the diagnostics were built to detect. Why the solver can represent those states, and what governs whether they can be trained at all, are separate and harder questions.

Three observations from that chapter remain unexplained. Two networks of the same size, trained the same way and landing on the same energy, need not be the same object. A natural-gradient step that is decisive in one training regime does nothing in another. And an importance-sampled objective reproduces a Markov chain’s answer in one corner of the phase diagram and departs from it in the next.

All three can be examined through the per-sample, column-centred log-derivative matrix

$$O_{ki} = \frac{\partial \log |\Psi_{\theta}(\mathbf{x}_k)|}{\partial \theta_i}, \quad k = 1, \dots, B, \quad i = 1, \dots, P, \quad (7.1)$$

and on its two Gram matrices, the quantum geometric tensor $S = O^{\top} O / B$ and the neural tangent kernel $K = O O^{\top}$ [30, 31]. That gives the chapter its structure:

Architecture determines the shape of the tangent space.

The *optimiser* acts on it through its metric.

The *sampling measure* determines how well that metric can be estimated.

The chapter takes them in that order. The instrument throughout is the *effective dimension* of the tangent space—the participation ratio of the QGT spectrum $\{\lambda_a\}$,

$$d_{\text{eff}}(S) = \frac{(\sum_a \lambda_a)^2}{\sum_a \lambda_a^2}, \quad (7.2)$$

which counts how many directions actually carry Fisher weight, whatever the parameter count says. Beside it sit the local-energy variance $\text{Var}(E_L)$, zero only for an exact eigenstate; the effective sample size; and the condition number $\kappa(S)$, which is reported in §7.2 and turns out to be large everywhere and therefore diagnostic of nothing by itself.

Architectures are compared on a common probe set—the same configurations for every model, never each model’s own $|\Psi|^2$ —and below $\omega = 0.1$ every relative error is measured against our own converged variational energies, so it reports internal consistency rather than accuracy. The tooling is anchored on the analytic two-electron ground state, where the ansatz returns $E = 3.000127$ Ha (+0.004%) at squared overlap 1.000000.

7.1 Message passing and the variational geometry

How do the implemented message-passing and separable model families differ after training? We ask this separately of the correlator and the backflow. The Slater reference, analytic cusp and training protocol are shared, but the correlators use different learned pair features, and the backflows differ in capacity, coordinate preprocessing and centre-of-mass projection (§3.5.3). The comparisons therefore identify differences between the archived implementations; they do not isolate message routing alone.

7.1.1 The correlator: tangent-space compression

At $N = 6$, $\omega = 1$ the two correlators are indistinguishable in energy. Trained to DMC quality across a 20k–164k parameter ladder, every model lands within $\pm 0.1\%$ of the reference, and the best raw energy belongs to a 164k separable network—twice the message-passing parameter count. Both have reached the same observed energy floor at this resolution. That agreement hides two differences developed below: the message-passing correlator has lower local-energy variance across the confinement range and, under the chosen parametrisations, concentrates its tangent weight differently.

Their tangent geometry is not indistinguishable. On a common probe set the message-passing correlator reaches a reference-quality state in $d_{\text{eff}} \approx 1.2\text{--}1.7$ over three seeds, where every separable network needs 2.9–3.6; the two sets do not overlap, and their closest approach is still a factor of 1.7. Capacity does not close the gap. Scaling the separable network eightfold, from 20k to 164k parameters, moves its effective dimension only from 3.6 to 2.9, and the 89k network that matches the message-passing one in size sits at 3.6. The gap moves by at most 0.26 whether scored on either family’s samples or on pooled points. With three and four runs respectively, we report the values rather than a sigma-separation.

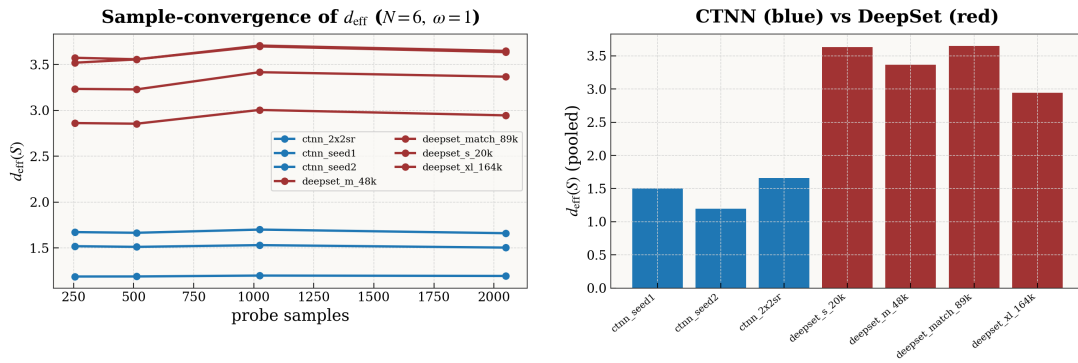


Figure 7.1: Effective dimension of the correlator’s tangent space ($N=6$, $\omega=1$) on a common pooled probe. *Left*: convergence in the number of probe samples. *Right*: per model. The CTNN (blue) sits at $d_{\text{eff}} \approx 1.2\text{--}1.7$ and every DeepSet (red) at $\approx 2.9\text{--}3.6$; scaling the DeepSet from 20k to 164k parameters does not close the gap. d_{eff} depends on the parametrisation (§7.1.1).

The dimension gap is a strong-confinement effect: toward the Wigner crystal both families reach ~ 3.7 , although they occupy different tangent subspaces. A single-seed inversion at the crystal reported in an earlier draft did not survive seeding. Because no backflow is present here, division of correlation between the correlator and backflow is not at issue.

One more limit applies to every tangent-space quantity in this chapter. The QGT spectrum depends on the parametrisation: rescaling a layer or changing a normalisation can reweight tangent directions without changing the state. The result is therefore specific and useful: under the parametrisations and normalisation conventions used here, the message-passing correlator concentrates its empirical Fisher weight in fewer directions. It does not show that the architecture represents the state with intrinsically fewer degrees of freedom. Only the energy and local-energy variance are state-level comparisons, and only the variance separates these families.

Local-energy variance. d_{eff} stops separating the families at the crystal, and rank will turn out to be gauge-like (§7.1.2). The local-energy variance does neither: it is defined on the state, vanishes only for an exact eigenstate, and is directly comparable between any two ansätze evaluated on their own $|\Psi|^2$. Measured at matched training protocol, $\text{Var}(E_L)$ is lower for the message-passing correlator at every confinement (Table 7.1), by factors between 1.2 and 3.4, while the two are indistinguishable in energy.

At the crystal the three-seed means differ by a factor of 1.8, rather than the 7.4 obtained by pairing single best runs. Capacity does not remove the effect: at $\omega = 1$ the capacity-matched 89.4k separable network widens the ratio to 3.4. Intermediate confinements ($\omega = 0.28, 0.05, 0.03$) preserve the ordering at ratios 1.2–1.8. The difference is not resolved at the smallest depth-analysis tier (12k against 9k), so the claim is about trained models at production width, not either architecture at every size.

Table 7.1: Local-energy variance at $N=6$, matched training protocol, both architectures at reference-quality energy ($|\text{err}| \leq 0.13\%$). Lower is better; $\text{Var}(E_L) \rightarrow 0$ only for an exact eigenstate. The basis column gives the evidential status of each row: the $\omega = 0.01$ row is a three-seed mean, the rest are single runs of the production cascade. The unseeded rows establish the ordering, which never reverses, and not the precise factor (§7.1.1). The last row repeats $\omega = 1$ against the capacity-matched 89.4k separable network.

ω	CTNN	DeepSet	ratio	basis
1.0	1.09×10^{-2}	2.51×10^{-2}	2.3	single run, 79.8k vs 47.7k
0.5	4.62×10^{-3}	1.01×10^{-2}	2.2	single run, 79.8k vs 47.7k
0.1	5.63×10^{-4}	1.14×10^{-3}	2.0	single run, 79.8k vs 47.7k
0.01	1.85×10^{-5}	3.35×10^{-5}	1.8	3-seed mean, 79.8k vs 47.7k
1.0	2.75×10^{-2}	9.38×10^{-2}	3.4	matched, 79.8k vs 89.4k

Relating the leading tangent directions to selected operators. At the exact $N = 2$ anchor the selected dictionary describes the leading mode with $R^2 = 1.00$, using a breathing mode (∂_ω), a correlation-hole mode, and the first relative excitation. This is not special to two bodies. Projecting the leading tangent mode of trained correlators onto a dictionary of selected operators (breathing/monopole, quadrupole, correlation-hole, pair-linear), seed-averaged over well-trained checkpoints (energy within 1.5% of reference), the CTNN reaches $R^2 \gtrsim 0.93$ at $N = 6$ and $R^2 \approx 1.00$ at $N = 12$ across the whole confinement range $\omega \in [0.01, 1]$, and both architectures reproduce the $N = 2$ naming at strong confinement (Fig. 7.2). For the message-passing correlator, the direction carrying most of the tangent weight is well described by this small dictionary.

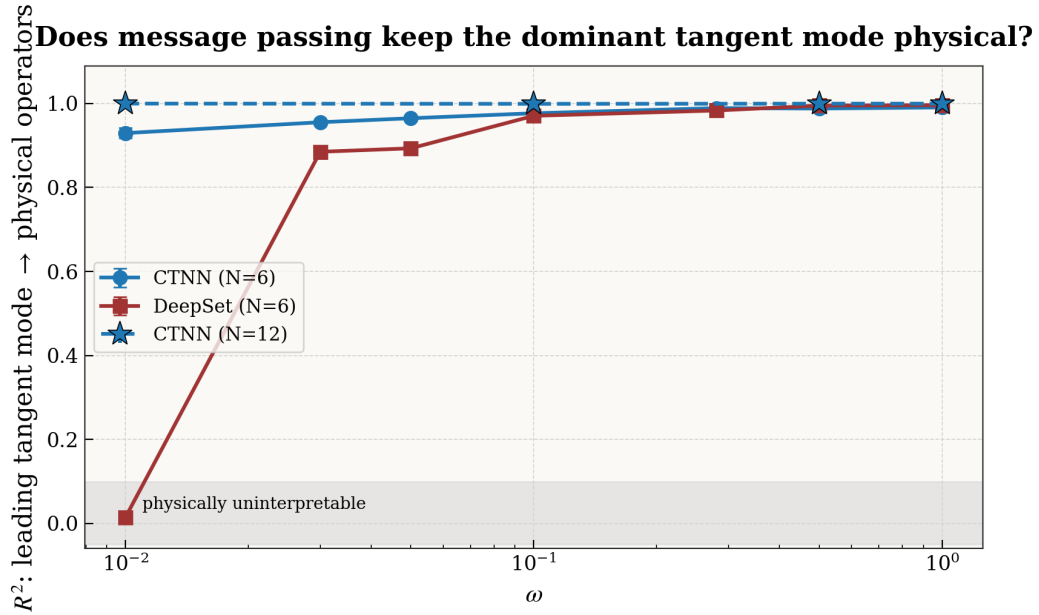


Figure 7.2: R^2 of the correlator’s leading tangent-kernel eigenmode against the selected four-operator dictionary, seed-averaged over well-trained checkpoints. The CTNN retains a high dictionary fit across the grid; the DeepSet fit falls at weak confinement. No random-dictionary null or held-out fit was run, so this is a relative comparison. Three seeds at $\omega = 0.01$.

The dominant mode is not the tangent space. Figure 7.2 scores the *dominant* eigenmode against the dictionary; Fig. 7.3 asks where the eigenvalue-weighted mass of the leading modes *as a whole* lies. A network can place its largest mode squarely inside the dictionary and spend most of the rest outside it, which is what the separable network does—so the leading-mode score is generous to it. At $\omega = 1$ it has the slightly higher leading-mode fit ($R^2 \approx 0.996$ against 0.991), with energies identical to four figures.

Write the eigenvalue-weighted fraction of the top tangent modes lying *outside* the selected-operator span as the *dictionary-orthogonal tangent mass*. The DeepSet carries a large excess at *every* confinement—0.44 against the CTNN’s 0.06 at $\omega = 1$, a gap of ~ 0.3 – 0.5 that persists from the strongly-confined limit down to the crystal (Fig. 7.3). Thus the separable network can reach the same energy while leaving nearly half its tangent weight along directions the selected dictionary does not resolve. Those directions are not necessarily unphysical; the result is only that these four operators account for one tangent space more completely than the other. Nor is this merely a consequence of the CTNN’s smaller d_{eff} : toward the Wigner regime the two effective dimensions converge to ~ 3.7 while the dictionary gap persists. At $N = 12$ the contrast is stronger still: the CTNN’s dictionary-orthogonal mass stays $\lesssim 0.02$ across ω , while a matched DeepSet trained to $+0.2\%$ sits at ≈ 0.77 at $\omega = 1$.

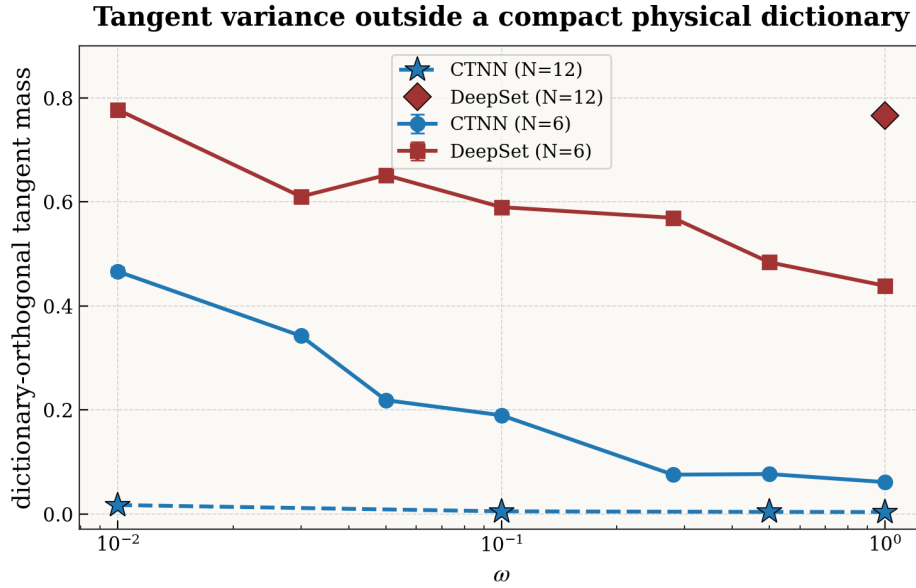


Figure 7.3: Dictionary-orthogonal tangent mass—the eigenvalue-weighted tangent variance outside the span of the selected four-operator dictionary—against ω . At $\omega = 1$, where the two correlators agree in energy to 0.001%, the DeepSet places 44% of its tangent variance outside the dictionary and the CTNN 6%; the excess persists at every ω .

7.1.2 The backflow: rank collapse and its absence

The contrast is sharper in the backflow, but the comparison is less controlled. The correlator, data, training protocol and optimiser are shared; capacity, coordinate preprocessing and centre-of-mass projection are not (§3.5.3). At the production width the copresheaf network has 182k parameters and the conventional network 51k, and no matched pair was trained. The results therefore compare the archived implementations, not message routing alone.

At $\omega = 1$ both implementations reach reference quality, but they diverge as the trap weakens (Table 7.2). The CTNN backflow stays within 0.01–0.12% across all N and ω , whereas the conventional backflow reaches $+0.58\%$ at $N = 6, \omega = 0.01$ and $+0.69\%$ at $N = 20, \omega = 0.01$, about six times the CTNN error there.

Table 7.2: Relative energy error (%) of the two backflow architectures, seed-averaged.

ω	$N = 6$		$N = 12$		$N = 20$	
	CTNN	conv	CTNN	conv	CTNN	conv
1.0	−0.03	+0.06	−0.01	+0.04	+0.06	+0.07
0.1	−0.05	+0.16	+0.01	+0.16	+0.02	+0.18
0.01	−0.04	+0.58	−0.01	+0.54	+0.12	+0.69

How component-level numbers are read. Energy and $\text{Var}(E_L)$ are state-level quantities; the backflow’s rank, displacement magnitude and component-wise d_{eff} are not. Because the correlator and backflow can substitute for one another, these component-level diagnostics record the training route as well as the state it reached.

This distinction matters here. An earlier rank collapse under joint, from-scratch training affected both architectures, moved with the seed and carried no energy penalty. The collapse reported below occurs under the staged protocol, affects only the conventional backflow, is seed-robust where replicated and accompanies an energy loss. The energetic change makes it more than a component-level curiosity.

We define the *backflow rank* as the participation ratio of the displacement field $\Delta\mathbf{x}$: the effective number of independent directions along which the backflow moves the electrons.

The columns of Table 7.3 have different ceilings: $2N - 2$ for the copresheaf network, which projects out the mean displacement, and $2N$ for the conventional network. Their absolute levels are therefore not directly comparable. The change with confinement is. At $\omega \geq 0.1$ the conventional field remains near its ceiling; at $\omega = 0.01$ it collapses to rank 1 for $N = 6, 12$ and 20 , while the CTNN field remains near full rank. The tangent dimension changes in the same direction, growing from ~ 4 to ~ 10 for the CTNN and falling from ~ 6 to ~ 1 for the conventional backflow. The rank-one result is reproduced by both seeds at $N = 6$ and $N = 12$; the $N = 20$ value is from a single re-run. Because the conventional field retains a centre-of-mass component, the rank-one endpoint cannot yet be identified with a relative collective mode.

What the comparison supports. The core observation is simple: in the Wigner regime the conventional implementation collapses to a rank-one displacement field and its energy degrades, whereas the CTNN implementation does neither. The cause is not isolated. The conventional network is a third the size, only the CTNN scales its inputs by $\sqrt{\omega}$, and only the CTNN removes the mean displacement. The same conventional network works above the Wigner regime, so the failure is confinement-dependent, but that does not distinguish capacity, preprocessing and architecture. Tables 7.2 and 7.3 therefore describe the production implementations used in Chapter 6; they are not an architecture ablation.

Table 7.3: Backflow displacement rank (participation ratio), evaluated on the field returned by each network. The ceilings are $2N - 2$ (10/22/38) for the copresheaf network and $2N$ (12/24/40) for the conventional one, so compare the collapse rather than the absolute levels. The networks are not capacity-matched (182k against 51k). Entries are two-seed means except at $N = 20, \omega = 0.01$, which is a single re-run whose per-run rank artefacts were not retained.

ω	$N = 6$		$N = 12$		$N = 20$	
	CTNN	conv	CTNN	conv	CTNN	conv
1.0	8.8	10.4	20.8	22.2	36.9	35.4
0.1	7.7	10.4	18.2	21.7	33.8	37.6
0.01	9.8	1.0	20.8	1.0	35.2	1.0

Pricing each component by deletion. We next ablate components on a common probe drawn from the full model’s $|\Psi|^2$. Holding the configurations fixed isolates the change in the kinetic term when a component is removed. It does not give the energy of a relaxed ablated state, so these numbers rank local contributions rather than energetic costs.

(i) On this probe the **backflow’s share shrinks and the Jastrow messages’ share grows** as the trap weakens. Deleting the backflow costs +5.6% at $\omega = 1$ and +1.4% at $\omega = 0.1$, but only +0.013–0.017% at $\omega = 0.01$ (three seeds), while deleting the Jastrow messages runs the other way: +3.9% at $\omega = 1$ rising to +9–15% at $\omega = 0.01$. A variational deletion on separately sampled production-width networks gives the opposite trend: removing the message-passing backflow costs 0.5% at $\omega = 1$, 1.0–1.2% at $\omega = 0.1$ and 1.9–2.0% at $\omega = 0.01$. The methods and networks differ, and the division of correlation between components is route-dependent, so neither result establishes a physical migration of relational work from nodes to amplitude. The conclusion that survives is:

(ii) **Within the backflow, the transport maps are the displacement:** zeroing the inter-particle transport maps ($\rho_{v \rightarrow e}, \rho_{e \rightarrow v}$) perturbs $\Delta\mathbf{x}$ by as much as deleting it outright—the mean $|\Delta\mathbf{x}_{\text{abl}} - \Delta\mathbf{x}|$ is comparable to the mean $|\Delta\mathbf{x}|$ at every (N, ω) . The displacement therefore comes primarily from the transported messages, not the per-particle path. Both architectures use relative coordinates, but the conventional backflow aggregates one round of pairwise messages per particle, whereas the CTNN

maintains an explicit edge state across the graph. Empirically, the former does not sustain a high-rank displacement field at the crystal; the latter does, and its transport maps carry almost the entire displacement.

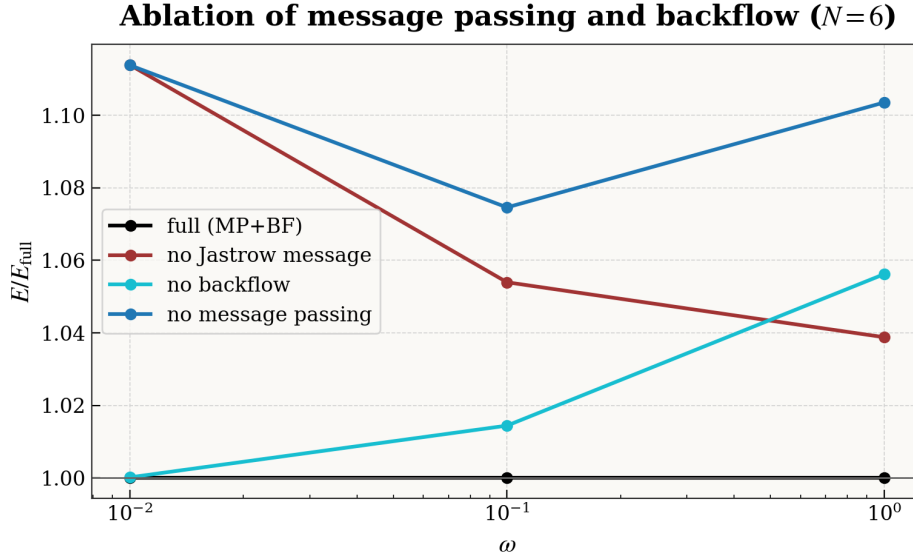


Figure 7.4: The *fixed-measure instantaneous ablation cost* of each component (trained networks, $N=6$). Configurations are drawn once from the full model’s $|\Psi|^2$ and every arm is re-evaluated on them, so $\Delta E_{\text{probe}}/E_{\text{full}}$ is the change in the kinetic term when a component is switched off, not the energy of a retrained network. Cyan: backflow removed; dark red: Jastrow messages removed; blue: all message passing removed. The curves cross near $\omega \approx 0.4$. The trend reverses when the ablated production-width networks are evaluated variationally on their own samples (text).

7.1.3 Evidence consistent with a common relational bottleneck

The two effects may share a cause in how relational information is routed. The separable correlator pools independently encoded pairs, and the conventional backflow aggregates one round of pairwise messages; neither retains a transported edge state. The CTNN variants do. To test whether the two diagnostics move together, we retained the conventional backflow in both arms, changed the correlator family, retrained each arm and followed the sweep in ω .

The backflow’s failure boundary moves with the correlator. With a separable correlator the two give way together near $\omega \approx 0.05$: correlation-hole capture falls from 0.68 to 0.07 and the displacement rank from ~ 10 to 1 across the same step. With a message-passing correlator both hold half a decade further, to $\omega \approx 0.01$ (Fig. 7.5).

Each arm is a separately trained model, and the correlators differ in primitive features as well as routing. The co-movement is therefore consistent with a shared relational bottleneck, but does not exclude feature scaling, optimisation path or capacity as common causes.

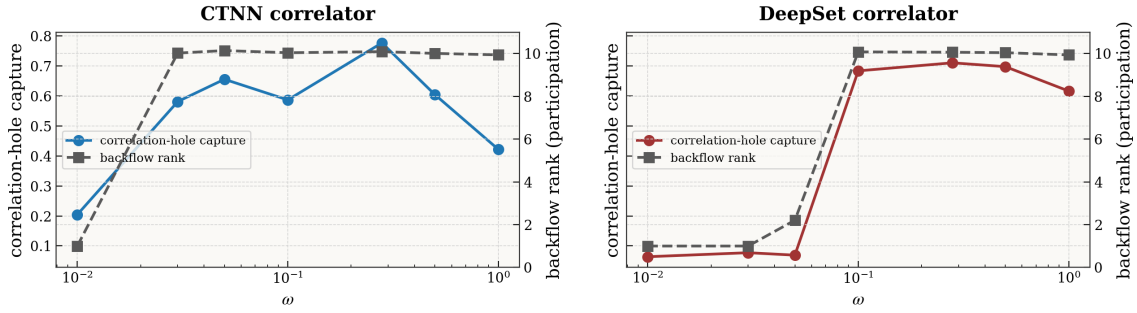


Figure 7.5: Correlator dictionary fit and backflow displacement rank for $N = 6$, with the conventional backflow used in both arms. For each correlator family, the correlation-hole capture (left axis) and the backflow displacement rank (right axis) give way at the same ω : $\omega \approx 0.05$ for the DeepSet, $\omega \approx 0.01$ for the CTNN. The co-movement is consistent with a shared bottleneck, but the correlator features are not otherwise matched.

7.1.4 What the trained representation looks like

One descriptive result belongs beside these controlled comparisons. Examining a single high-quality trained solution shows the correlator occupying a feature space of one to three effective dimensions across the entire grid and never expanding: at $\omega = 10^{-3}$, where the real-space density has developed rings and discrete occupancies, the latent rank is no larger than at $\omega = 1$. What changes across the crossover is which branch the leading axis draws on, not how many axes there are. The displacement field is the opposite kind of object, spanning nearly all the spatial directions available to it and leaving the correlator’s feature geometry almost unperturbed.

These are component-level measurements on one run. They are gauge-like—the correlator and backflow are partial substitutes, so the division of labour between them records the training route as much as the state—and no architectural conclusion above depends on them. They are recorded because a low effective covariance rank is not what one would predict for a state whose real-space structure is this elaborate, and because the reorganisation happens near $\omega \approx 0.1$, the same confinement at which the energy partition crosses over (§6.2.3). Whether that coincidence reflects one mechanism or two we cannot say from a single run. Appendix E gives the full analysis.

The implemented correlator families differ before their energies do. Under the parametrisations used here, the CTNN concentrates its Fisher weight in fewer directions at strong confinement and is better described by the selected dictionary; its local-energy variance is also lower by factors of 1.2 to 3.4. In the backflow sweep, the conventional implementation collapses to rank one and loses half a percent where the CTNN does not. Since the comparisons are not matched in every feature, scale and constraint, these are findings about the archived implementations rather than message passing alone; their co-movement motivates the shared-bottleneck hypothesis. The state-level result is the variance: correlators that agree in energy to four figures but differ in variance have reached different *states*. Their overlap was not measured, so the variance does not quantify that difference.

7.2 Conditioning and the sampling measure

Two design choices remain, and the tangent geometry connects them. One asks what a natural-gradient step buys over a diagonal one. The other asks what changes when the loss is defined by an analytic proposal instead of by $|\Psi|^2$. They are the same question twice over: stochastic reconfiguration inverts the metric S , and the sampling measure is what builds S in the first place.

7.2.1 Natural-gradient preconditioning

Stochastic reconfiguration preconditions the gradient by the inverse QGT, $\theta \leftarrow \theta - \eta S^{-1} \nabla_{\theta} E$; geometrically it *whitens* the tangent kernel. At $N = 2$ this is exact: the SR step aligns with the imaginary-time direction to numerical precision. The question is where whitening changes the outcome.

Fixing the ansatz and varying only the optimiser from a common stage-3 base: **for $|\Psi|^2$ -VMC, SR and Adam are indistinguishable**—exact at $N = 2$ (var $\sim 10^{-9}$), and at $N = 6$ the energy gap $E_{\text{Adam}} - E_{\text{SR}}$ is $\lesssim 0.002$ across ω and both seeds; in the matched pair at $\omega = 1$ the two agree to four decimals in relative error. **For weak-form collocation from the same common base, SR is decisive:** there $E_{\text{Adam}} - E_{\text{SR}}$ is large and positive (e.g. +2.9 at $N = 6, \omega = 1$), and plain Adam does not converge the objective.

This controlled comparison does not conflict with §7.3.1, where Adam is the production default. With a bare objective and fixed proposal, the collocation gradient stalls under a diagonal step and SR rescues it. The production pipeline instead controls the gradient at its source through proposal refitting, oversampling, reward normalisation, rollback and cascade warm-starting; once these are in place, Adam suffices, although their individual effects were not isolated (§7.3.4). Below $\omega \simeq 0.01$, the samples used to build S collapse and SR becomes unstable, so Adam is the more robust default. Its large speed advantage in our code mainly reflects the per-sample loop used to construct the score matrix: vectorising that calculation makes it 25 times faster at $N = 6$ and 82 times faster at $N = 20$ (Appendix G). The comparison here concerns what SR buys, not what it costs.

A large condition number alone does not explain the difference. Table 7.4 reports $\kappa(S) = \lambda_{\max}/\lambda_{\min}$ measured on the trained states under $|\Psi|^2$ sampling, and it is enormous, 10^7 to 10^{10} , for *both* architectures and at every capacity in the ladder, including precisely the models on which Adam and SR agree to four decimals. Whatever makes SR unnecessary under $|\Psi|^2$ sampling, it is not that the metric is well conditioned.

Table 7.4: Condition number of the empirical quantum geometric tensor at $N=6$, $\omega = 1$, measured on each trained state’s own $|\Psi|^2$ samples at fixed batch $B = 768$. Since $B < P$, $\lambda_{\min}$ is the smallest sample-supported eigenvalue and depends on batch size, regularisation and the numerical threshold; κ should therefore be read by order of magnitude. The d_{eff} values use the same finite batch rather than the converged common probe of §7.1.1, and are included only to show that κ does not track them. Adam and SR reach the same energy on every row.

architecture	params	$\kappa(S)$	d_{eff}
CTNN (seed 1)	79.8k	6.9×10^7	1.46
CTNN (seed 2)	79.8k	5.2×10^7	1.39
DeepSet (small)	20.2k	3.2×10^9	3.56
DeepSet (medium)	47.7k	3.4×10^7	4.35
DeepSet (matched)	89.4k	3.2×10^{10}	3.61
DeepSet (large)	164k	1.4×10^7	4.76

For the tested $N = 2$ and $N = 6$ cases, the data rule out condition number alone. Our working hypothesis is that SR matters when the tangent metric is anisotropic and still statistically estimable; a noisy or rank-deficient estimate can make S^{-1} amplify rather than suppress noise. The instability of CG-SR at ultra-weak confinement is consistent with this account (§7.3.3). The criterion remains a hypothesis because the collocation-space spectrum was not measured at matched batch size; neither the optimiser comparison nor the spectrum measurement was extended to $N = 12$ or 20.

7.2.2 Consequences of the sampling measure

VMC and importance-sampled collocation use the same Slater–Jastrow–backflow form, but the reported models are not capacity matched. Their training also changes the loss and its sampling measure. The following three consequences concern that measure change; they are not controlled architecture comparisons.

Target sampling aligns the estimator with the metric; a covering proposal does not. Sampling from $|\Psi|^2$ makes the estimators of S and of the gradient share the measure that defines them, so the directions carrying Fisher weight are also the directions the batch populates. That is an alignment between estimator and objective, not preconditioning in the sense of applying S^{-1} . A fixed covering

proposal instead drives a monotone *ESS collapse* as N and $1/\omega$ grow: at $N = 6, \omega = 0.01$ the effective sample size, measured against a converged state, falls to 0.11% of the batch—about 5 of the 4096 retained points—and the gradient estimator is then degenerate. (Table 7.10 finds 1.5 for the same mixture against a different converged checkpoint at the same (N, ω) ; at a handful of effective points the ESS is itself a noisy number, and the two are the same collapse.) In this setting the result diagnoses severe estimator failure; continued training with the same static proposal does not address the lost overlap. (Two ESS normalisations appear in this chapter, differing by the oversampling factor; each figure states which.)

§7.3.4 below nonetheless reports usable collocation energies at the same (N, ω) —+0.22–0.26% across three seeds, Table 7.7—and the two situations differ in the sampler and in the quantity being described. The sampler first: the collapse measured above is for the *static* origin-centred mixture, while the campaign’s robust recipe refits its proposal every 30 epochs and oversamples $16\times$. The campaign’s *baseline* rows, which keep the static proposal, are the ones that blow up (+3.9 to +7.2% at $\omega = 0.001$). Second, and more interesting, the two statements are not about the same quantity. The energy stays within a third of a percent while the modulus overlap ceases to be resolvable—which is the sharpest illustration in this thesis of why energy alone does not certify a state. We stop short of saying the overlap *is* zero: $\mathcal{S}^2$ is a ratio estimator built from the same degenerate weights, so at this ESS it reports *unresolved* and not orthogonal.

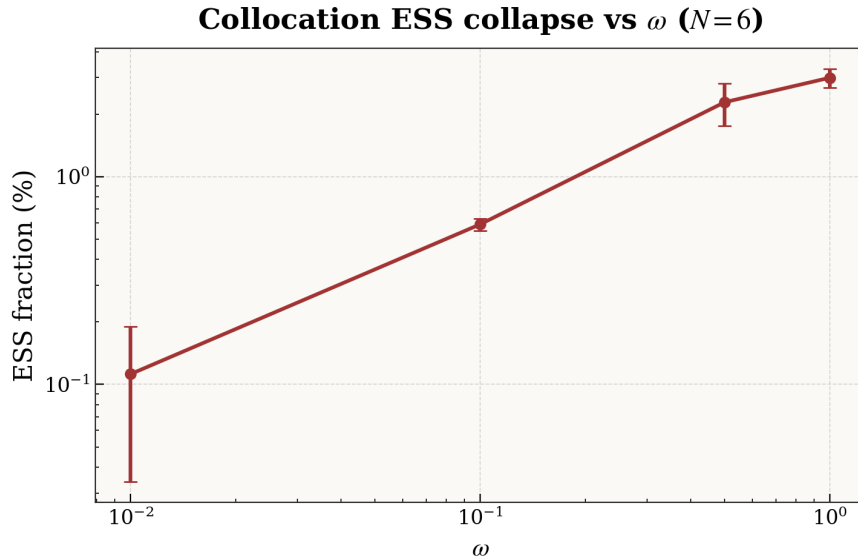


Figure 7.6: The naive collocation proposal collapses in the Wigner regime ($N=6$). The effective sample size (as a percentage of the batch) falls by more than an order of magnitude from $\omega=1$ to $\omega=0.01$, where it reaches $\sim 0.1\%$ —about five usable points in a batch of 4096. This diagnoses proposal–target mismatch for the sampled state and motivates the physics-informed ring proposal of §7.4, which is designed to repair it.

Amplitude agreement degrades with proposal overlap. We test whether the two paradigms reach similar wavefunction amplitudes, rather than merely similar energies, using the symmetric modulus-overlap diagnostic $\mathcal{S}^2 = \langle \Psi_B / \Psi_A \rangle_{|\Psi_A|^2} \langle \Psi_A / \Psi_B \rangle_{|\Psi_B|^2}$. At $N = 2$ the amplitudes agree closely ($\mathcal{S}^2 > 0.997$ for every ω , both optimisers, both seeds): the paradigm is efficiency-only, collocation paying $\sim 5\times$ the variance for essentially the same amplitude. At $N = 6$ they *diverge*: under Adam $\mathcal{S}^2$ falls from 0.85 at $\omega = 1$ to 0.77 at $\omega = 0.1$ (both seeds), tracking the ESS collapse, and at $\omega = 0.01$ the estimator returns values near zero but is no longer resolved—at an ESS of about five points it is a ratio of two degenerate importance-weighted averages, and we record it as *unresolved* rather than as a measured orthogonality. SR partially rescues the $\omega = 0.1$ value ($\mathcal{S}^2 \approx 0.95$) but does not close the gap. Weak-form collocation reaches essentially the same amplitude as VMC at the two-electron anchor. At $N = 6$ the amplitudes already differ at strong confinement and become harder to compare as proposal overlap deteriorates. In

these tests, the route that benefits from SR is also the one that becomes unreliable as the proposal loses overlap. An adaptive proposal that tracks $|\Psi|^2$ is the next remedy tested in §7.4.

Why the collocation energy readout is biased. For an exact eigenstate the local energy $E_L = \hat{H}\Psi/\Psi$ is constant over configuration space, so $\text{Var}(E_L) \rightarrow 0$ as the ansatz approaches the ground state—the zero-variance principle, and the reason $\text{Var}(E_L)$ is a sharp quality measure. That cancellation is a property of the *full* local energy, in which the kinetic (Laplacian) and potential terms combine to a constant; it is not a property of the Laplacian term alone. The collocation objective does *not* discard the Laplacian: it is a score-function (REINFORCE) estimator that *forms* the full E_L as a detached reward and differentiates only $\log|\Psi|$ (§7.3.1). What it gives up is the *measure*, and three things then spoil the readout, in increasing order of severity. The estimator is self-normalised, since $|\Psi_\theta|^2$ is unnormalised, so it carries an $\mathcal{O}(1/n)$ finite-sample bias that vanishes with sample size. The tail controls deliberately change the target: tempering $w \rightarrow w^\alpha$ and clipping the log-weights trade bias for variance on purpose, so the estimated quantity is no longer $\langle E_L \rangle_{|\Psi|^2}$. And, dominant in the regimes that matter, proposal–target mismatch inflates the variance beyond any affordable sample size (§7.3.4). Importance sampling is not biased merely for being importance sampling; it is the last two that make the readout unusable as a stopping rule. The reliable readouts are $\text{Var}(E_L)$ and the modulus overlap, both on $|\Psi|^2$ samples, and for the same reason no energy reported in this thesis is an importance-sampling estimate: every one is a Metropolis evaluation of $|\Psi|^2$.

The practical question is therefore how far an analytic proposal can preserve usable gradient statistics without a Markov chain.

7.3 Training with MCMC-free gradient updates

The conditioning story has a practical edge. If the sampling measure is what makes the metric estimable, then choosing that measure is a design decision, not a fact of nature—and the obvious thing to try is a measure with no Markov chain in it at all. “MCMC-free” refers to the gradient throughout: no configuration entering a parameter update comes from a chain, but a Metropolis probe still selects checkpoints and certifies the final energy, and the cascades start from reference-guided pretrained states (§8.2 states the claim exactly).

Two distinct limits. The method encounters proposal degeneracy and failure to transfer a checkpoint between confinements. The results below separate them.

The first limit is statistical. A fixed analytic proposal loses overlap with $|\Psi_\theta|^2$ as the trap opens; the importance weights grow heavy-tailed; the gradient estimator comes to be dominated by a handful of draws; the empirical metric becomes unreliable and preconditioning can then amplify noise. The evidence is consistent with a sampling limit, and a better proposal moves the observed boundary—from $\omega \approx 0.01$ with an origin-centred mixture to $\omega = 0.0035$ with one built out of the measured Wigner geometry.

The second limit concerns transfer. Below $\omega = 0.0035$ a stable trained state fails to survive transfer to a weaker trap even when the sampler is initialised directly on the correct shell. Diagnostics point to a near-balance between the confining determinant and the compensating Jastrow, which is disturbed by the 23% change in ω . Correct shell initialisation alone does not repair the transferred state. It limits the cascade, which carries trained states down in small steps of ω , and not for MCMC-free gradient updates as such: started instead from a pretrained checkpoint, the reliability campaign of §7.3.4 trains $N=6$ at $\omega = 10^{-3}$ to within 0.35% of our SR-VMC energy.

§7.3 documents the proposal-degeneracy evidence; §7.4 moves that observed boundary and then encounters the transfer failure.

A Markov chain is what makes neural variational Monte Carlo sequential. Chains must equilibrate, successive samples are correlated, and mixing times grow with particle number and with correlation strength. They grow fastest, in other words, exactly where the physics gets interesting.

Draw the configurations instead from a fixed analytic proposal and reweight them, and the whole gradient step becomes embarrassingly parallel. The question is what that costs.

Two ways of posing the problem behave very differently, and the difference decides everything downstream. The *strong form* asks the wavefunction to satisfy the local Schrödinger equation pointwise, minimising the variance of E_L and differentiating through the Laplacian to do it. The *weak form* uses a

score-function (REINFORCE) gradient in which the Laplacian enters only as a reward and is never differentiated. Appendix C shows why this is difficult for a backflow: E_L generally has a pole at a simple trial node, and differentiating through a node that moves with the parameters makes the empirical gradient more singular. None of the tested combinations of sampling, loss and preconditioning improved beyond an error near 0.3%. Model sizes were not matched to production VMC, so the experiment motivates the change of objective without identifying the objective as the sole cause. Everything below uses the weak form.

Using the same Slater–Jastrow–backflow ansatz form, and with no Markov chain in any gradient, the weak form comes within about a tenth of a percent of DMC at strong confinement and within about a third of a percent of our lowest SR–VMC energies well into the correlated regime for $N \in \{6, 12\}$ (Table 7.5). The energies are collected here, where the method that produced them is explained, rather than beside the SR–VMC energies of §6.1: without the failure mechanism of §7.3.4 in hand, the pattern of where they hold and where they give way is not interpretable.

Table 7.5: Metropolis energies of states trained with MCMC-free gradients. Parentheses in the reported-energy column give standard errors. For $N=6$, both the best result and the mean $\pm$ standard deviation of three protocol-controlled seeds are shown. The $N=2$ and $N=12$ rows are single runs; † marks continuation runs using the corrected proposal density. References are external where available and otherwise the lowest SR–VMC energy, so those deviations measure internal agreement. Energies are in effective Hartree.

N	ω	Reference	Reported energy	relative deviation (%)	
				best	mean $\pm$ s.d.
2	1.0	3.00000	3.00105(44)	+0.035	—
	0.5	1.65977(1)	1.65945(80)	−0.019	—
	0.1	0.44079(1)	0.440944(56)	+0.035	—
	0.01	0.073839(2)	0.073911(11)	+0.098	—
	0.001	0.013778	0.013889(9)	+0.806	—
6	1.0	20.15932(8)	20.1731(18)	+0.068	+0.082 $\pm$ 0.014
	0.5	11.78484(6)	11.7952(12)	+0.088	+0.098 $\pm$ 0.009
	0.1	3.55385(5)	3.56393(24)	+0.284	+0.293 $\pm$ 0.009
	0.01	0.69036	0.691895(80)	+0.222	+0.244 $\pm$ 0.020
	0.001	0.140832	0.141175(4)	+0.244	+0.282 $\pm$ 0.059
12 †	1.0	65.7001(1)	65.7207(17)	+0.031	—
	0.5	39.1596(1)	39.1727(12)	+0.034	—
	0.1	12.26984(8)	12.28361(32)	+0.112	—

7.3.1 The weak-form objective

The collocation training used here is exactly the MCMC-free gradient pipeline specified in the Methods: a weak-form (REINFORCE [49]) objective in which the local energy enters as a detached reward (§4.5, Eq. (4.8)), an analytic Gaussian-mixture proposal reweighted to $|\Psi_\theta|^2$ with ESS / Pareto- $\hat{k}$ monitoring [50] and tail controls (§4.5.1), and—in the Wigner regime—the physics-informed ring proposal with cascade warm-starting (§4.5.2). Two methodological findings shape what follows. First, among the optimisers (§4.4), **Adam** was the robust default across all regimes, whereas **CG-SR** converged faster at $\omega \geq 0.1$ but diverged for $\omega \leq 0.01$, where the importance weights collapse. This is consistent with an unresolved empirical metric, although that metric was not measured there directly. Second, **cascade warm-starting**, initialising each confinement from a stable checkpoint at a stronger one, was decisive for reaching the deep-Wigner regime, exploiting the transfer of correlation structure across ω .

Three ansatz families were tried within this framework. The Jastrow-only and backflow variants share a Slater–Jastrow core; the neural Pfaffian changes the antisymmetric part. Their definitions and sizes are in §3.6. The **backflow+Jastrow** family produced the best results at $N=6$ and $N=12$ and carries every number below. The **Jastrow-only** family was the fastest and most robust to train, served as the initialiser for the backflow runs, and becomes the better choice at $N=20$ for the memory reason of §7.3.5. The **neural Pfaffian** [51], which replaces the determinant with a more general antisymmetric form and in principle offers richer nodal topology, reached only +0.43–0.50% at $N=6$, $\omega=1$ against +0.07–0.10%

for backflow+Jastrow. Its larger parameter space coincided with noisier importance-sampled gradients, but the comparison does not separate parameter count, optimisation and nodal flexibility.

Why the weak form is stable where the strong form is not. The catch-22 analysis (Appendix C) identifies a nodal heavy-tail problem for direct residual training. Near a simple trial node, write $\Psi_\theta \sim ad_\theta$, $E_L \sim c/d_\theta$ and, for a moving node, $\partial_\theta \log |\Psi_\theta| \sim 1/d_\theta$. A strong squared residual sampled from a smooth covering proposal is then non-integrable. Differentiating it also propagates through the Laplacian.

The weak-form estimator does not remove nodal heavy tails, but it reduces the singularity it must handle and keeps the Laplacian out of the gradient path. Its per-sample energy-gradient factor scales as $(E_L - E)\partial_\theta \log |\Psi| \sim 1/d_\theta^2$; the mean is locally integrable under $|\Psi|^2 \sim d_\theta^2$, although its ideal variance can diverge. This distinction matches the empirical result: the weak form trains the tested backflows where direct residual minimisation fails, without establishing that all weak-form estimators are well behaved near nodes.

The better performance of pure REINFORCE ($\beta=0$) over the hybrid objective ($\beta>0$) has a related possible explanation. The direct weak-form term $\tilde{E} = \frac{1}{2}\|\nabla \log |\Psi|\|^2 + V$ is cheaper to evaluate (no Laplacian needed). Averaged under the current $|\Psi|^2$ measure, it equals the variational energy by integration by parts. The implemented direct path, however, differentiates $\tilde{E}(R)$ at retained samples without the score contribution from the changing sampling measure; $V(R)$ then has no explicit parameter derivative and the update behaves as a kinetic-only pathwise gradient. At nonzero β , the observed energy oscillations are consistent with a conflict between this kinetic-only pathwise update and the REINFORCE estimate of the variational gradient.

7.3.2 Collocation performance

$N=6$: within a few tenths of a percent across all ω . For six electrons, collocation training comes close to the reference wherever one exists, but not to within it. At $\omega \geq 0.5$ every run lands 0.07–0.10% above DMC, with evaluation errors of 0.005–0.013%, so the gap is resolved and several times the deviation of the SR-VMC route (Table 6.2). An earlier reading of these runs put the best of them at +0.013%, statistically consistent with DMC; that number came from a single 30 000-sample evaluation of the best of three seeds, and the independent re-evaluation shows it to have been the lucky tail of a noisier estimate. At $\omega = 0.1$ the robust recipe sits at +0.29%. At the most extreme confinements, $\omega \in \{0.01, 0.001\}$, where the electrons are separated by $\sim 30\text{--}120 a_0^*$ (a few oscillator lengths) and no DMC reference is available, the energies stay within 0.22–0.35% of our lowest variational energies; this is agreement between two independent training paradigms using the same ansatz form, not a certificate of absolute accuracy.

These results are achieved without any Markov chain in the training gradients. MCMC enters only outside the gradient loop: a small Metropolis probe periodically selects the best checkpoint, and the final heavy evaluation certifies the energy. The training itself uses 4096 importance-resampled collocation points per epoch with an oversampling factor of 8–16, so the importance weights need 32–65k amplitude evaluations per gradient step. The SR route evaluates the amplitude about as often: its chain runs a hundred Metropolis sweeps of 1200 walkers per step. What collocation removes is therefore not evaluations but their sequential dependence and the equilibration of the chain. Timed per step (Appendix G), neither is where the time goes. At $N=6$ a production SR-VMC step takes 516 s, of which the chain is 8 s and building the SR score matrix one sample at a time is 495 s; a collocation epoch takes 21 s. At the same optimiser the two paradigms cost within about ten percent of each other per step, because the exact-Laplacian local energy dominates both. Per-step cost does not see burn-in, autocorrelation between successive samples or slow mixing between configurations, which is where the chain’s real cost lies and which we did not measure.

Staged continuation is what makes those budgets pay, and an exploratory run at $\omega=1.0$ shows the effect cleanly: an initial BF+Jastrow joint REINFORCE stage (900 epochs, $\text{lr}=5\times 10^{-4}$), a low-LR continuation (750 epochs, $\text{lr}=3\times 10^{-4}$) and a hard-focus stage (500 epochs, $\text{lr}=2\times 10^{-4}$), each inheriting the previous checkpoint, improve monotonically through +0.08%, +0.05% and +0.010%. That run was hand-tuned and is not among the protocol-controlled numbers of Table 7.5; we report it because it is the observation that put cascade warm-starting into the campaign’s robust recipe (Table 7.6).

$N=12$: a few hundredths of a percent at strong confinement. For twelve electrons, the BF+Jastrow collocation results track DMC closely at $\omega \geq 0.5$, at +0.031% and +0.034% for $\omega=1.0$ and 0.5 (Table 7.5), resolved above the reference at evaluation errors of 0.003%. No reliability campaign was run at this size, so these are single continuation runs and not protocol-controlled in the way the

$N=6$ rows are. At $\omega=0.1$ the error increases to $+0.11\%$, reflecting the growing difficulty of importance sampling in 24 dimensions at weak confinement. The ESS at $\omega=0.1$ is typically 5–15 per epoch (out of 4096 kept points), indicating that the Gaussian mixture proposal still maintains usable overlap with $|\Psi|^2$.

Attempts to extend to $\omega \leq 0.01$ at $N=12$ were unsuccessful: both direct starts and cascade warm-starts from $\omega=0.1$ failed to converge, with the ESS collapsing to 1 and the rollback mechanism rejecting all gradient updates. This marks the boundary of the current collocation approach for $N=12$.

$N=20$ exposes an implementation-level memory–sampling trade-off. Twenty electrons in two dimensions push the tested collocation pipeline against its memory limit. The backflow’s edge-feature tensor scales as $\mathcal{O}(N^2 h_{\text{bf}})$, which at $N=20$ forces the hidden dimension, the collocation budget or the oversampling factor down far enough to degrade the gradient—so far that a Jastrow-only ansatz, freed of that cost, trains better than the backflow ansatz it should in principle beat. These runs show a limit of the tested allocation, not that the architecture has a conceptual $N=20$ boundary. The results are reported in Appendix D.2.

7.3.3 Ablations and failed variants

Most of the method space we explored did not work. The failures are reported here beside the successes because they narrow the useful design space and expose assumptions that the successful recipe alone would leave implicit. The table records the observed outcome and the interpretation supported by the available diagnostics.

Table 7.6: Outcomes of tested collocation variants. Errors are relative to the reference of Table 7.5. The three marked † are taken up in prose below; the rest are recorded for completeness.

Attempt	Outcome	Interpretation
Pure REINFORCE ($\beta=0$) over the hybrid objective	adopted	The direct path omits the sampling-measure score contribution; at $\beta > 0$ it contributes a kinetic-only pathwise gradient that conflicts with REINFORCE, and the two gradients oscillate against each other.
Cascade warm-start from higher ω	essential	Cold starts never converged at $\omega=0.01$ for $N \geq 6$. Cascading from $\omega=0.1$ cut the initial error from $\sim 30\%$ to $\sim 5\%$ and made $+0.15\%$ reachable in ~ 1000 epochs.
ESS-adaptive oversampling with instability rollback	essential	Raises the proposal budget when overlap degrades and reverts on energy spikes. Without both, $\omega=0.001$ diverged within a few hundred epochs.
CG-SR below $\omega \simeq 0.01$ †	abandoned	$+4\text{--}36\%$ against $+0.15\text{--}0.24\%$ from Adam, coincident with severe importance degeneracy; the metric was not measured directly in this regime.
Langevin proposal refinement †	abandoned	Worse in every configuration tested ($+153\%$ against $+5.7\%$ at $N=20$, $\omega=0.1$). The weights are computed against a proposal the samples have stopped following; see below.
Finite-difference loss	residual abandoned	$+0.22\%$ at $N=6$, $\omega=1$ against $+0.010\%$ from REINFORCE. Sensitive to the step size and to noise in the Laplacian estimate.
Neural Pfaffian ansatz [51]	abandoned	$+0.43\%$ at $N=6$, $\omega=1$, forty times the BF+Jastrow error. The larger parameter space coincides with noisier gradients; size, optimisation and nodal flexibility are not separated.
Backflow at $N=20$ †	abandoned	$+18\%$ against $+1.32\%$ for Jastrow-only under a memory-forced capacity and sample reduction; see §7.3.5.

Why CG-SR becomes unreliable at weak confinement. CG-SR gave the fastest convergence and the best final energies at $\omega \geq 0.1$, and then failed outright below it. Ill-conditioning is a plausible part of the explanation: at weak confinement, parameters controlling short-range correlation and the orbital envelope act on scales separated by $1/\sqrt{\omega}$. The Fisher matrix $F_{ij} = \mathbb{E}[\partial_i \log |\Psi|^2 \partial_j \log |\Psi|^2]$ has

eigenvalues spanning many orders of magnitude in the measured VMC case, although it was not measured under the collocation distribution at the failed points.

Ill-conditioning alone does not account for the failure, because anisotropy is what makes natural gradients useful in the first place; a well-conditioned metric is one Adam handles unaided. A plausible additional limit is estimation: F is built from importance-resampled points, and when the ESS is low a handful of high-weight samples dominate it. This can leave the empirical matrix noisy and rank-deficient. Tikhonov damping stabilises the inversion but also suppresses curvature information.

This suggests three regimes. Where the metric is well conditioned, SR may be unnecessary. Where it is anisotropic and still estimable, SR can be valuable. And where importance-sampling collapse makes the empirical metric unreliable, inverting it can amplify estimation noise. The failures above are consistent with the third regime, but do not establish the three-way classification. The practical implication is about the estimator rather than the curvature: a second-order method should check that the metric it inverts is statistically resolved at the sample sizes in use.

Why pushing the samples uphill makes things worse. Langevin refinement was motivated by an intuition that is hard to argue with. If the Gaussian proposal overlaps $|\Psi|^2$ poorly, evolve the samples up the score $\nabla_{\mathbf{R}} \log |\Psi|^2$ for K steps before reweighting, and the overlap should improve. It failed in every configuration tested, and the reason is standard in the MCMC literature and easy to miss in an importance-sampling setting.

The weights $w_i = |\Psi(\mathbf{R}_i)|^2/q(\mathbf{R}_i)$ are valid only if the $\mathbf{R}_i$ came from q . After K steps from an out-of-equilibrium start the samples follow some intermediate $\tilde{q}$ that depends on K and ε , while the weights still divide by q . The resulting bias is not noise. It pushes the wavefunction toward the regions the Langevin dynamics favours, which moves $\tilde{q}$ further the same way, and the loop closes on itself—which is what the +153% in the table is. Ten to twenty steps in $Nd=40$ dimensions do not equilibrate. Fixing it means running to equilibrium, adding a Metropolis–Hastings step, or computing annealed weights along the non-equilibrium path. The first two are MCMC under another name; the third costs enough to forfeit the advantage that made the route attractive.

A common reading of the failures. Four failures—CG-SR, Langevin refinement, the larger Pfaffian and the $N=20$ backflow—are compatible with one finite-sample account. Natural gradients need an estimate of the metric. Langevin needs samples that have actually equilibrated. A larger parameter space needs a correspondingly better-resolved gradient. The experiments do not isolate this account from model and optimisation differences, except that the Langevin variant has a known proposal–weight mismatch. Section 7.3.4 therefore follows the measurable quantities from proposal overlap to final evaluation without assigning every failure a unique cause.

7.3.4 Reliability analysis: gradient-quality diagnostics

The results in Table 7.5 report the *best* energies obtained by collocation training, selected from multiple runs on different seeds. A natural question is whether these results are *typical* or whether they reflect rare fortunate outcomes from a high-variance process. This subsection reports a systematic 30-run reliability campaign designed to answer that question and test how proposal overlap, gradient noise and final accuracy move together.

Experimental design

All runs use the BF+Jastrow ansatz ($\sim 49k$ parameters) initialised from the same pretrained checkpoint. The grid is $N=6$ across $\omega \in \{1.0, 0.5, 0.1, 0.01, 0.001\}$ with three random seeds per cell (42, 137, 314), giving $5 \times 2 \times 3 = 30$ runs. Two training recipes are compared head-to-head:

- **Config A (baseline):** Adam optimiser with $\text{lr}=5 \times 10^{-4}$, $\text{lr}_{\text{jas}}=5 \times 10^{-5}$, 4096 collocation points, $8 \times$ oversampling, local-energy clipping at 5σ , no reward normalisation, no rollback, fixed Gaussian mixture proposal (Eq. (4.9)).
- **Config B (robust):** Identical to Config A plus four stabilisation mechanisms: $16 \times$ oversampling (doubling the proposal budget), reward normalisation (standardising the REINFORCE reward to zero mean and unit variance before the gradient step), instability rollback (reverting to the previous checkpoint when the energy jumps exceed 3σ , with learning-rate decay of 0.95), and an adaptive Gaussian mixture proposal (8 GMM components refitted every 30 epochs to track the evolving $|\Psi|^2$).

All runs train for 1200 epochs. A 20 000-sample Metropolis probe of $|\Psi|^2$ every 50 epochs selects the checkpoint, and the selected checkpoint was evaluated at the end of training by a fresh 30 000-sample Metropolis run. Every number below is instead an independent re-evaluation of that saved checkpoint, made for this revision with four separately seeded chains of 102 400 samples each and blocking; it reproduces the original evaluations within their errors on average, but not the best-of-three values selected on them (§7.3.4).

Main results

Table 7.7 gives all thirty runs, three seeds per cell.

Table 7.7: Reliability campaign: relative energy deviations for $N=6$ across five trap frequencies, scored against the reference column of Table 7.5—external at $\omega \geq 0.1$, and our own converged SR-VMC energy at $\omega \leq 0.01$, where the deviation then measures internal consistency rather than absolute accuracy. Each cell is one $(\omega, \text{recipe}, \text{seed})$ run, evaluated by independent blocked Metropolis sampling of the saved checkpoint (errors 0.003–0.02%, 0.05–0.09% for the baseline at $\omega = 10^{-3}$). Bold marks the best run at each ω . The last two columns give the mean $\pm$ standard deviation and the worst of the three seeds. The standard deviation is reported in place of a coefficient of variation because the latter is uninformative when the mean deviation approaches zero.

ω	Recipe	Seed 42	Seed 137	Seed 314	mean $\pm$ s.d.	worst
1.0	Baseline	+0.067	+0.066	+0.078	0.070 ± 0.006	0.078
	Robust	+0.068	+0.081	+0.096	0.082 ± 0.014	0.096
0.5	Baseline	+0.107	+0.112	+0.104	0.108 ± 0.004	0.112
	Robust	+0.103	+0.103	+0.088	0.098 ± 0.009	0.103
0.1	Baseline	+0.479	+0.597	+0.486	0.520 ± 0.066	0.597
	Robust	+0.284	+0.302	+0.293	0.293 ± 0.009	0.302
0.01	Baseline	+0.368	+0.455	+0.661	0.494 ± 0.150	0.661
	Robust	+0.246	+0.262	+0.222	0.244 ± 0.020	0.262
0.001	Baseline	+4.012	+6.906	+4.628	5.182 ± 1.525	6.906
	Robust	+0.252	+0.350	+0.244	0.282 ± 0.059	0.350

At strong confinement the stabilisation mechanisms make little difference: the two recipes land within a few hundredths of a percent of each other and the best runs are indistinguishable. At weak confinement they matter much more. At $\omega = 0.1$ and 0.01 the robust recipe halves the mean error and shrinks its seed spread about sevenfold; at $\omega = 10^{-3}$ the mean error falls by a factor of eighteen, from +5.2% to +0.28%, and the *worst* robust seed beats the *best* baseline seed by an order of magnitude. The diagnostics below examine why the difference appears where it does.

The gradient-quality chain

The reliability campaign records a diagnostic sequence associated with the quality of collocation-trained wavefunctions:

$$\begin{array}{c}
 \boxed{\begin{array}{l}
 \text{proposal mismatch} \longrightarrow \hat{k} \text{ inflation} \longrightarrow \text{gradient noise} \\
 \phantom{\text{proposal mismatch}} \longrightarrow \text{optimisation stochasticity} \longrightarrow \text{run-to-run variance}
 \end{array}}
 \end{array} \tag{7.3}$$

Each quantity is checked against the training logs below. One caveat governs how strongly to read the arrows: the robust recipe changes four things at once (oversampling, reward normalisation, rollback and adaptive proposal refitting), so the campaign shows that the package works, not which element causes which improvement. The chain is a working account consistent with the measurements, not a set of individually isolated causal links. We did not ablate the four stabilisers separately; a factorial or stepwise ablation is what would be needed to establish the chain link by link.

Step 1: proposal–target mismatch. The REINFORCE estimator of Eq. (4.8) requires samples from $|\Psi_\theta|^2$, but collocation draws from the analytic mixture $q(\mathbf{R})$ of Eq. (4.9) and reweights. Everything depends on how well q overlaps the target. At $\omega = 1$ the electrons occupy a compact cloud of radius

$\ell = \omega^{-1/2}$ and the three-component mixture covers it. At $\omega = 10^{-3}$ the cloud extends to $\sim 100 a_0^*$ against $\ell \approx 31.6 a_0^*$, the pair correlation develops sharp shell structure (§6.2), and the mass concentrates in a thin correlated region an isotropic Gaussian covers badly.

The mismatch shows directly in the importance weights. At $\omega=1.0$, weights are relatively uniform: the effective sample size is typically $\text{ESS} \sim 5000\text{--}6000$ out of the $\sim 65\text{k}$ *raw proposal draws* ($16\times$ oversampling of 4096 points)—the second of the two normalisations noted in §7.2—and the Pareto shape parameter $\hat{k}$ of the weight tail is 0.4–0.5, indicating a well-behaved thin-tailed distribution. At $\omega=0.001$, the ESS collapses to 1–6 out of the same 65k draws, and $\hat{k}$ rises above 1.0, indicating a catastrophically heavy fitted tail and a sample far from the regime in which the reweighted estimator is reliable.

Step 2: $\hat{k}$ inflation and gradient quality. The tail index $\hat{k}$ of the importance weights turns that mismatch into a number (§4.5.1): below 0.7 the reweighted estimator is trustworthy, above 1.0 the fitted tail is heavy enough that a handful of samples dictates the update. Two qualifications keep the reading honest. $\hat{k}$ is fitted to the *weights*, not to the full gradient integrand $w(E_L - b) \nabla_\theta \log |\Psi_\theta|$, whose tail index we do not measure; and because w is a ratio of proper densities its mean is finite in the limit, so $\hat{k} > 1$ diagnoses a sample far from that limit rather than a gradient that fails to exist. What the campaign establishes is severe proposal–target mismatch and a gradient estimator dominated by a few draws. That provides a plausible common explanation for the downstream behaviour, and it is all we claim.

Table 7.8 summarises the average $\hat{k}$ across the campaign. The correspondence between $\hat{k}$ and final accuracy is striking where $\hat{k}$ exceeds about 0.7, and absent below it.

Table 7.8: Average Pareto $\hat{k}$ of the importance weight distribution during training, by regime and recipe. Lower is better. The fitted tail has finite variance only for $\hat{k} < 1/2$ and a finite mean only for $\hat{k} < 1$; $\hat{k} < 0.7$ is the practical reliability threshold, and $\hat{k} > 1$ indicates a sample so far from the asymptotic regime that the estimator behaves as though no finite mean existed (§4.5.1).

ω	$\hat{k}$ (avg)		Mean error (%)	
	Baseline	Robust	Baseline	Robust
1.0	0.75	0.50	0.070	0.082
0.5	0.83	0.54	0.108	0.098
0.1	1.48	0.70	0.520	0.293
0.01	2.63	1.39	0.494	0.244
0.001	3.13	1.86	5.182	0.282

The robust recipe reduces $\hat{k}$ by a factor of 1.5 to 2.1, with the larger reductions at the confinements where the baseline is worst. At $\omega=1.0$ this takes $\hat{k}$ from 0.75 (borderline) to 0.50 (reliable), with no measurable effect on the energy. At $\omega=0.1$ it takes $\hat{k}$ from 1.48 (broken) to 0.70 (borderline), corresponding to the transition from mediocre (+0.52%) to marginal (+0.29%) accuracy. At $\omega=0.001$ even the robust recipe has $\hat{k}=1.86$, but the adaptive proposal and rollback mechanisms prevent the catastrophic divergence seen in the baseline (+5.2%).

Step 3: from gradient noise to optimisation stochasticity. When $\hat{k}$ is large, most of the gradient signal comes from a few samples with extreme importance weights. Which samples are extreme depends on the random proposal draw at each epoch—a function of the random seed. Two seeds that happen to draw different extreme samples will follow different optimisation trajectories, even though they share the same architecture and hyperparameters.

This differs from an i.i.d. finite-variance setting, where an unbiased minibatch gradient concentrates at the usual square-root rate as the batch grows. The self-normalised, resampled estimator used here is not unconditionally unbiased. When $\hat{k} > 1$ the weight distribution is so heavy-tailed that the realised estimator behaves as though its variance were unbounded: increasing the batch size reduces the spread sublinearly, and extreme draws continue to dominate the update at every batch size we could afford. The result is optimisation trajectories that diverge qualitatively across seeds, explaining the large run-to-run variance observed at low ω .

Step 4: where the reported energy can mislead. The final energy is not an importance-sampling estimate. It is a Metropolis evaluation of $|\Psi_\theta|^2$ at the checkpoint the probe selected, and every value

in Tables 7.5 and 7.7 is an independent re-evaluation of that checkpoint with four separately seeded chains and blocking. Where the reported number can mislead is one step earlier, in the selection. The checkpoint is the best of about two dozen probe readings of 20 000 samples each, and the minimum of noisy readings is biased low. Across all thirty runs the probe that chose a checkpoint read it as better than the independent evaluation finds, by 0.07% on average and in 26 of 30 cases; the original single end-of-training evaluation, by contrast, is unbiased on average (+0.001%) but noisy enough that the best of three seeds selected on it came out optimistic. Table 7.9 shows representative runs.

Table 7.9: Selection bias of the checkpoint probe. “Probe” is the Metropolis probe reading that selected the checkpoint during training; “original” the single 30 000-sample evaluation made at the end of training; “independent” the blocked re-evaluation used in the tables (error in parentheses, last digits). The gap is independent minus probe. All values are deviations in percent.

ω	Recipe	Seed	probe	original	independent	gap
1.0	Baseline	314	+0.001	+0.060	+0.078(12)	+0.077
1.0	Robust	137	+0.002	+0.013	+0.081(6)	+0.079
0.5	Robust	42	+0.037	+0.002	+0.103(12)	+0.066
0.1	Robust	137	+0.136	+0.263	+0.302(8)	+0.166
0.01	Baseline	42	+0.320	+0.500	+0.368(17)	+0.048
0.001	Baseline	314	+4.114	+4.733	+4.628(52)	+0.514

The first two rows are the instructive ones. A probe reading of +0.001% looked like an essentially exact wavefunction; the state sits at +0.08%, and it was the probe that was lucky. The second row was, until this re-evaluation, the headline strong-confinement number of this chapter. Selection on a noisy estimate inflates any “best run”, which is why the tables report independent re-evaluations and why the reliability campaign is summarised by seed means rather than by its best seeds.

Summary of the chain. The chain in Eq. (7.3) provides a unified account of the full landscape of collocation results. At high ω , the proposal matches the target well ($\hat{k} \sim 0.5$), gradients are reliable, optimisation is stable, and all seeds converge to within about a tenth of a percent of DMC. At low ω , the proposal-to-target mismatch grows, $\hat{k}$ inflates, gradients become noisy, and optimisation becomes seed-dependent. The robust recipe acts on multiple links of this chain simultaneously: higher oversampling improves the raw ESS, reward normalisation stabilises the gradient scale, rollback prevents catastrophic divergence, and the adaptive proposal reduces the fundamental mismatch. Each is aimed at a different link in the chain; which of them is necessary was not tested, and the campaign shows only that together they suffice.

7.3.5 The memory wall at $N=20$

The reversal of the backflow advantage at $N=20$ highlights an important architectural trade-off. The message-passing backflow (CTNNBackflowNet, §3.5.2) stores edge features of size $\mathcal{O}(N^2 \times h_{\text{bf}})$, which at $N=20$ with $h_{\text{bf}}=128$ requires $\sim 10\times$ the memory of the Jastrow-only network. This forces a choice: reduce h_{bf} (losing expressivity), reduce the collocation budget (losing gradient quality), or reduce the oversampling factor (losing ESS). All three options degrade the result.

The Jastrow-only network, with $\sim 25\text{k}$ parameters and $\mathcal{O}(N^2)$ edge cost (no hidden-dimension multiplier), permits full-size collocation budgets ($n_{\text{coll}}=4096$, oversample=8–16) even at $N=20$. The resulting gradient is less noisy, and the simpler loss landscape (fewer parameters) makes optimisation more reliable.

This does not mean that backflow is unnecessary at $N=20$; it means that the current training pipeline cannot supply enough gradient signal to train backflow effectively at this scale. The implication is that scaling collocation to larger systems requires either (i) memory-efficient backflow architectures (e.g. sparse attention, low-rank message passing), or (ii) improved sampling that provides higher ESS at constant computational cost.

7.4 A physics-informed proposal and its limit

The ESS collapse of §7.2 is a limitation of the *proposal*, not of collocation itself, and it can be pushed back a long way. A physics-informed proposal carries collocation with MCMC-free updates deep into the Wigner regime; what limits it, and the wall that survives, follow from the same physics.

7.4.1 The origin-centred proposal misses the Wigner shell

Every proposal in the standard pipeline is a mixture of *origin-centred* Gaussians,

$$q(\mathbf{R}) = \frac{1}{K} \sum_k \mathcal{N}(\mathbf{R}; 0, \sigma_{f,k}^2/\omega). \quad (7.4)$$

But as $\omega \rightarrow 0$ the electrons localise into a Wigner molecule on a *shell* far from the origin. The classical $N = 6$ minimum is the (1, 5) configuration: one central electron and five on a ring of radius $1.334\omega^{-2/3}$ [3, 4]. In oscillator units that ring sits at 1.96, 2.87 and 4.22ℓ for $\omega = 0.1, 0.01$ and 0.001 , moving outward as the trap weakens and past the reach of a width-2 ℓ origin Gaussian. The proposal aims where $|\Psi|^2$ has no mass; hence the collapse.

7.4.2 The Wigner-ring proposal

Before the numbers, the scope of “MCMC-free” needs to be exact. The Wigner geometry enters in three roles, and only the third sits inside the training loop. (i) A proposal can be *evaluated* against a trusted state obtained by MCMC, which is how Table 7.10 is produced—a measurement, not a training step, and one that uses a chain by design. (ii) Its geometry can be *fitted* to $|\Psi_\theta|^2$ samples as a diagnostic. (iii) In production it is refitted from the current importance-weighted batch alone, with no chain anywhere; that is the MCMC-free proposal adaptation used in the cascade of §7.4. Throughout, a Metropolis probe still runs outside the gradient loop to select checkpoints and certify the final energy, and no sample it produces ever reaches a parameter update. We replace the origin mixture with a *Wigner-ring* proposal: a radial Gaussian at the shell radius times an angular distribution for the ring electrons, plus a central Gaussian. On trusted states this lifts the effective sample size by 15–32 $\times$ over the origin mixture (Table 7.10) without changing the network. The gain is large, although the fitted proposal still reaches only 47.5 effective samples of 4096. Fitting the ring to $|\Psi|^2$ samples (the simplest adaptive form) doubles the ESS again; a (1, 5) + (0, 6) mixture *hurts* at $N = 6$, since the state is (1, 5) and the extra channel dilutes the proposal mass.

Table 7.10: ESS (of 4096) against a trusted $N = 6, \omega = 0.01$ state. The origin mixture is already collapsed; the ring proposal covers the shell. The checkpoint differs from the one in Fig. 7.6, which finds about 5 for the same mixture; at this ESS the difference is noise.

proposal	ESS
origin-Gaussian mixture (standard)	1.5
Wigner ring (1, 5), hand-tuned	22.5
Wigner ring (1, 5), fitted to $ \Psi ^2$	47.5

7.4.3 Angular ordering reduces overlap with a uniform-angle proposal

Why does the ring work at $\omega = 0.01$ but fail at $\omega = 0.001$? The molecule *crystallises angularly* as the trap weakens. The nearest-neighbour angular-gap standard deviation of the five ring electrons is $40.8^\circ/33.3^\circ/25.5^\circ/6.7^\circ$ at $\omega = 1/0.1/0.01/0.001$: a “ring liquid” at moderate coupling becomes an angular crystal (electrons pinned near 72° apart) at $\omega = 0.001$. A proposal with independent, uniform angles almost never lands on the crystalline configuration—so the angular proposal must reflect the spacing too. We model the five gaps, which sum to 2π , as Dirichlet(α). Here $\alpha = 1$ gives uniform simplex gaps, $\alpha < 1$ favours uneven gaps and $\alpha > 1$ favours regular spacing. Fitting α to the measured gap variance provides a plausible mechanism for the observed loss of overlap.

7.4.4 An MCMC-free cascade reaches a stable checkpoint at $\omega = 0.0035$

Combining the ring proposal (radial Gaussian $\times$ Dirichlet gaps) with importance oversampling, a *continuous* adaptive re-fit of the proposal to the current importance-weighted batch (every few steps, no MCMC), and a gentle ω -cascade seeded from a stable $\omega = 0.01$ state, MCMC-free gradient updates produce states that lie on the Wigner energy-scaling curve down to $\omega = 0.0035$ —four cascade rungs, with heavy-VMC energies within 2% of the scaling estimate, and within 1% on three of the four rungs (Table 7.11), well past where the origin proposal collapses (ESS ≈ 1.5 of 4096 at $\omega = 0.01$, Table 7.10).

Three things bound that result. The agreement in Table 7.11 is against a scaling curve fitted to our own $\omega = 0.01$ energy, so it establishes continuity rather than accuracy; two structural checks support it—the radial density peaks at 3.34ℓ against a classical ring at 3.42ℓ , and a sampler initialised on the shell converges to the same peak, not a second basin. No *angular* diagnostic was run at $\omega = 0.0035$: the bond-order and Lindemann machinery of §6.2.5 ran on the production grid, not the cascade rungs, and those states were not retained. So a Wigner molecule at $\omega = 0.0035$ rests here on energy continuity and radial geometry, not on demonstrated angular order. And the MCMC-free claim is a claim about the gradient—no Metropolis sample reaches a parameter update at any rung—not about ancestry: each rung is warm-started from a stronger confinement, and that ladder runs back to a pretrained checkpoint whose residual objective used an external energy as its target. The frontier parameters therefore inherit structure shaped several confinements upstream by reference-guided pretraining, the same inheritance the SR-VMC states carry (§8.1). The essential ingredient is the continuous re-fit: it keeps ESS healthy ($\sim 10^2$) *while* the energy descends, where a static or slowly-updated proposal collapses to ESS ~ 1 as the wavefunction moves away from it.

Table 7.11: Wigner-ring cascade with MCMC-free gradient updates, $N = 6$: heavy-VMC energy vs. the fitted Wigner scaling curve $E(\omega) \approx 0.690 (\omega/0.01)^{0.689}$, whose exponent is close to the classical $2/3$ but whose prefactor is anchored at $\omega = 0.01$, where no external reference exists. The agreement is therefore internal scaling consistency, not validation, and it is retained through $\omega = 0.0035$.

ω	E	scaling est.	ratio
0.0077	0.577	0.577	1.00
0.0059	0.481	0.480	1.00
0.0045	0.400	0.398	1.01
0.0035	0.340	0.335	1.02

7.4.5 The remaining limit: transfer between confinements

Below $\omega = 0.0035$ the cascade becomes unstable, and—informatively—the instability is *non-monotonic* across proposal variants: wide-exploratory, tempered-tracking, and outward-anchored proposals each converge some rungs and diverge others. That pattern points away from initial proposal placement as the sole explanation and toward the warm start. Four transfer diagnostics, taking the stable $\omega = 0.0035$ rung down to $\omega = 0.0027$ (a 23% step), narrow the possibilities.

Where the collapse lives. First, the frontier state is a genuinely Wigner-localised one on the evidence available: at its own $\omega = 0.0035$ its radial density peaks at 3.34ℓ against a classical ring at 3.42ℓ , and it occupies a *single* basin—initialising the sampler on the shell converges to the same 3.34ℓ . We stop short of calling it a Wigner molecule in the sense Chapter 6 defines, because that definition requires registered angular order and no angular diagnostic was run on this state. Transferred to $\omega = 0.0027$, however, the wavefunction collapses to $\sim 1\ell$, and two tests constrain *where* the collapse lives. It is not explained by the walkers’ initial placement: initialising the Metropolis walkers directly on the classical shell (3.57ℓ) still relaxes inward to 1.1ℓ , so the collapsed density is retained by the wavefunction rather than imposed by sampler initialisation. The measured backflow displacement is also small ($|\Delta\mathbf{x}| \approx 0.13\ell$, radially inward) and essentially *preserved* across the transfer ($0.15\ell \rightarrow 0.13\ell$)—it scales correctly with ℓ but is far too small to bridge the 2.4ℓ gap between the Slater core ($\sim 1\ell$) and the Wigner shell.

A working explanation. The bare harmonic-oscillator determinant suppresses the shell by $\exp(-\omega r^2/2) \sim \exp(-5.6)$ in amplitude ($\sim 10^{-5}$ in density); the Wigner peak exists only because the Jastrow amplifies it back by a comparable factor, with the backflow contributing a small inward nudge. A 23% change in ω perturbs both the Slater confinement and the Jastrow correlation length.

The resulting collapse is consistent with that balance being disturbed, although the legacy backflow scaling discussed in §3.5 provides another possible contribution. Initialising on the shell shows that sampler placement alone is insufficient; it does not show that every coordinate-scaled warm start must fail. Nor does it rule out representing the $\omega = 0.0027$ state, and the reliability campaign suggests the ansatz can represent weaker traps still: started from a pretrained checkpoint rather than from a stable neighbouring state, and trained with an adaptively refitted Gaussian mixture, it reaches $\omega = 10^{-3}$ at $N=6$ within 0.35% of our SR-VMC energy on every seed (Table 7.7). The observed obstacle is carrying a stable state down this cascade, not representing the weaker confinement itself. Continuing the cascade below $\omega = 0.0035$ would need per- ω re-optimisation of that kind, possibly against a shell-*amplitude*-anchored objective, or a genuinely ℓ -covariant parametrisation. The contribution here is concrete: a physics-informed proposal that carries a cascade with MCMC-free updates from the wall of the fixed origin-centred proposal down to a radially localised Wigner-regime state at $\omega = 0.0035$, and evidence that angular ordering and the Slater–Jastrow balance both matter near its present boundary.

Part V

Discussion and Conclusions

Life must be understood backwards. But they forget the other proposition, that it must be lived forwards.

SØREN KIERKEGAARD
Journals

Chapter 8

Discussion

8.1 Geometry between architecture and optimiser

The three parts of this thesis converge on one object. Architecture fixes the shape of the variational tangent space; the optimiser acts on that space through its metric; and the sampling measure decides how well the metric can be estimated. Read in that order, results that arrived separately form a connected account.

Its first link is the one the energy cannot see. Two correlator families reach the same reference-quality energy at strong confinement while carrying their Fisher weight differently. Both the effective dimension and the leading tangent modes depend on the parameter coordinates. Under the implemented parametrisations, the message-passing models use fewer effective directions and are more completely described by the selected four-operator dictionary. No random-dictionary null or held-out fit was run, so this is a relative description, not a test of whether the remaining directions are physical. The state-level result is simpler: the message-passing correlator has lower local-energy variance at every confinement. That establishes that the trained states differ, although the unmatched learned features prevent attributing the difference to message passing alone.

The second link is where the optimiser meets that space. The result there is narrower than the usual one about natural gradients. Under $|\Psi|^2$ sampling the empirical metric is badly conditioned, $\kappa \sim 10^7$ – 10^{10} on precisely the models where Adam and SR agree to four decimals, so conditioning cannot be the criterion. What we propose separates the regimes is whether a batch resolves the metric at all. Under $|\Psi|^2$ the samples come from the measure that defines S . Under a fixed collocation proposal, SR is required in the tested configurations, while CG-SR becomes unstable once proposal overlap collapses. The negative half of that is measured. The positive half—that collocation under a fixed proposal occupies an anisotropic but still estimable middle regime—is inferred from the estimator, since κ was never obtained under the collocation measure at matched batch size.

Whether the chain generalises is a fair question and the honest answer is that we cannot settle it here. Message-passing neural quantum states have been adopted for extended fermion systems on the argument that relational structure scales [12, 13]; here the tangent diagnostics identify where the implemented families first differ. The dictionary gap widens over the sizes measured, but three sizes do not establish a scaling law, and the comparison is seeded at $N=6$.

What agreement with the benchmarks supports. The energies agree closely with the available fixed-node references over the benchmarked grid, and that agreement is neither exactness nor a meeting of two blind calculations. Three separate things limit what it shows. The first is ancestry: where a DMC value exists it is also the target of residual pretraining. The pure-energy SR tail that follows minimises $\langle \hat{H} \rangle$ under $|\Psi_\theta|^2$ and no longer sees the target, which strengthens the comparison considerably, but the final states still descend from reference-guided training. The second is what an energy can certify. The

variational bound keeps the reported expectation above E_0 whatever the training aimed at, but it says nothing about how close to E_0 it lies, and a flexible ansatz can reproduce an approximate reference without describing every other property of the exact state. Our evidence beyond the energy is partial: the local-energy variance, which vanishes only for an eigenstate, and a squared overlap of 1.000000 with the exact state where one is available, at $N=2$. The third concerns the cells where our energies sit below the reference. The reference is variational only with respect to its own imposed nodes, so there is no contradiction, and better nodes in the network could explain it. A convention difference in the Hamiltonian would not: the exact two-electron value is reproduced, and a convention mismatch would shift every cell. That leaves better nodes against an extrapolation or sampling error on either side, or an error in the chain of compilations through which the reference reached us, and we cannot tell these apart. A fixed-node DMC calculation that uses the network’s nodes would be the cleanest test. The tables accordingly report deviation, not error, and what we claim is close agreement with the available fixed-node references, not a validation of the wavefunctions. Below $\omega = 0.1$ for $N \geq 6$ we have no external value to compare with, and those states are predictions.

8.2 Physics returned to the solver

A particularly useful result is that measurements from the trained state improve the proposal used for later training.

The order matters. A generic three-component Gaussian proposal covers the strong-confinement droplet adequately and collapses in the Wigner regime, where the effective sample size falls to about 1.5 of 4096 and the gradient is dominated by a few samples. Nothing in the objective or the architecture diagnoses that. What diagnoses it is the structure measured from the trained states themselves: a single ring at $N=6$ with a well-determined radius, angular order that appears well after radial stiffening, and shell occupancies that are read rather than assumed. Rebuilding the proposal as a radial Gaussian at the measured shell radius times a Dirichlet angular model lifted the effective sample size by 15–32 $\times$ with no network training, although the absolute ESS remained low. It carried a cascade with MCMC-free updates from $\omega \approx 0.01$ to a stable checkpoint at $\omega = 0.0035$.

Together these observations support a working mechanism: proposal–target mismatch produces heavier weights, noisier gradients and less stable optimisation, while inversion of an unreliable metric can amplify that noise. The interventions change several links at once, so the present experiments do not establish the sequence causally. At $N=20$, memory limits also force smaller models or batches and prevent the implementation from separating those effects. A better proposal nevertheless moves the observed frontier.

Below 0.0035 the obstacle changes character. Metropolis walkers started on the classical shell relax inward regardless; the displacement is too small to bridge the gap; and the collapse is non-monotonic across proposal variants. The evidence therefore points to the warm start. At these confinements, the component decomposition shows Jastrow amplification comparable to the determinant’s suppression of the Wigner shell. After a 23% change in ω , the transferred density falls back towards the non-interacting shell. This is consistent with an imbalance between the Slater and Jastrow factors during transfer, although the backflow unit convention remains another unresolved source of non-covariance. We have not shown the ansatz cannot represent the $\omega = 0.0027$ state, and there is evidence that it can represent weaker traps: trained at $\omega = 10^{-3}$ directly from a pretrained checkpoint, not carried down the cascade, it comes within 0.35% of our SR-VMC energy (Table 7.7). The limit at 0.0035 belongs to the cascade, and the remedy we suggest, per- ω re-optimisation, possibly against a shell-amplitude-anchored objective, is what that campaign already does.

The same failure mode may extend beyond the quantum dot: any importance-sampled scheme whose proposal increasingly misses a strongly correlated target should monitor whether the metric it inverts remains statistically resolved at the available sample size.

The staged crossover. The physics that drove the loop stands in a particular relation to what was known. That parabolic dots form Wigner molecules with shell structure is established [2, 3, 5, 7, 14]. Nor is the separation of radial and angular ordering new: earlier simulations and symmetry-restored calculations report radial order before orientational correlations [6, 14]. The contribution here is a topology-conditioned measurement within the retained fixed- S_z states, using finite- n nulls and a held-out shell reconstruction across the selected grid. The $N=6$ ring follows the classical $1.334\omega^{-2/3}$ scaling and

reaches within 1.7% at the weakest trap. Its bond order remains near the random-angle null through $\omega = 0.1$ and becomes large only at $\omega = 10^{-3}$. Whether the same staging survives a total-spin scan we cannot say.

What “MCMC-free” means here. The claim is about the gradient: no configuration entering a parameter update comes from a Markov chain. A small Metropolis probe runs outside the gradient loop to select checkpoints and certify the final energy, and no sample it produces reaches an update. It is not a claim of freedom from Monte Carlo, since importance sampling and resampling are themselves Monte Carlo; nor of freedom from the reference, since where a DMC value exists the residual pretraining anneals toward it and the rollback monitors against it. Nor, at the deep-Wigner rungs where we have no reference, is the state reference-free in ancestry: its warm starts descend from benchmark-guided states at stronger confinement. Against the SR route, collocation is several times less accurate across the benchmarked range—within about a tenth of a percent of DMC at strong confinement and three tenths at $N=6$, $\omega=0.1$ —and not competitive at $N=20$. The two look complementary rather than rival: collocation for fast, parallel exploration and pretraining with MCMC-free gradient updates; SR for final accuracy where correlation is strong.

8.3 Scope, and what to do next

Five limits bound everything above. All states are the lowest found within a fixed spin-projection sector, and total spin is not scanned; that matters most in the deep-Wigner regime, where spin transitions are plausible and the structural conclusions are strongest. The architecture comparison is seeded at $N=6$, rests on fewer models at $N=12$ and on a single production state at $N=20$, and the error bars on the production energies do not include the variation between independent training runs. The backflow variants are not matched in capacity, input scaling or centre-of-mass handling, so neither the cross-architecture energy gap nor the rank collapse isolates architecture. The correlators also use different primitive pair features. The tangent-space quantities are properties of the networks as parametrised, not of the states they represent. And one entry in the backflow rank table, at $N=20$ and $\omega = 0.01$, comes from a re-run whose diagnostics were not retained.

Five things follow that we would do next.

Make the proposal learned, not designed. Up to the frontier the proposal is the bottleneck and not the objective, which argues for a better sampler rather than a better loss: normalising flows trained against $|\Psi|^2$, or the shell-informed construction of §7.4 generalised beyond ring geometries. Past the frontier the target changes: what is needed there is an ℓ -covariant parameterisation, or an objective anchored on the shell amplitude, so that a stable state survives being carried to a weaker trap.

Make rotation equivariance exact. Permutation equivariance holds by construction and the production CTNN projects out the centre of mass, but its displacement head reads a d -vector from a network that has seen absolute coordinates, so $\text{SO}(2)$ equivariance is only encouraged—by relative-vector edge features and by the random in-plane rotation applied to every batch—and never enforced. In a rotationally invariant trap whose weak-confinement physics is a freely rotating molecule, that is the wrong symmetry to leave approximate. The fix is local to one module: build the displacement as $\Delta \mathbf{r}_i = \sum_{j \neq i} g_\beta(\text{invariants}_{ij}) \mathbf{r}_{ij}$, a learned scalar gain on each relative vector, so the output rotates exactly with the input and the augmentation can be dropped. We did not run it.

Break the quadratic backflow. At twenty electrons the tested implementation is limited by the $\mathcal{O}(N^2 h_{\text{bf}})$ edge tensor. Sparse or low-rank message passing, a local backflow without all-to-all edge features, or the sample-side reductions of minSR and the kernel formulation [47, 48] would extend the reach without giving up the nodal correction that is the reason for having a backflow at all.

Match the backflow comparison. A widened conventional backflow at 182k parameters, with the same trap-scaled inputs, displacement unit and centre-of-mass projection, is the control needed to turn the rank-collapse observation into an architecture ablation.

Measure the metric where the hypothesis lives. The estimability account of §7.2 rests on a spectrum measured under $|\Psi|^2$ and inferred under collocation. Measuring the QGT under the collocation measure at matched batch size, with the stability of its leading eigenvalues across independent batches, a bootstrap on the resolvable part, and its sensitivity to batch size and regularisation, would turn that account from a hypothesis into a criterion or refute it.

*All things are implicated with one another, and the bond
is holy; and there is hardly anything unconnected with
any other thing.*

MARCUS AURELIUS
Meditations

Chapter 9

Conclusions

The work formed a feedback cycle between solver and physics. We constructed a variational state and compared it with external references wherever they were available; the energies agree closely. We then used the state as an instrument to resolve the Wigner regime: rings at definite occupancies, n -fold angular order that becomes large only in the last decade of confinement, and a shell radius approaching its classical $\omega^{-2/3}$ scaling. Those are the measurements a sampler needs in order to place its mass correctly. Feeding the measured geometry back into the proposal carried a cascade with MCMC-free updates from $\omega \approx 0.01$, where the fixed origin-centred sampler fails, to a stable checkpoint at $\omega = 0.0035$.

Architecture. The implemented model families differ before their energies do (§7.1). Under the parametrisations used here, the message-passing correlator concentrates its Fisher weight in fewer effective directions and is better described by the selected operator dictionary. Its lower local-energy variance is a state-level difference. The correlators also use different primitive features, so the present comparison does not attribute that difference to routing alone. At weak confinement the conventional backflow falls to rank one and loses half a percent of the energy, but it is not matched to the CTNN in capacity, input scaling or centre-of-mass handling. This is an implementation result until those controls are supplied. The correlator swap moves the same boundary and is consistent with, rather than proof of, a shared relational bottleneck.

Optimiser. Conditioning alone does not decide whether natural-gradient preconditioning earns its cost (§7.2). Under $|\Psi|^2$ sampling the empirical metric is badly conditioned on the very models where Adam and stochastic reconfiguration agree to four decimals. Under a fixed collocation proposal, SR is required in the tested configurations and becomes unstable after proposal overlap collapses. We read that pattern as saying that estimability, rather than conditioning, separates the regimes. It is a hypothesis the failures support, not yet a criterion: the metric was never measured under the collocation measure itself.

Paradigm. An accurate wavefunction can be trained with no Markov chain in the gradient, inside a regime whose edges we found by crossing them (§7.3). The claim is about the gradient only: a Metropolis probe still selects checkpoints and certifies the final energy, and the cascades start from reference-guided pretrained states. The weak-form objective comes within about a tenth of a percent of DMC at $N \in \{6, 12\}$ under strong confinement and within about a third of a percent of our own variational energies well into the correlated regime, as measured by independent Metropolis evaluation of every trained state. The failures are consistent with proposal mismatch, heavier weights, gradient noise and an unreliable empirical metric, but the interventions do not isolate every link. A better proposal moves that limit. The cascade then meets a second failure, of transfer between confinements; trained directly from a pretrained checkpoint instead, the same ansatz form reaches $\omega = 10^{-3}$.

Physics. The trained state serves as an instrument (§6.2). The crossover it resolves is staged: the cloud expands and its shells stiffen radially well before the electrons acquire angular order within them. The shell geometry that emerges at the weakest traps is resolved topology by topology, and its measured radius approaches the classical value $1.334 \omega^{-2/3}$ to within 1.7%. The staging itself is better established

than where the transition sits: a grid with one point per decade can order the stages but not locate them. And these are the lowest states found at fixed spin projection, in the regime where a change of ground-state spin is most plausible.

9.1 Outlook

Three papers based on this work are in preparation. The steps that would strengthen the method itself are set out in §8.3; what follows concerns the physics the same machinery could reach. Each direction changes the Hamiltonian or the question while keeping the variational principle, the optimiser and diagnostics that act only on samples of $|\Psi|^2$.

The most immediate is spin. Every state here is the lowest found at fixed spin projection, and a scan over total spin would settle whether the ground state changes spin in the deep Wigner regime, where the structural conclusions are strongest. Larger N follows once the quadratic memory cost of the backflow is removed, and would test the shell sequences of classical clusters across several filled shells.

The confinement itself can be changed. A multi-well potential, as in coupled dots [60] and the spin qubits built from them, enters the Hamiltonian directly, but it breaks the rotational symmetry on which both the shell diagnostics and the collocation proposal rely, and both would have to be rebuilt around several centres. A perpendicular magnetic field breaks time-reversal symmetry instead. The harmonic-oscillator orbitals become Fock–Darwin orbitals, the ansatz needs a phase as well as an amplitude, and the ground state passes through a sequence of angular-momentum transitions as the field grows. The Wigner crossover under a field is a natural test for the structural diagnostics developed here.

Two further directions change the question rather than the system. The quantum geometric tensor studied in Chapter 7 is the metric that the time-dependent variational principle inverts, so real-time evolution of a neural state [8], such as the response of a Wigner molecule to a sudden change of the trap, would inherit the conditioning and estimability questions raised here directly. And Rényi entanglement entropies between regions or shells of the dot can be estimated from samples of two independent copies of the trained state, giving the crossover a measure that neither the energy nor the density provides.

9.2 Concluding perspective

All four answers are bounded in the same way. Every state is the lowest found within a fixed spin-projection sector; the architecture comparison is seeded at $N=6$ and thinner above it, and its tangent-space measures belong to the networks as parametrised. The backflow variants are not matched in capacity, input scaling or centre-of-mass handling, and the correlators use different primitive features, so neither comparison isolates message passing. §8.3 sets out what would move each of those.

For the well-resolved $N \leq 12$ studies, the observed failures were more closely associated with alignment, conditioning and sampling than with raw parameter count. This does not make capacity irrelevant. At $N=20$ the memory cost of the backflow’s edge tensor forces a capacity cut and coincides with degraded collocation gradients. Across architecture, optimisation and sampling, performance is repeatedly associated with alignment to the collective structure of the problem: breathing, correlation-hole and relative modes at strong confinement, and the shell geometry of the Wigner molecule at weak confinement. Whether this is one mechanism or several related ones remains open.

Several of these results were reached only after an initial reading of the same data proved wrong: a rank collapse that survived seeding where a dimension inversion did not, a transfer failure first attributed to sampler placement and later found to persist in the retained wavefunction, an importance-sampling error whose signature was indistinguishable from an architectural deficiency, and a strong-confinement collocation energy that looked statistically exact until an independent evaluation showed it to be the lucky tail of a best-of-three selection. Each was identified by a diagnostic that could separate the competing explanations. That is the methodological point these chapters have tried to make concrete: a variational calculation is only as trustworthy as the tests that can tell its failure modes apart.

Part VI

Appendices

Chapter A

Laplacian conditioning

A.1 Residual training for Schrödinger

We consider the stationary many-body Schrödinger equation on configuration space $\mathbf{R} = (\mathbf{r}_1, \dots, \mathbf{r}_N) \in \Omega \subset \mathbb{R}^D$ with $D = dN$ (for our quantum dots, $d = 2$):

$$\hat{H}\Psi(\mathbf{R}) = E\Psi(\mathbf{R}), \quad \hat{H} = \hat{T} + \hat{V}, \quad \hat{T} = -\frac{1}{2} \sum_{i=1}^N \Delta_i, \quad (\text{A.1})$$

where $\hat{V}$ collects external and interaction potentials.

In *strong residual minimisation* one introduces

$$L := \hat{H} - E \quad \Rightarrow \quad L\Psi = 0, \quad (\text{A.2})$$

and minimises a squared residual loss of the schematic form

$$\mathcal{L}_{\text{res}}(\theta) = \frac{1}{2} \|L\Psi_\theta\|_{L^2(\Omega; w)}^2 = \frac{1}{2} \int_{\Omega} |(L\Psi_\theta)(\mathbf{R})|^2 w(\mathbf{R}) d\mathbf{R}, \quad (\text{A.3})$$

for a nonnegative weight w induced by the sampling strategy. A key message of the numerical-analysis viewpoint is that strong residual losses can be *severely ill-conditioned*, and training speed is tightly linked to spectral properties of an operator/matrix induced by L [43].

Relation to the objectives used in this thesis. Stage I pretraining (§4.3) draws configurations from an adaptive proposal q_t and minimises an empirical version of

$$\mathcal{L}_I(\theta) = \int q_t(\mathbf{R}) |E_L(\mathbf{R}) - E_{\text{eff}}|^2 d\mathbf{R} \quad (\text{A.4})$$

$$= \int |(\hat{H} - E_{\text{eff}})\Psi_\theta|^2 \frac{q_t(\mathbf{R})}{|\Psi_\theta(\mathbf{R})|^2} d\mathbf{R}, \quad (\text{A.5})$$

where the second line holds away from nodes. It therefore induces the weight $w_{\theta,t} = q_t/|\Psi_\theta|^2$, rather than the constant weight used in the simplest form of Eq. (A.3). Because this weight changes with both the proposal and the wavefunction, and because the implementation reuses samples and trims outliers, the fixed-weight analysis below is not the exact Stage I Hessian. It isolates the more general curvature amplification caused by squaring a residual that contains the Laplacian.

Stage II instead minimises the variational energy (§4.4), with updates preconditioned by the sampled quantum geometric tensor. The later collocation route (§4.5) uses a weak, score-function gradient and does not differentiate through the local energy. Neither objective has the normal-equation geometry derived below, although both still require the Laplacian when evaluating E_L .

A.2 Linearized geometry and the matrix A

To expose the mechanism, follow the standard linearization argument used in conditioning analyses [43]. Fix an expansion point θ_0 and define tangent (feature) functions

$$q_j(\mathbf{R}) := \partial_{\theta_j} \Psi_{\theta}(\mathbf{R})|_{\theta=\theta_0}, \quad j = 1, \dots, P. \quad (\text{A.6})$$

For small parameter increments $\delta\theta$, we have the first-order model

$$\Psi_{\theta_0+\delta\theta}(\mathbf{R}) \approx \Psi_0(\mathbf{R}) + \sum_{j=1}^P q_j(\mathbf{R}) \delta\theta_j, \quad \Psi_0 := \Psi_{\theta_0}. \quad (\text{A.7})$$

Applying the residual operator gives

$$(L\Psi_{\theta_0+\delta\theta})(\mathbf{R}) \approx (L\Psi_0)(\mathbf{R}) + \sum_{j=1}^P (Lq_j)(\mathbf{R}) \delta\theta_j. \quad (\text{A.8})$$

Introduce the weighted inner product (complex-valued wavefunctions allowed)

$$\langle f, g \rangle_w := \int_{\Omega} f(\mathbf{R})^* g(\mathbf{R}) w(\mathbf{R}) d\mathbf{R}. \quad (\text{A.9})$$

Let the Jacobian J be the linear map whose j th column is Lq_j . Then the loss (A.3) becomes (to second order in $\delta\theta$)

$$\mathcal{L}_{\text{res}}(\theta_0 + \delta\theta) \approx \frac{1}{2} \|r_0 + J\delta\theta\|_w^2, \quad r_0 := L\Psi_0. \quad (\text{A.10})$$

with normal-equation matrix

$$A := J^* J, \quad A_{ij} = \langle Lq_i, Lq_j \rangle_w. \quad (\text{A.11})$$

Here A is Hermitian positive semidefinite; in practice it can be close to singular, so one often effectively works with a damped variant $A + \lambda I$ (implicit or explicit).

Now consider gradient descent on the quadratic model (A.10) (with step size η). The error in parameter space evolves as

$$\delta\theta^{(t+1)} - \delta\theta^* = (I - \eta A) (\delta\theta^{(t)} - \delta\theta^*), \quad (\text{A.12})$$

so each eigen-direction of A contracts by a factor $(1 - \eta\lambda)$. Hence convergence speed is governed by the eigenvalue spread of A , typically summarized by the (effective) condition number

$$\kappa(A) = \frac{\lambda_{\max}(A)}{\lambda_{\min}(A)} \quad (\text{A.13})$$

(over the nonzero/regularised spectrum). Large $\kappa(A)$ implies stiffness: a step size small enough to be stable for $\lambda_{\max}$ yields slow progress along directions associated with $\lambda_{\min}$. This is precisely the conditioning bottleneck emphasized in physics-informed training analyses [43].

A.3 Bi-Laplacian scaling

The defining feature of Schrödinger is that L contains a second-order differential operator. To isolate the kinetic contribution, consider the regime where the residual is dominated by the Laplacian part,

$$L \approx -\frac{1}{2} \Delta_{\mathbf{R}}. \quad (\text{A.14})$$

From L to L^*L . The matrix A in (A.11) is a Gram matrix in the feature space $\{Lq_j\}$. The corresponding operator-level object is the *Hermitian square* L^*L (with adjoint taken in the $L^2(\Omega; w)$ inner product) [43]. For general non-constant w , the adjoint L^* differs from the formal differential adjoint by weight-induced lower-order terms; the cleanest picture is obtained for constant w and standard boundary conditions (periodic/Dirichlet), where the Laplacian is self-adjoint.

In the Laplacian-dominant regime (A.14), one has (up to a constant factor)

$$L \approx -\frac{1}{2}\Delta \quad \Rightarrow \quad L^*L \approx \frac{1}{4}\Delta^2, \quad (\text{A.15})$$

i.e. a bi-Laplacian scaling. The prefactor does not affect conditioning ratios.

Fourier/eigenmode derivation of k^4 . On a periodic domain (or, more generally, in a Laplacian eigenbasis) $-\Delta\phi_{\mathbf{k}} = |\mathbf{k}|^2\phi_{\mathbf{k}}$, so $\Delta^2\phi_{\mathbf{k}} = |\mathbf{k}|^4\phi_{\mathbf{k}}$. If an error $\delta\Psi$ expands as $\delta\Psi = \sum_{\mathbf{k}} c_{\mathbf{k}}\phi_{\mathbf{k}}$ and $L \approx -\frac{1}{2}\Delta$, then

$$\|L\delta\Psi\|_{L^2}^2 \approx \frac{1}{4} \sum_{\mathbf{k}} |\mathbf{k}|^4 |c_{\mathbf{k}}|^2. \quad (\text{A.16})$$

Thus the squared residual penalizes high-frequency components with a *quartic* weight $|\mathbf{k}|^4$. If the relevant spectrum spans $|\mathbf{k}| \in [k_{\min}, k_{\max}]$, the induced eigenvalue ratio scales like

$$\kappa \sim \left(\frac{k_{\max}}{k_{\min}} \right)^4, \quad (\text{A.17})$$

which is a sharp mathematical expression of “Laplacian-driven stiffness”. This quartic growth is consistent with the Laplacian/Poisson toy examples and frequency-dependent conditioning effects emphasized in [43].

Length-scale interpretation. Let ℓ denote the smallest length scale that must be represented. Then $k_{\max} \sim 1/\ell$ and (A.17) implies

$$\kappa \text{ grows rapidly as } \ell \downarrow 0. \quad (\text{A.18})$$

In many-body Schrödinger problems, small ℓ arises from nodal structure, strong localisation, and short-range correlation features; in all such regimes, the Laplacian makes the strong residual extremely curvature-sensitive.

A.4 Barron/spectral perspective: the Laplacian

The conditioning story above can be rephrased in approximation terms. In spectral Barron-type viewpoints [40], function classes are controlled by norms that weight the Fourier transform by powers of frequency (or related spectral weights). A key implication is that applying derivatives systematically increases the importance of high-frequency tails [43]. Since Δ multiplies a Fourier mode by $|\mathbf{k}|^2$, and the squared residual effectively involves L^*L (hence Δ^2 in Laplacian-dominant regimes), the residual can require substantially more “spectral budget” than the wavefunction itself. In practical terms: even if Ψ is approximable in a given network class, *approximating its strong residual* can be much harder, because the loss implicitly demands accurate high-frequency curvature (cf. (A.16)). This is one of the motivations, in numerical analysis of PINNs, for considering weak/variational formulations that reduce derivative order when the strong form is too stiff [43].

A.5 Implications for Stochastic Reconfiguration

Stochastic reconfiguration (SR) updates parameters using a natural-gradient-like preconditioner [27, 29, 61]. Writing $O_i = \partial_{\theta_i} \log |\Psi_{\theta}|$, one common SR update is

$$S(\theta) \delta\theta = -g(\theta), \quad S_{ij} = \text{Cov}_{p_{\theta}}[O_i, O_j], \quad g_i = 2 \text{Cov}_{p_{\theta}}[E_L, O_i], \quad (\text{A.19})$$

where $p_{\theta} \propto |\Psi_{\theta}|^2$ and $E_L = (\hat{H}\Psi_{\theta})/\Psi_{\theta}$. SR improves conditioning in parameter space by using the (quantum) information geometry of the variational family [27, 61]. However, the Laplacian-induced

stiffness described above is an *operator-level* phenomenon: strong residuals magnify curvature errors via Δ and Δ^2 (cf. (A.16)), which remains a fundamental challenge even when parameter updates are geometrically preconditioned.

Chapter B

Coulomb conditioning

B.1 Coulomb structure and singularity

For our two-dimensional electronic systems ($\mathbf{r}_i \in \mathbb{R}^2$), the interaction contains Coulomb terms of the form

$$\hat{V}_C(\mathbf{R}) = \sum_{1 \leq i < j \leq N} \frac{1}{r_{ij}}, \quad r_{ij} = |\mathbf{r}_i - \mathbf{r}_j|. \quad (\text{B.1})$$

This is a *singular multiplication operator*: it diverges at particle coalescence $r_{ij} \rightarrow 0$. The exact eigenfunction remains finite (or vanishes by antisymmetry) but develops non-smooth short-distance structure (a cusp) so that divergences in kinetic and potential contributions cancel in physically meaningful quantities such as the local energy [29].

Coulomb does not merely add “another term” to the residual; it introduces *singular short-range physics* that forces delicate cancellations with the Laplacian. Strong residual objectives square these effects and become extremely sensitive to tiny cusp errors.

B.2 2D cusp conditions: antiparallel vs parallel spins

We present the 2D cusp logic in a way that separates the two spin channels. The cleanest statement (especially for practical ansätze) is in terms of the short-distance behaviour of a *Slater–Jastrow* factorization

$$\Psi(\mathbf{R}) = D(\mathbf{R}) J(\mathbf{R}), \quad J(\mathbf{R}) = \exp\left(\sum_{i < j} u(r_{ij})\right), \quad (\text{B.2})$$

because (i) the Slater part D enforces antisymmetry and determines whether Ψ vanishes at coalescence, and (ii) the Jastrow slope $u'(0)$ is what cancels the $1/r$ Coulomb singularity in the local energy [29].

Fix a pair (i, j) and introduce relative/centre-of-mass coordinates in 2D:

$$\mathbf{R}_{ij} := \frac{\mathbf{r}_i + \mathbf{r}_j}{2}, \quad \mathbf{r} := \mathbf{r}_i - \mathbf{r}_j, \quad r := |\mathbf{r}|. \quad (\text{B.3})$$

A direct computation gives, in any dimension,

$$\Delta_i + \Delta_j = \frac{1}{2} \Delta_{\mathbf{R}_{ij}} + 2 \Delta_{\mathbf{r}}, \quad (\text{B.4})$$

hence the singular short-distance kinetic behaviour is controlled by the relative term $-\Delta_{\mathbf{r}}$.

A key 2D identity is

$$\Delta_{\mathbf{r}} r = \frac{1}{r} \quad (r > 0, \mathbf{r} \in \mathbb{R}^2). \quad (\text{B.5})$$

Antiparallel spins (distinguishable in the Slater part). If the two electrons have opposite spin, they enter different determinants in D and generically D does *not* vanish at $\mathbf{r} \rightarrow 0$. Then $\Psi(0) \neq 0$ and a direct cusp expansion in 2D is legitimate:

$$\Psi(\mathbf{r}, \dots) = \Psi(0, \dots) \left(1 + a r + \mathcal{O}(r^2) \right), \quad r \rightarrow 0. \quad (\text{B.6})$$

Using (B.5), the singular part of the relative Laplacian is

$$\Delta_{\mathbf{r}} \Psi \approx \Psi(0) a \Delta_{\mathbf{r}} r = \Psi(0) \frac{a}{r}, \quad \frac{\Delta_{\mathbf{r}} \Psi}{\Psi} \approx \frac{a}{r}. \quad (\text{B.7})$$

Since the pair kinetic contribution contains $-\Delta_{\mathbf{r}}$, the singular part of the local energy is

$$E_L(\mathbf{R}) = \frac{\hat{H}\Psi}{\Psi} \supset \left(-\frac{a}{r} + \frac{1}{r} \right) = \frac{1-a}{r}. \quad (\text{B.8})$$

Thus finiteness of E_L at coalescence requires

$$a = 1, \quad \text{i.e.} \quad \partial_r \log |\Psi|_{r=0} = 1 \quad (2\text{D, antiparallel spins}). \quad (\text{B.9})$$

Equivalently, in a Slater–Jastrow form with $D(0) \neq 0$, this is the same as $\partial_r \log J|_0 = 1$ for the ij -pair.

Parallel spins (identical fermions, node at coalescence). For same-spin electrons, antisymmetry forces D (and hence Ψ) to vanish as $\mathbf{r} \rightarrow 0$. Generically one has a *linear* zero:

$$D(\mathbf{r}, \dots) = \boldsymbol{\eta}(\dots) \cdot \mathbf{r} + \mathcal{O}(r^2), \quad r \rightarrow 0, \quad (\text{B.10})$$

with some nonzero vector $\boldsymbol{\eta}$ depending on the other coordinates. In this case, the cancellation of the Coulomb singularity is controlled by the Jastrow factor through cross terms in $-\Delta_{\mathbf{r}}(DJ)/(DJ)$. A standard short-distance analysis (carried out explicitly for 2D quantum dots) shows that the divergent kinetic contribution behaves like

$$-\frac{\Delta_{\mathbf{r}}(DJ)}{DJ} \supset -\frac{(2k+d-1)}{r} \partial_r \log J \quad \text{with} \quad d=2, \quad k=1 \text{ (identical particles)}, \quad (\text{B.11})$$

while the Coulomb potential contributes $+1/r$ to E_L as before. Hence finiteness of the local energy requires

$$\partial_r \log J|_{r=0} = \frac{1}{2k+d-1} = \frac{1}{3} \quad (2\text{D, parallel spins}). \quad (\text{B.12})$$

This 1 (antiparallel) versus 1/3 (parallel) 2D cusp statement is the standard quantum-dot/Jastrow–Slater cusp condition [62].

What to remember. In 2D:

$$\text{antiparallel spins:} \quad \partial_r \log |\Psi|_{r=0} = 1 \iff \partial_r \log J|_0 = 1, \quad (\text{B.13})$$

$$\text{parallel spins:} \quad \Psi(0) = 0 \text{ (linear node)}, \quad \partial_r \log J|_0 = \frac{1}{3}. \quad (\text{B.14})$$

B.3 Why Coulomb is ill-conditioned (2D emphasis)

The cusp conditions above show *how* Coulomb creates stiffness: it forces kinetic and potential contributions to be individually large and singular, while their *sum* is finite only under a short-distance constraint. Strong residual minimisation is therefore a *cancellation problem*.

Sensitivity amplification in the local energy. If the learned antiparallel-spin cusp slope is $a_\theta = 1 + \delta a$, then (B.8) implies

$$E_L(\mathbf{R}) \supset -\frac{\delta a}{r}, \quad (\text{B.15})$$

so an arbitrarily small cusp mismatch produces an arbitrarily large local-energy spike as $r \rightarrow 0$.

For parallel spins, if the learned Jastrow slope is $\partial_r \log J|_0 = \frac{1}{3} + \delta b$, then the same short-distance analysis leading to (B.12) implies

$$E_L(\mathbf{R}) \supset -\frac{3\delta b}{r}. \quad (\text{B.16})$$

Thus both channels generate $1/r$ spikes unless the cusp is satisfied.

Why squaring is especially dangerous in 2D. A $1/r$ defect in E_L (or in $(\hat{H} - E)\Psi$) is concentrated near coalescence. But in 2D the squared singularity is not integrable—it diverges logarithmically:

$$\int_{r < \varepsilon} \left(\frac{1}{r}\right)^2 d^2\mathbf{r} = \int_0^\varepsilon \frac{1}{r^2} (2\pi r dr) = 2\pi \int_0^\varepsilon \frac{dr}{r} = +\infty.$$

Therefore, without effectively resolving the cusp, *strong* squared-residual objectives can be dominated (or even made ill-defined in the continuum limit) by the coalescence region. In Monte Carlo practice, this manifests as heavy-tailed fluctuations and extreme sensitivity to rare close-pair samples.

Residual expansion and cancellation structure. Writing $L = \hat{T} + (\hat{V} - E)$ and expanding the squared residual norm:

$$\|L\Psi\|_w^2 = \|\hat{T}\Psi\|_w^2 + \|(\hat{V} - E)\Psi\|_w^2 + 2 \operatorname{Re}\langle \hat{T}\Psi, (\hat{V} - E)\Psi \rangle_w, \quad (\text{B.17})$$

we see that near coalescence both $\hat{T}\Psi$ and $\hat{V}_C\Psi$ are large. The exact solution relies on cancellation between these contributions (cf. (B.8) and (B.12)), which means (B.17) demands high *relative* accuracy in the cusp region: if either term is off by a small fraction, the net residual can be large in absolute magnitude. This is a classic recipe for ill-conditioning: the loss landscape contains directions where small parameter changes cause huge residual changes in a small region.

Operator-level picture (L^*L). From the same viewpoint as Appendix A, strong residual training induces matrices/operators tied to L^*L [43]. For $L = \hat{T} + (\hat{V} - E)$, one has (formally)

$$L^*L = \hat{T}^2 + (\hat{V} - E)^2 + \hat{T}(\hat{V} - E) + (\hat{V} - E)\hat{T}. \quad (\text{B.18})$$

Coulomb enters here in two destabilizing ways:

1. the positive term $(\hat{V} - E)^2$ involves a $1/r^2$ -type singular weight;
2. the mixed terms contain derivatives of V through product rules, e.g. $\hat{T}(V\Psi)$ generates $\nabla V \cdot \nabla\Psi$ and ΔV contributions, which are even more singular/distributional at coalescence.

This formal expansion explains why the Coulomb singularity can severely inflate the effective spectral spread of the training operator, aligning with the operator-conditioning viewpoint [43].

B.4 Barron/spectral perspective: the Coulomb singularity

Coulomb forces the *true* 2D solution to have cusp-like, non-smooth short-range structure, so approximating the *strong residual* requires representing high-frequency content that is not well captured by globally smooth/low-complexity function classes.

In spectral/Barron-style approximation theories [40], approximation quality is governed by spectral decay and norms that weight the Fourier tail [43]. A cusp (or a linear node times a cusp in J) implies slower spectral decay than a globally smooth function. Strong residual training then *further* amplifies the high-frequency tail because it applies Δ (and effectively L^*L), which multiplies Fourier modes by powers of $|\mathbf{k}|$ (Appendix A). Hence, even if Ψ can be approximated reasonably in function value, the strong residual can remain large unless the cusp structure is represented accurately. This is exactly the type of regularity mismatch that motivates weak/variational formulations in numerical analysis of PINN-type methods, since weak forms reduce the derivative order demanded of the approximation [43].

Practically, QMC wavefunctions therefore incorporate explicit short-range correlation structure (e.g. cusp-enforcing Jastrow factors), precisely to remove these singular residual pathologies and reduce the variance of local quantities [29, 62].

B.5 Implications for Stochastic Reconfiguration

SR performs a natural-gradient step using a (quantum) Fisher/metric matrix [27, 29, 61]. In one common formulation,

$$S_{ij} = \operatorname{Cov}_{p_\theta}[O_i, O_j], \quad g_i = 2 \operatorname{Cov}_{p_\theta}[E_L, O_i], \quad S \delta\theta = -g, \quad (\text{B.19})$$

with $O_i = \partial_{\theta_i} \log |\Psi_\theta|$.

Coulomb-induced cusp mismatch produces E_L spikes of the form (B.15)–(B.16), which has two consequences:

1. **Heavy-tailed Monte Carlo forces.** The estimator of $g_i = \text{Cov}[E_L, O_i]$ becomes noisy when E_L has rare but extreme spikes (short-range events). This increases variance and can destabilize the linear solve unless one uses damping/regularisation, as standard in SR practice [29].
2. **Coupling to curvature sensitivity.** Even if SR corrects anisotropy in parameter space (natural gradient) [27, 61], it does not remove the underlying operator-driven stiffness: the spikes originate from imperfect kinetic–potential cancellation enforced by Coulomb cusp physics.

Summary. Laplacian dominance creates stiffness through k^4 -type amplification of curvature errors; Coulomb creates stiffness by enforcing cusp constraints (in 2D: 1 vs 1/3 depending on spin channel) and cancellation against the Laplacian. Both mechanisms worsen conditioning of strong residual objectives and increase statistical difficulty of SR through heavy-tailed local-energy fluctuations [29, 43, 62].

Chapter C

The collocation–backflow catch-22

This appendix examines a difficulty encountered when backflow was trained with a *strong-form* collocation loss: the pointwise variance of the local energy E_L . Near a generic simple node of an approximate wavefunction, E_L has a pole. A smooth collocation proposal therefore overweights precisely the region in which the residual is singular, while screening that region changes the objective and may remove configurations that are especially sensitive to motion of the node. Across the interventions tested for the $N=6$, $\omega=0.5$ dot, this pipeline did not improve beyond an error of about 0.3%.

The local analysis explains why the strong form is difficult, but it does not prove that the observed floor is irreducible or that nodal singularities uniquely caused every failed run. The experiments also use smaller models than the production VMC calculation. The weak-form collocation method used in the main results avoids differentiating through E_L and performs substantially better. This appendix records the strong-form result and the reason for changing the objective.

C.1 Setup and notation

We consider a Slater–Jastrow–backflow ansatz of the form

$$\Psi_\theta(\mathbf{R}) = \det \tilde{S}_\uparrow(\mathbf{R}; \theta_{\text{BF}}) \det \tilde{S}_\downarrow(\mathbf{R}; \theta_{\text{BF}}) \exp(J(\mathbf{R}; \theta_J)), \quad (\text{C.1})$$

where $\tilde{S}_\sigma$ is the Slater matrix evaluated at backflow-shifted coordinates $\tilde{\mathbf{r}}_i = \mathbf{r}_i + \Delta\mathbf{x}_i(\mathbf{R}; \theta_{\text{BF}})$, J is a neural Jastrow correlator, and $\theta = (\theta_J, \theta_{\text{BF}})$ are trainable parameters. A CTNN message-passing network produces the backflow shifts $\Delta\mathbf{x}_i$.

We train by minimising the *screened collocation* objective

$$\mathcal{L}(\theta) = \text{Var}_{\mathcal{S}}[E_L(\mathbf{R}; \theta)], \quad E_L = \frac{\hat{H}\Psi_\theta}{\Psi_\theta}, \quad (\text{C.2})$$

where the collocation set $\mathcal{S}$ is obtained by importance-sampling from a Gaussian proposal q and retaining the top- K points ranked by $|\Psi_\theta|^2/q$ (“screened collocation”). This top- K selection naturally suppresses near-node configurations where $|\Psi|^2 \approx 0$.

C.2 The divergence of E_L near nodes

The local energy decomposes as

$$E_L = -\frac{1}{2} \frac{\nabla^2 \Psi}{\Psi} + V = -\frac{1}{2} \left[\nabla^2 J + |\nabla J|^2 + 2 \nabla J \cdot \frac{\nabla D}{D} + \frac{\nabla^2 D}{D} \right] + V, \quad (\text{C.3})$$

where $D := \det \tilde{S}_\uparrow \det \tilde{S}_\downarrow$ and the last term $\nabla^2 D/D$ contains second derivatives of the backflow shifts $\nabla^2 \Delta\mathbf{x}_i$ amplified by the inverse determinant.

For the exact wavefunction, E_L is constant. If Ψ is an eigenstate, $E_L = E$ everywhere. The cancellation is a property of the *full* local energy $\hat{H}\Psi/\Psi$ with $\Psi = De^J$, in which the Jastrow cross term $2\nabla J \cdot \nabla D/D$ of Eq. (C.3) participates alongside $\nabla^2 D/D$; we isolate the determinant terms below because they are the ones the backflow parameters move, not because the others are absent. For an exact eigenstate, the potentially singular contributions in the full expression $\nabla^2(De^J)/(De^J)$ cancel at a simple node. This is not a property of $\nabla^2 D/D$ in isolation: for a generic smooth determinant with $D \sim d$ near the nodal surface $\mathcal{N}$, $\nabla^2 D$ may remain $\mathcal{O}(1)$. The exact eigenstate satisfies

$$\frac{\nabla^2(De^J)}{De^J} \text{ finite on } \mathcal{N}, \quad (\text{C.4})$$

through cancellation among the determinant, Jastrow-cross, and remaining Jastrow terms in Eq. (C.3).

For an approximate wavefunction, the cancellation need not hold. Let d_θ be signed distance from a simple node of the trial wavefunction. Locally, $\Psi_\theta \sim ad_\theta$, whereas $\hat{H}\Psi_\theta$ is generically nonzero on that surface. The local energy then behaves as

$$E_L(\mathbf{R}) = \frac{\hat{H}\Psi_\theta}{\Psi_\theta} \sim \frac{c}{d_\theta} \quad \text{as } d_\theta \rightarrow 0, \quad (\text{C.5})$$

with a coefficient c that vanishes only when the required cancellation happens. Equation (C.3) shows how the pole can arise through the determinant terms, but no comparison with the unknown exact nodal surface is needed: it is a local property of the trial wavefunction and its residual.

C.3 Local power counting near a trial node

The sampling measure determines whether the pole in Eq. (C.5) is integrable. In local normal coordinates, a tubular neighbourhood of a codimension-one nodal surface has volume element $d\sigma dd_\theta$. For a simple node,

$$|\Psi_\theta|^2 \sim d_\theta^2, \quad E_L^2 \sim d_\theta^{-2}. \quad (\text{C.6})$$

Under the VMC measure $p_\theta \propto |\Psi_\theta|^2$, the local contribution to the second moment of E_L is therefore finite:

$$\int_{-\varepsilon}^{\varepsilon} p_\theta E_L^2 dd_\theta \sim \int_{-\varepsilon}^{\varepsilon} dd_\theta < \infty. \quad (\text{C.7})$$

The same second moment under a proposal with a smooth, nonzero density at the node scales as $\int d_\theta^{-2} dd_\theta$ and is not locally integrable. Directly differentiating the empirical strong loss is more singular still. For a parameter that moves the node, $\partial_\theta d_\theta \neq 0$ and

$$\partial_\theta E_L \sim \frac{\partial_\theta c}{d_\theta} - \frac{c \partial_\theta d_\theta}{d_\theta^2}. \quad (\text{C.8})$$

In the determinant representation, the same behaviour appears in terms such as

$$\frac{\partial E_L}{\partial \theta_{\text{BF}}} \supset \frac{1}{D} \frac{\partial(\nabla^2 D)}{\partial \theta_{\text{BF}}} - \frac{\nabla^2 D}{D^2} \frac{\partial D}{\partial \theta_{\text{BF}}}, \quad (\text{C.9})$$

which diverge as $D \rightarrow 0$. Finite batches regularise these expressions through their nearest sampled distance to the node, but the resulting updates can be dominated by rare points. This predicts severe heavy tails, not the impossibility of optimisation in every strong-form implementation.

C.4 The catch-22

The preceding analysis gives the practical tension behind the name of this appendix:

1. **Backflow moves the node.** Parameters that displace a simple node have a logarithmic derivative $\partial_{\theta_{\text{BF}}} \log |\Psi| \sim d_\theta^{-1}$, so near-node configurations are unusually sensitive to those parameters.

2. **The strong residual is singular at a generic trial node.** As shown in §C.2–C.3, E_L generally has a first-order pole, and differentiating it with respect to a node-moving parameter introduces a second-order pole.
3. **Screening controls the pole by suppressing its neighbourhood.** Under screened collocation, the top- K selection ranks points by $|\Psi|^2/q \propto D^2 e^{2J}/q$. Near-node points therefore tend not to enter the training batch. This stabilises the estimator, but it also weakens its sensitivity to the part of the wavefunction that changes most sharply when the node moves.

A finite implementation must therefore balance two errors:

Including points arbitrarily close to a trial node exposes the strong-form gradient to heavy tails. Screening them improves stability but changes the effective objective and can reduce the signal for moving the node.

The experiments below are consistent with this account, but do not isolate it from capacity, proposal design, or optimisation effects.

C.5 Why VMC and the weak form are less singular

VMC samples from $p_\theta \propto |\Psi_\theta|^2$. For a simple node, the probability mass within normal distance ϵ scales as ϵ^3 , so configurations closest to the node are suppressed without being removed entirely. The variational force for a real wavefunction is

$$g_i = 2 \operatorname{Cov}_{p_\theta}[E_L, O_i], \quad O_i = \partial_{\theta_i} \log |\Psi_\theta|. \quad (\text{C.10})$$

For a parameter that moves the node, $E_L \sim d_\theta^{-1}$ and $O_i \sim d_\theta^{-1}$. The per-sample product therefore scales as d_θ^{-2} . Its mean under $p_\theta \sim d_\theta^2$ is locally finite, although its second moment need not be. VMC can thus retain a usable nodal signal while remaining exposed to heavy-tailed finite-sample noise.

The decisive difference from the strong form is that the force in Eq. (C.10) does not differentiate E_L . The weak-form collocation estimator used later has the same property: it treats E_L as a detached reward and differentiates the score. SR additionally preconditions the VMC force with the sampled metric, but that does not remove the underlying tails. Robust trimming, damping, and the step cap in §4.4 control their finite-sample effect.

None of the strong-form interventions in §C.6 closed the observed gap. This supports the change of objective, while leaving open whether a different strong-form estimator or a fully matched experiment could do so.

C.6 Experimental evidence

We present a systematic study for $N=6$ electrons in a 2D harmonic trap with $\omega=0.5$ (DMC reference: $E_{\text{DMC}} = 11.78484(6)$ [55, 56]). The ansatz uses a neural Jastrow correlator ($\sim 15,000$ parameters) and a CTNN backflow network ($\sim 8,600$ parameters) with `bf_scale` = 0.7. All energies are evaluated via MCMC with 15 000 samples after burn-in.

C.6.1 Baseline: correlator alone vs. correlator + backflow

The backflow reduces the error from 0.42% to 0.32–0.33%, far short of the $<0.01\%$ reached by VMC + SR on the same ansatz *family* (Table 6.2). The two are not matched in size: the networks here total $\sim 24\text{k}$ parameters against $\sim 236\text{k}$ for the production model, so the comparison bounds what this objective achieves at this capacity rather than isolating the objective alone. The interventions below did not move the floor, but the capacity difference prevents us from attributing the full gap to the objective.

Table C.1: Baseline collocation results for $N=6$, $\omega=0.5$. “Correlator only” trains without backflow; “joint” trains both networks from scratch for 300 epochs with $\text{bf_scale} = 0.7$. The backflow provides only a marginal improvement of ~ 0.1 percentage points.

Configuration	E (Ha)	σ_E	err (%)
Correlator only (300 ep)	11.834	0.003	0.42
Correlator + BF, joint (300 ep)	11.823	0.003	0.33
Correlator + BF, separate opt	11.822	0.003	0.32
DMC reference	11.78484	—	0.00

Table C.2: Effect of sampling strategy on collocation accuracy. All experiments use 50-epoch fine-tuning from the best collocation checkpoint (0.35%), except “hard-smooth (scratch)” which trains 300 epochs from random initialisation.

Method	Description	err (%)	Outcome
Top- K baseline	standard screened collocation	0.35	stable
Hard mining + Huber	top- K of residuals	0.41	neutral
Hard mining + smoothness	+ $\ \nabla^2 \Delta x\ ^2$ penalty	0.38	slight help
SIR (single proposal)	$ \Psi ^2$ -reweighted resampling	0.47	worse
SIR (mixture)	Gaussian mixture proposal	0.44	worse
Gumbel top- K [63]	stochastic top- K selection	5.00	catastrophic

C.6.2 Sampling strategies

We tested multiple strategies to bring near-node information into the collocation batch, hoping to give the backflow a stronger learning signal.

Neither importance-resampled nor stochastic top- K selection improved the result. Gumbel top- K , which replaces deterministic selection with stochastic Gumbel-softmax, triggered early stopping after rapid energy divergence. These tests alter both coverage and selection, but we did not record the resulting distribution of distances to the node. They are therefore consistent with the heavy-tail account without directly establishing that greater near-node coverage caused the failures.

C.6.3 Loss modifications

Table C.3: Effect of loss function modifications. The hybrid loss adds a mean-energy (variational) term; the det filter discards points with small $|\det \tilde{S}|$.

Method	Description	err (%)	Outcome
Variance loss (baseline)	Huber on $E_L - \bar{E}$	0.33	best collocation
Hybrid anneal	$\beta \cdot \text{Var} + (1-\beta) \cdot \text{Mean}$, $\beta: 0.7 \rightarrow 0.15$	2.59	catastrophic
Gumbel + hybrid	stochastic top- K + hybrid	5.00	catastrophic
Hybrid + det filter	$\beta=0.2$ fixed + $ \det $ filter	24.75	broken

The mean-energy gradient $\nabla_\theta \langle E_L \rangle_S$ under biased collocation sampling does *not* equal the VMC gradient—the sampling distribution is not $|\Psi|^2$ —so the variational principle does not hold, and the backflow drifts to large, unphysical shifts ($|\Delta \mathbf{x}| \approx 0.83$, versus ~ 0.23 under variance-only training). The variance loss’s implicit constraint on backflow magnitude is *protective*, not pathological.

C.6.4 Gradient regularisation

We attacked the divergence sources in E_L directly:

Soft Coulomb trains on wrong physics (V is no longer the true potential). Regularising the determinant distorts the $\log |\Psi|$ landscape. Kinetic energy clipping is neutral because the clipped points are exactly those that carried the (noisy) backflow signal. None of the regularisations improve upon the baseline.

Table C.4: Regularisation of the E_L singularity (50-epoch fine-tune from 0.35% checkpoint). Each modification smooths or clips one of the divergent contributions to E_L .

Method	Modification	err (%)	Outcome
Soft Coulomb	$1/r \rightarrow 1/\sqrt{r^2 + \varepsilon^2}$, $\varepsilon=0.2$	1.41	much worse
Regularised det	$\log D \rightarrow \log \text{addexp}(\log D , \log \delta)$	0.49	worse
T clipping	$\text{sign}(T) \min(T , T_{\max})$, $T_{\max}=30$	0.43	neutral

C.6.5 Gradient preconditioning

Table C.5: Gradient preconditioning experiments (300-epoch scratch training unless noted). K-FAC [64] uses block-diagonal Kronecker-factored Fisher on the backflow network only.

Method	err (%)	Outcome
Vanilla Adam	0.33	baseline
K-FAC on CTNN	0.34	no improvement
K-FAC + smoothness	0.51	worse
Variance-minimising pretrain	0.54	worse
Adam fine-tune (30 ep, lr=1e-4)	0.34	no improvement
Adam burst (15 ep, lr=3e-4)	0.56	worse

K-FAC provides a block-diagonal approximation to the Fisher matrix and should, in principle, help with the ill-conditioned parameter-space geometry. Here it does not improve the energy. This is consistent with the problem arising partly from the differentiated local energy (Appendices A–B), which parameter-space preconditioning does not remove. The experiment alone does not separate that explanation from an inaccurate K-FAC approximation or poor tuning.

C.6.6 Summary of all interventions

The pattern is striking:

- Neither SIR nor Gumbel top- K improves the error. Because nodal distances were not logged, these runs do not show whether a change in near-node coverage was responsible.
- The interventions that changed the loss most directly (the hybrid mean-energy term and determinant filter) produced the largest errors. Because the collocation measure is not $|\Psi|^2$, their mean-energy terms are not variational; the experiments do not isolate this fact from the other changes made by those losses.
- The smoothness penalty $\lambda \|\nabla^2 \Delta \mathbf{x}\|^2$ helps modestly (from 0.59% to 0.38% without it vs. with it). This is consistent with reduced curvature sensitivity, but does not identify which contribution to the local energy changed.
- Among the differentiated-residual variants tested here, none approaches the production VMC + SR result. The later weak-form collocation method also trains backflow without differentiating E_L .

C.7 Connection to Appendices A and B

The catch-22 is a specific manifestation of the general conditioning difficulties analysed in Appendices A–B:

Laplacian stiffness (Appendix A). The k^4 -type amplification of curvature errors (§A.3) is compounded by backflow, because $\nabla^2(\det \tilde{S})$ introduces second derivatives of $\Delta \mathbf{x}$ that are not present in a fixed-node ansatz. These additional high-frequency modes inflate the spectral spread of the training operator L^*L .

Table C.6: Complete summary of all collocation-based backflow experiments for $N=6$, $\omega=0.5$. No intervention breaks the $\sim 0.32\%$ accuracy floor. The final row is the same ansatz *family* under VMC + SR at production width (Table 6.2), an order of magnitude larger in parameter count; it is included to show the size of the gap, not as a matched comparison.

Experiment	err (%)	Category
Correlator + BF, sep opt (best)	0.32	baseline
Correlator + BF, joint 300 ep	0.33	baseline
K-FAC on CTNN	0.34	preconditioner
Adam fine-tune 30 ep	0.34	optimiser
Top- K baseline (retrained)	0.35	sampling
Hard mining + smoothness	0.38	sampling
Hard mining + Huber	0.41	sampling
Correlator only	0.42	no backflow
T clipping	0.43	regularisation
SIR (mixture)	0.44	sampling
SIR (single)	0.47	sampling
Regularised det	0.49	regularisation
K-FAC + smoothness	0.51	preconditioner
Var-min pretrain	0.54	pretrain
Adam burst lr=3e-4	0.56	optimiser
Soft Coulomb	1.41	regularisation
Hybrid loss (annealed)	2.59	loss
Gumbel top- K + hybrid	5.00	sampling + loss
Hybrid + det filter	24.75	loss + filter
Production family, VMC + SR	<0.01	variational

Coulomb singularity (Appendix B). The $1/r$ Coulomb potential (§B.1) adds a singular multiplication operator to $\hat{H}$. Near both the coalescence point ($r_{ij} \rightarrow 0$) and the node ($D \rightarrow 0$), E_L contains products of these two singularities, and the cusp cancellation (§B.2) must be maintained simultaneously with the nodal cancellation (C.4). In the implemented strong objective $\mathbb{E}_q[(E_L - E_{\text{eff}})^2]$, these compound singularities create an extremely stiff landscape.

The backflow-specific pathology. What distinguishes the backflow catch-22 from the general conditioning problem is the *self-referential* nature of the singularity: the backflow parameters θ_{BF} determine *where* the node is, so updating θ_{BF} moves the singularity in configuration space. Under the collocation loss, the gradient is determined by E_L at the current collocation points while the update itself moves the node. Reused points may therefore cease to describe the most relevant neighbourhood after the update. VMC+SR instead refreshes samples from the current $|\Psi_\theta|^2$, allowing its sampling distribution to track the moving node more closely. This difference is consistent with the observed stability gap, but was not isolated from model size or the other estimator choices.

Chapter D

Additional collocation results

This appendix records the collocation results that Chapter 7 refers to but does not tabulate: the $N=20$ energies, and the variants that were tried and failed. Both bear on the same question—what limits the MCMC-free update route—from the side of what does not work.

D.1 What was varied

Four axes were explored. The *sampling*: Gaussian and Gaussian-mixture proposals scaled by $1/\sqrt{\omega}$, oversample-then-resample with ESS diagnostics, ESS-adaptive oversampling, weight tempering and log-weight clipping. The *objective*: finite-difference collocation residuals, a hybrid weak form, and pure REINFORCE, which became the default, with the Laplacian kept in the reward path but never differentiated. The *preconditioner*: Adam, diagonal-Fisher natural gradient, and full SR in Woodbury and CG modes with damping anneals and parameter-step caps. And the *training style*: single sweeps, wave-based cascades with checkpoint inheritance, and asynchronous campaigns relaunching from the best available checkpoint.

Two patterns held across this unmatched search. Moderately sized backflow+Jastrow models with stable training outperformed the larger variants tested here, without isolating whether size, architecture or optimisation caused the difference. At low ω , regularising the resampling distribution (ESS control, top-mass reduction) helped where hard cusp gating did not—a controlled $2 \times 2 \times 3$ matrix over gating, resampling regularisation and three seeds showed no measurable gain from gating and a consistent gain from regularisation.

Warm-starting along the confinement cascade $\omega : 1.0 \rightarrow 0.5 \rightarrow 0.1 \rightarrow 0.01 \rightarrow 0.001$ was necessary throughout; cold starts at low ω did not converge. Representative cascade results are $N=6$ at $\omega=1.0$ ($E = 20.16167 \pm 0.00181$, +0.012%), $\omega=0.5$ ($E = 11.78466 \pm 0.00199$, -0.002%) and $\omega=0.1$ ($E = 3.56525 \pm 0.00046$, +0.32%), and $N=12$ at $\omega=1.0$ ($E = 65.70699 \pm 0.00595$, +0.010%). These are single end-of-run evaluations of selected runs and were not independently re-evaluated; the selection effect documented in §7.3.4 makes the strong-confinement values among them optimistic by several hundredths of a percent. For $N=2$, the raw Metropolis deviations are +0.035%, -0.019%, +0.035%, +0.098% and +0.806% at $\omega = 1, 0.5, 0.1, 0.01, 0.001$. The archived zero-variance regression intercepts give +0.012%, +0.023%, -0.004%, +0.002% and +0.154%, respectively. Those intercepts are not variational estimates, and no regression interval was retained, so they are not used as the headline energies in Table 7.5.

D.2 The $N = 20$ collocation energies

At $N=20$ the backflow’s edge-feature tensor, which scales as $\mathcal{O}(N^2 h_{\text{bf}})$, forces the hidden dimension down to $h_{\text{bf}}=48\text{--}64$ to fit in memory. The best backflow run at $\omega=1.0$ then reached only $\sim+18\%$ error, while a Jastrow-only ansatz—freed of that tensor, and so able to afford a larger collocation budget and higher oversampling—reached +1.32% ($E = 157.934 \pm 0.018$). The more expressive architecture is beaten by the simpler one, which is consistent with the memory-forced capacity cut degrading gradient quality more than its extra flexibility repays.

Polishing the Jastrow checkpoints at reduced learning rate improved the $\omega=1.0$ error from $\sim+2.6\%$ to $+1.32\%$ over ~ 400 epochs, and the $\omega=0.5$ error from $\sim+7.0\%$ to $+2.74\%$ ($E = 96.447 \pm 0.015$); the best $\omega=0.1$ result was $+5.53\%$ ($E = 31.637 \pm 0.008$). The gap to the sub-percent accuracy reached at $N \in \{6, 12\}$ tracks the capacity cut and the gradient noise it brings. A more memory-frugal backflow is the direct next intervention; these experiments do not isolate the collocation objective as the source of the remaining gap.

D.3 Negative results worth recording

Scalar final energy is not sufficient for method development on this route: ESS, top-mass concentration, rollback counts and the probe-to-final gap are what distinguish a run that converged from one that stopped moving. Several intuitively attractive interventions delivered nothing in the tested range—hard cusp gating most notably—and recording that was necessary to keep the search from circling. Stricter SR step caps, heavier damping, ESS-adaptive sampling and rollback controls improved operational stability without closing the accuracy gap at the hardest targets. In the tested $N=6$ low- ω campaign, changing the proposal was the intervention associated with closing that gap (§7.4).

Chapter E

The learned representation of one trained solution

This appendix records what the internal representation of *one* high-quality trained solution looks like. It is descriptive, it is measured on a single training run, and—as §7.1.2 sets out—most of the quantities in it are component-level and therefore gauge-like: the correlator and the backflow are partial substitutes, so how the labour is divided between them records the route the training took as much as the state it reached. Nothing in the architectural argument of Chapter 7 rests on anything here. It is reported because the regularities are striking and because they are ones the network was free *not* to adopt, not because they establish a property of the architecture.

Three findings survive that qualification well enough to be worth stating. Across $N \in \{2, 6, 12, 20\}$ and $\omega \in \{1, 0.5, 0.1, 10^{-2}, 10^{-3}\}$ the correlator feature space stays on one to three effective dimensions and never expands, even where the real-space density develops elaborate shell structure; what changes across the crossover is which branch the leading axis is built from, not how many axes there are. The displacement field is the opposite kind of object, occupying nearly all the spatial directions available to it. And the coupling between the two changes character near $\omega \approx 0.1$, the same place the energy partition crosses over (§6.2.3).

Every diagnostic below is evaluated on fresh samples drawn from $|\Psi_\theta|^2$, with the estimators and stability protocol of §5.2.

E.1 Correlator geometry: a low-rank representation that reorganises

We characterise the correlator feature space $Z = \bar{\phi} \|\bar{\psi}\| \mathbf{g}$ by its entropy-based effective rank $r_{\text{eff}}(Z)$, the correlation $|\rho(\text{head}, \text{PC1})|$ between the head output and its first principal component, and the power of PC1 decomposed across architectural branches. Table E.1 reports these quantities across the grid.

The manifold is low-dimensional everywhere. The effective rank stays below 3.1 across the entire grid, and the head output is almost perfectly reconstructed from PC1 alone in all but one analysed cell ($|\rho| \geq 0.94$ except $N=20, \omega=0.01$, where $|\rho| = 0.79$). This holds even at strong correlation where the real-space density develops pronounced shell structure—the correlator does not respond by expanding its latent dimensionality. The modest peak at $N=12, \omega=0.1$ ($r_{\text{eff}} = 3.08$) coincides with the intermediate-correlation regime where neither the shell picture nor the crystalline picture provides a clean single-mode description.

A $\phi \leftrightarrow \psi$ transition in the leading axis. What varies across regimes is *which branch dominates PC1*. At moderate-to-weak correlation ($\omega \geq 0.1, N \geq 6$), PC1 is predominantly ψ -loaded: the pair branch carries 60–73% of the variance, meaning the correlator’s primary degree of freedom tracks pair correlations. At strong correlation the loading shifts toward ϕ (orbital) or $\mathbf{g}$ (auxiliary) branches for $N=6$ and $N=12$. The $N=20$ run does not follow the same branch-switching pattern: its pair branch remains dominant at both weakest confinements. Given that these are single-run, component-level diagnostics,

Table E.1: Correlator geometry (CTNN). r_{eff} : entropy-based effective rank of Z . $|\rho|$: absolute correlation between head output and PC1 reconstruction. $\phi/\psi/g$: fraction of PC1 variance from the orbital, pair, and auxiliary branches. The ψ -dominated entries ($\psi > 0.5$) are shown in bold.

N	ω	r_{eff}	$ \rho $	ϕ	ψ	g
2	0.001	1.00	1.00	0.00	1.00	0.00
2	0.01	1.04	0.66	0.00	1.00	0.00
2	0.1	1.31	0.99	0.00	0.53	0.46
2	0.5	1.25	1.00	0.00	0.58	0.42
2	1.0	1.53	0.98	0.00	0.35	0.65
6	0.001	1.52	0.97	0.01	0.12	0.87
6	0.01	1.38	0.99	0.05	0.44	0.50
6	0.1	2.73	0.94	0.66	0.21	0.14
6	0.5	2.69	0.98	0.15	0.63	0.22
6	1.0	2.69	0.99	0.03	0.62	0.35
12	0.001	1.66	0.96	0.54	0.13	0.33
12	0.01	1.54	0.97	0.43	0.09	0.48
12	0.1	3.08	0.99	0.12	0.63	0.26
12	0.5	2.32	0.99	0.12	0.69	0.19
12	1.0	1.76	0.99	0.08	0.73	0.19
20	0.001	2.59	0.99	0.18	0.74	0.08
20	0.01	1.09	0.79	0.04	0.94	0.01
20	0.1	2.20	0.97	0.36	0.44	0.20
20	0.5	2.30	1.00	0.17	0.70	0.13
20	1.0	2.32	0.99	0.18	0.71	0.11

we do not interpret this discrepancy as a size-dependent physical transition. The transition is sharpest at $N=12$, where ψ goes from 0.09 at $\omega=0.01$ to 0.63 at $\omega=0.1$.

These loadings track the physics. When interactions are weak relative to the trap ($\omega \gtrsim 0.1$), the non-interacting shell structure provides a good zeroth-order description, and the correlator’s primary role is refining pair correlations on top of that. When interactions dominate ($\omega \lesssim 0.01$), the leading correlator coordinate shifts away from pair features toward the single-particle and auxiliary branches. The network reorganises the features through which it represents the state, and that reorganisation—not any growth in dimension—is how it adapts. Nothing in the architecture imposes this behaviour: the $N=6$ and $N=12$ regime changes appear as a rotation of a low-dimensional axis across branches rather than as growth in latent dimensionality.

E.2 Backflow: a high-rank field with a low-rank footprint

Table E.2 characterises the message-passing displacement field. We report the normalised displacement magnitude $\|\Delta\mathbf{x}\|/\ell$, the effective rank of $\Delta\mathbf{x}$ (viewed as a $B \times 2N$ matrix of flattened displacements), and the CKA similarity between the correlator features with and without backflow.

Displacement magnitude scales with ω but saturates near $\|\Delta\mathbf{x}\|/\ell \sim 0.1$. At strong correlation ($\omega \leq 0.01$), displacements are small relative to the harmonic length, $\|\Delta\mathbf{x}\|/\ell \sim 0.02\text{--}0.05$. As the trap tightens and the system enters the moderate-to-weak correlation regime, the magnitude grows and saturates at $\sim 8\text{--}10\%$ of ℓ for $\omega \gtrsim 0.1$. The message-passing backflow has small displacements throughout this grid. Its tanh output bound limits their scale; the present data do not isolate any additional effect of the message-passing structure. The conventional backflow can exhibit displacements comparable to the system scale in poorly converged regimes (e.g. $\|\Delta\mathbf{x}\|/\ell \sim 0.95$ for $N=6$, $\omega=0.001$ with the conventional network).

Full-rank displacements, near-unity CKA. The effective rank of the displacement field is consistently $\approx 2N - 2$, saturating the available spatial degrees of freedom modulo centre-of-mass conservation (confirmed by COM shifts of order $10^{-14}\text{--}10^{-17}$). Under the stated cutoff, the sampled displacement

Table E.2: CTNN backflow geometry. $\|\Delta\mathbf{x}\|/\ell$: mean displacement magnitude normalised by harmonic length. Rank: effective rank of the displacement covariance. $2N-2$: maximum possible rank after centre-of-mass projection. CKA: linear centred kernel alignment between correlator features with and without backflow (1.0 = identical feature manifolds).

N	ω	$\ \Delta\mathbf{x}\ /\ell$	Rank	$2N-2$	CKA
2	0.001	0.003	1.9	2	1.000
2	1.0	0.029	2.0	2	1.000
6	0.001	0.017	10.0	10	0.999
6	0.01	0.035	9.9	10	0.998
6	0.1	0.096	10.0	10	0.995
6	0.5	0.093	10.0	10	0.991
6	1.0	0.089	10.0	10	0.979
12	0.001	0.017	21.9	22	1.000
12	0.01	0.030	21.9	22	0.998
12	0.1	0.096	21.9	22	0.993
12	0.5	0.096	21.9	22	0.993
12	1.0	0.091	21.9	22	0.995
20	0.001	0.023	37.7	38	0.999
20	0.01	0.046	37.4	38	1.000
20	0.1	0.075	37.8	38	0.997
20	0.5	0.099	37.6	38	0.990
20	1.0	0.099	37.6	38	0.991

matrix is not dominated by one or two collective directions such as breathing or rigid rotation. Its effective rank reaches the nominal COM-free ceiling. Rank alone does not show that every particle depends on every other particle. The field stays near that ceiling across the whole grid, including the deepest Wigner points, where one might have expected the correction to become collective. Whether that is a property of *this* architecture or of trained backflows generally is not something the present measurement can decide. Chapter 7 compares the two implemented backflow families, but their capacity, preprocessing and centre-of-mass handling are not matched (§7.1.2).

Yet the CKA between correlator features with and without backflow exceeds 0.99 in nearly all cases (and exceeds 0.979 everywhere). The correlator’s measured feature geometry is therefore similar with and without the displacement field on these samples. This is a statement about that geometry, not about the subspace the backflow occupies. Whether the two components are genuinely complementary, rather than merely uncorrelated on this measure, is tested by ablation in §7.1.2.

E.3 Correlator–backflow alignment

The previous subsection found that the correlator feature geometry changes little when backflow is present in this trained solution. We next ask whether the displacement aligns with quantities derived from the correlator. Three diagnostics address that narrower question: (i) the mean cosine similarity $\langle \cos(\nabla_{\mathbf{x}} f, \Delta\mathbf{x}) \rangle$ between the correlator’s gradient in configuration space and the displacement field (alignment implies the backflow pushes along directions the correlator is sensitive to); (ii) the fraction of configurations where $\Delta\mathbf{x}$ is aligned with ∇f ; and (iii) $|\delta f|/\sigma_f$, where $\delta f = f(\tilde{R} + \Delta\mathbf{x}) - f(\tilde{R})$ is the change in the correlator output when it is evaluated on the backflow-shifted coordinates, in units of the correlator’s own standard deviation σ_f . Two cautions attach to this last quantity. It is a *counterfactual sensitivity*: in the production ansatz the correlator sees the unshifted $\tilde{R}$ (only the determinant sees $\tilde{R} + \Delta\mathbf{x}$; §3.5), so δf measures how much the correlator *would* respond to the displacement, not an actual term in the wavefunction. And it is a typical relative change, not a variance decomposition: $|\delta f|/\sigma_f = 0.7$ means the counterfactual change is 0.7 standard deviations, not that 70% of the variance is attributable to backflow. Table E.3 reports these alongside the fraction of particles pushed toward their nearest neighbour.

An alignment change near $\omega \approx 0.1$ on the sampled grid. The data show their largest change between two adjacent sampled confinements. At $\omega \leq 0.01$ the coupling is weak and inconsistent in

Table E.3: Correlator–backflow coupling diagnostics (message-passing ansatz). $\langle \cos \rangle$: mean cosine between ∇f and $\Delta \mathbf{x}$. Aligned: fraction of configs with $\nabla f \cdot \Delta \mathbf{x} > 0$. $|\delta f|/\sigma_f$: backflow-induced change in the correlator output, relative to its standard deviation. $\rightarrow \text{NN}$: fraction of particles displaced toward their nearest neighbour (0.5 = isotropic). Only $N \geq 6$ shown; $N=2$ displacements are near-zero for most ω .

N	ω	$\langle \cos \rangle$	Aligned	$ \delta f /\sigma_f$	$\rightarrow \text{NN}$
6	0.001	0.14	0.63	0.03	0.43
6	0.01	0.18	0.77	0.05	0.28
6	0.1	0.49	0.99	0.25	0.16
6	0.5	0.62	1.00	0.33	0.19
6	1.0	0.77	1.00	0.41	0.18
12	0.001	−0.48	0.01	0.03	0.61
12	0.01	0.22	0.82	0.05	0.29
12	0.1	0.56	1.00	0.39	0.12
12	0.5	0.65	1.00	0.49	0.20
12	1.0	0.75	1.00	0.55	0.21
20	0.001	0.32	0.97	0.06	0.44
20	0.01	0.22	0.89	0.07	0.45
20	0.1	0.39	1.00	0.25	0.17
20	0.5	0.65	1.00	0.58	0.22
20	1.0	0.78	1.00	0.69	0.25

sign: $\langle \cos \rangle$ runs from -0.48 to 0.32 across the six cells, the alignment fraction is erratic, and backflow barely affects the correlator output ($|\delta f|/\sigma_f < 0.07$ everywhere). The $N=12$, $\omega=0.001$ cell is strongly *anti-aligned* ($\langle \cos \rangle = -0.48$, alignment fraction 0.01), which we record without an account of it: at that confinement the displacement field is small enough (Table E.2) that its direction relative to the correlator gradient carries little energy either way, but we cannot say why the sign is systematic rather than scattered. In this regime the backflow operates in a subspace largely orthogonal to the correlator’s gradient—it modifies the wavefunction in directions that W_θ is insensitive to, consistent with its role being to adjust the Slater determinant’s orbital positions rather than the correlator.

At $\omega \geq 0.1$, the picture changes qualitatively. The gradient cosine jumps to 0.5 – 0.8 , the alignment fraction reaches 1.0 for $N \geq 6$, and the counterfactual output change reaches $|\delta f|/\sigma_f = 0.25$ – 0.69 (i.e. up to ~ 0.7 standard deviations of the correlator). Here the displacement is aligned with directions to which the correlator is especially sensitive. This is compatible with cooperative refinement, but the diagnostic does not show that it corrects a particular residual error. The largest sampled change lies between $\omega = 0.01$ and $\omega = 0.1$, coinciding with the crossover in $\Gamma = V_{\text{int}}/T$ from $\Gamma > 10$ (interaction-dominated) to $\Gamma \sim 3$ – 6 (moderate correlation).

Directional structure: correlation-hole deepening. The nearest-neighbour direction fraction provides a complementary signature. At $\omega \geq 0.1$ for $N \geq 6$, only 12–25% of particles are displaced toward their nearest neighbour. The displacement points away from the nearest neighbour more often than toward it, which is consistent with a deepening of the correlation hole. This directional bias is strongest at $N=12$, $\omega=0.1$ ($\rightarrow \text{NN} = 0.12$), exactly where the correlator effective rank peaks and the $\phi \leftrightarrow \psi$ transition occurs. At strong correlation ($\omega \leq 0.01$, large N), the fraction rises toward 0.5 (isotropic). This is consistent with the larger equilibrium separations reducing the directional correlation-hole signal.

E.4 Limits of component-level interpretation

Taken together these measurements describe two components of very different character. The correlator occupies one to three directions, whose leading axis tracks the dominant global coordinate of the cloud and changes identity rather than growing in number as the physics changes. The displacement field is high-rank and configuration-dependent, and its presence leaves the correlator’s feature geometry almost unchanged ($\text{CKA} > 0.98$). The coupling between them is weak and inconsistent in sign at strong correlation, and strong and systematically aligned at moderate coupling, where the displacements follow the

correlator's gradient; the change is sharp and coincides with the crossover in Γ .

These measurements describe one trained solution and do not by themselves establish which properties are architecture-specific. A different training route could divide the same labour differently and arrive at the same energy, which for two partial substitutes is what one should expect. Deciding the question requires a matched experiment that varies only the architecture.

Chapter F

Structural diagnostics in full

Chapter 6 reports the two radial quantities the argument turns on— r_{mode} and the width-to-scale ratio γ . The complete diagnostics are collected here: the mean radius, width, FWHM and decile quantiles for $N=2$, and the ring-angle spread and pair Lindemann ratio for $N=6$. Estimators and stability protocol are defined in §5.1. All lengths are in Bohr; angles in radians.

F.1 Fitted energy and radius exponents

Section 6.2.3 states two conclusions from the energy partition—that interactions come to dominate, and that the trap–interaction balance approaches its classical form—and defers the fitted exponents that support them to here.

Log–log fits $E(\omega) \propto \omega^\alpha$ over the grid $\omega \in \{10^{-3}, 10^{-2}, 10^{-1}, 0.5, 1\}$ give a kinetic exponent $\alpha_T \simeq 1.00$ at every N , an interaction exponent $\alpha_{V_{\text{int}}} \simeq 0.66$, and trap exponents $\alpha_{V_{\text{trap}}} \simeq 0.81, 0.74, 0.72$ for $N = 2, 6, 12$. The kinetic energy therefore scales essentially linearly with ω while both potential terms grow sublinearly with nearly N -independent exponents; the ratio $\Gamma \propto \omega^{\alpha_{V_{\text{int}}} - \alpha_T} \approx \omega^{-0.34}$ predicts a factor $10^{3 \times 0.34} \approx 10.5$ across three decades, against the measured 9.3, 10.2 and 11.4.

From $V_{\text{trap}} = \frac{1}{2}\omega^2 \sum_i \langle r_i^2 \rangle$ the rms radius is $r_{\text{rms}} = \sqrt{2V_{\text{trap}}/(N\omega^2)}$, and fitting $r_{\text{rms}} \propto \omega^{\alpha_r}$ gives $\alpha_r \simeq -0.60$ to -0.64 , equivalently $\langle r^2 \rangle \propto \omega^{-\beta}$ with $\beta \simeq 1.19$ – 1.28 . The cloud therefore expands faster than the non-interacting ω^{-1} law for $\langle r^2 \rangle$.

These are averaged quantities and should not be confused with the shell-resolved radius of §F.4, which is the one compared against the classical Wigner prefactor; the two differ because r_{rms} and r_{mode} mix the central electron of a $(1, n)$ topology with the ring, and the shell-resolved radius does not.

Table F.1: Radial diagnostics for $N=2$ across ω (Bohr units).

ω	r_{mode}	$\langle r \rangle$	σ_r	FWHM	q_{10}	q_{50}	q_{90}	γ
1.00	1.435	1.632	0.705	1.743	0.735	1.581	2.581	0.491
0.50	2.303	2.475	1.007	2.466	1.209	2.409	3.829	0.437
0.10	3.601	3.830	1.728	4.449	1.640	3.714	6.127	0.480
0.01	14.191	14.593	5.615	13.602	7.390	14.322	22.038	0.396
0.001	64.292	63.953	18.479	43.977	40.392	63.814	87.714	0.287

Table F.2: Radial and angular diagnostics for $N=6$ across ω (Bohr; angles in radians).

ω	r_{mode}	σ_r	γ	$\sigma(\text{ring angles})$	$\gamma_{r_{ij}}$
1.00	2.0907	0.9249	0.4424	0.7046	0.3538
0.50	3.0702	1.3648	0.4445	0.6943	0.3397
0.10	8.1686	3.5025	0.4288	0.6842	0.3149
0.01	33.1538	13.9094	0.4195	0.6497	0.2889
0.001	113.5542	47.8076	0.4210	0.6368	0.2664

F.2 Shell-resolved structure across the whole grid

Chapter 6 quotes shell occupancies, radii, bond order and Lindemann numbers for individual (N, ω) cells in running prose. Table F.4 collects them for the whole grid from a single analysis pass, so that the trends can be read off instead of assembled from the text. Shell radii are given in trap units $\tilde{r} = \sqrt{\omega} r$; multiply by $\ell = \omega^{-1/2}$ for Bohr.

Reading $|\Phi_n|$ against its null. One column of that table needs a baseline to be interpretable, and its absence would make the bond-order numbers read far more favourably than they should. For n particles placed at *independent uniform* angles on a ring, $n\Phi_n = \sum_k e^{in\phi_k}$ is a two-dimensional random walk of n unit steps, so

$$\langle |\Phi_n| \rangle_{\text{random}} = \frac{1}{n} \int_0^\infty \frac{1 - J_0(t)^n}{t^2} dt \xrightarrow{n \rightarrow \infty} \frac{\sqrt{\pi}}{2\sqrt{n}} \approx \frac{0.886}{\sqrt{n}}, \quad \langle |\Phi_n|^2 \rangle_{\text{random}} = \frac{1}{n}. \quad (\text{F.1})$$

The exact values at the occupancies that occur are 0.525, 0.402, 0.366, 0.338, 0.298, 0.269, 0.257 and 0.204 for $n = 3, 5, 6, 7, 9, 11, 12, 19$; the Rayleigh limit undershoots them by 0.4–2.6%, which is why it is not used here. A finite $|\Phi_n|$ is therefore *not* by itself evidence of angular order: a five-particle ring with no order at all returns $|\Phi_5| \approx 0.40$. The ratio $|\Phi_n|/\langle |\Phi_n| \rangle_{\text{random}}$ is the quantity that carries information, and Table F.4 reports it.

Read that way the result is sharper than the raw values suggest, and it needs stating with the right threshold. From $\omega = 1$ down to $\omega = 0.1$ the ratio is 0.99–1.01 at every particle number and on every ring. At $\omega = 0.01$ it lifts, to 1.08 at $N=6$ and 1.02 at $N=12$, while $N=20$ stays at 1.00. Only at $\omega = 10^{-3}$ does it become large: 1.63 on the $N=6$ ring, 1.16 and 1.55 on the two rings of the $N=12$ (3, 9) molecule, and 1.18 and 1.46 on the outer two rings of the $N=20$ (1, 7, 12) molecule.

Whether the small departures are *resolved* is a separate question from whether they are *large*, and here both can be answered. Because the shells are assigned by radial rank alone and no angular cut is applied, Eq. (F.1) is also the null of this particular pipeline; generating it numerically instead—keeping every frame’s radii and redrawing all angles uniformly, then running the identical shell assignment—reproduces it within its error in every cell. Block-bootstrap errors on the ratio are 0.002–0.007. On that scale the $N=6$ ring at strong confinement sits resolvable *below* the null, at 0.989 ± 0.002 for $\omega = 1$ and 0.990 ± 0.002 for $\omega = 0.5$. That is physics rather than artefact. Since $\langle |\Phi_n|^2 \rangle = n^{-1} + n^{-2} \sum_{j \neq k} \langle \cos n(\theta_j - \theta_k) \rangle$, a depletion of close angular pairs—a correlation hole along the ring—makes the cross terms negative: what the ring shows at strong confinement is short-range angular repulsion without n -fold order. At $N=12$ and $N=20$ the ratio stays within 0.5% of the null through $\omega = 0.1$.

The conclusion that survives is the one that matters: statistical detectability is not what separates the regimes here, magnitude is. On that criterion n -fold order is absent through $\omega = 0.1$, where the resolved departures are all within about one percent; weakly present at $\omega = 0.01$ and clearly so only at $N=6$; and unambiguous at 10^{-3} , where every ring exceeds the null by 16–63%.

The outer-shell relative widths generally decrease as the trap weakens, while several inner-shell widths are nonmonotonic when the topology changes. On these diagnostics, radial localisation is already visible before the large rise in n -fold bond order anticipated in §1.4.

The values in this table are for the *modal* topology over all frames. The sector-conditioned values quoted for $N=6$ in §6.2.5 are computed only on the frames assigned to a given sector, and in an earlier analysis pass whose inputs we could not match to the runs used here. They are not uniformly higher, and the differences are not small (Table F.3). At $\omega = 0.01$ the excess over the null ranges from 8% on the modal estimator to 13% and 27% on the two sectors; at 10^{-3} the modal value lies between the two sector values. The staging does not depend on the choice: every estimator places $N=6$ modestly above the null at $\omega = 0.01$ and far above it at 10^{-3} . The size of the $\omega = 0.01$ excess does, and should be quoted as a range.

Table F.3: Two estimators of the $N=6$ bond order, as ratios to the exact random-angle null (0.402 for $n=5$, 0.366 for $n=6$). The modal estimator (this appendix) assigns every frame to (1, 5); the sector-conditioned estimator (§6.2.5) splits the frames between (1, 5) and (0, 6) and comes from an earlier analysis pass.

ω	modal, all frames		sector-conditioned	
	$ \Phi_5 $ on (1, 5)	$ \Phi_5 $ on (1, 5) frames	$ \Phi_6 $ on (0, 6) frames	
0.01	1.08	1.13 (69% of frames)	1.27 (31%)	
0.001	1.63	1.49 (80%)	1.81 (20%)	

Table F.4: Shell-resolved diagnostics at the modal topology, all frames, for every (N, ω) . Occupancies are $(n_{\text{in}}, \dots, n_{\text{out}})$; $\tilde{r}_s$ are mean shell radii in trap units; $w_s = \sigma(r_s)/\langle r_s \rangle$ are per-shell radial relative widths; $|\Phi_{n_s}|$ is the rotation-invariant bond-order modulus on shell s ; and the final column is $|\Phi_{n_s}|$ divided by its exact random-angle expectation, Eq. (F.1). A value of 1.00 in that column means the ring is equal to the independent-angle null within quoted precision; see the text for why departures of about a percent are at the resolution of this comparison. Single-particle inner shells are omitted from the angular columns, where $|\Phi_1| = 1$ trivially.

N	ω	occ.	$\tilde{r}_s$	w_s	$ \Phi_{n_s} $	ratio to null
6	1.0	(1, 5)	0.70, 1.66	0.43, 0.32	0.397	0.99
6	0.5	(1, 5)	0.76, 1.78	0.42, 0.31	0.397	0.99
6	0.1	(1, 5)	0.93, 2.15	0.41, 0.28	0.405	1.01
6	0.01	(1, 5)	1.33, 2.96	0.42, 0.22	0.434	1.08
6	0.001	(1, 5)	1.12, 4.29	0.63, 0.15	0.654	1.63
12	1.0	(1, 11)	0.55, 2.00	0.47, 0.34	0.268	1.00
12	0.5	(1, 11)	0.60, 2.18	0.47, 0.33	0.268	1.00
12	0.1	(1, 11)	0.74, 2.68	0.45, 0.31	0.270	1.00
12	0.01	(1, 11)	1.04, 3.77	0.41, 0.28	0.275	1.02
12	0.001	(3, 9)	2.52, 6.04	0.61, 0.24	0.611, 0.462	1.16, 1.55
20	1.0	(1, 19)	0.55, 2.32	0.44, 0.36	0.204	1.00
20	0.5	(1, 19)	0.60, 2.52	0.44, 0.35	0.203	1.00
20	0.1	(1, 19)	0.75, 3.14	0.42, 0.34	0.204	1.00
20	0.01	(1, 19)	1.05, 4.47	0.49, 0.32	0.205	1.00
20	0.001	(1, 7, 12)	1.27, 4.19, 7.83	0.47, 0.24, 0.11	0.398, 0.375	1.18, 1.46

F.3 Shell-model fit to the pair distribution

The reconstruction quoted in §6.2.5 tests whether the measured pair distribution is explained by the detected shell geometry. Two versions of that test exist and they are not equally informative.

The weaker version decomposes the pairs of two-shell frames into inner–inner, inner–outer and outer–outer histograms and recombines them with the combinatorial weights of the ideal topology. Because the sector histograms and the total are accumulated from the same frames, this recombination is close to exact by construction—cosine similarities exceed 0.99999 at every (N, ω) —and it establishes only that the sector bookkeeping is self-consistent.

The stronger version is a *generative* one and is the test reported here. Shell radii and widths are fitted on one half of the frames; each shell is then modelled as a regular n_s -gon carrying a per-frame global rotation and a fitted angular jitter; and the resulting pair-distance density is compared with the empirical density on the *held-out* half. Against it we run an angle-randomised null with the same fitted radial distributions but uniform angles, which isolates what polygonal order contributes over and above shell radii alone. The split tests the fitted parameters conditionally: the shell topology and broad analysis design were selected using the full data, so this is not held-out model selection. Table F.5 reports both.

The polygon model reproduces the held-out pair distribution to cosine similarity 0.987–0.999 across the whole grid. The null reaches 0.936–0.996, and the gap between them, $\Delta \cos$, grows monotonically as the trap weakens at every particle number—from 0.012 to 0.058 at $N=6$, from 0.006 to 0.026 at $N=12$, and from 0.003 to 0.010 at $N=20$. Shell radii alone account for most of the pair structure at strong

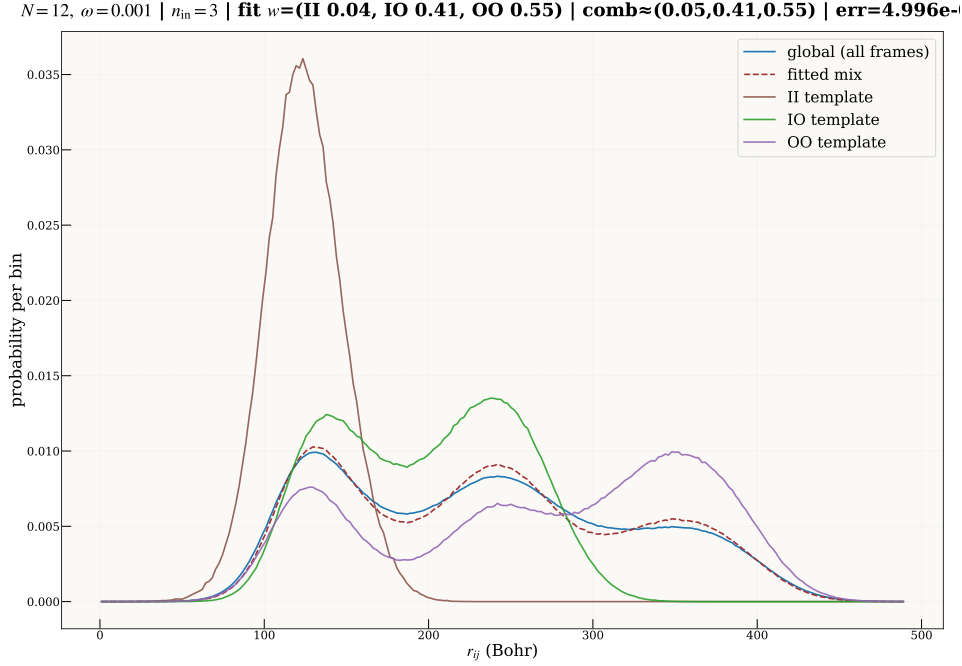


Figure F.1: Shell-resolved reconstruction of $g(r)$ at $\omega = 10^{-3}$ for $N=12$. The global pair distribution is reproduced by a convex mixture of inner–inner (II), inner–outer (IO) and outer–outer (OO) histograms with weights set by the observed ($n_{\text{in}}, n_{\text{out}}$) statistics.

confinement; angular order accounts for a growing remainder as the Wigner molecule forms. This is the same transition the bond-order ratios of Table F.4 record, measured on a different quantity and on held-out data.

How firmly $\Delta \cos$ is determined. The analysis code, restored from the notebook that produced this table, reproduces every published $\Delta \cos$ to six decimals, and two further tests were run on it. The first is a block bootstrap: contiguous blocks of both the training and the held-out frames are resampled, the shell model is refitted and a fresh model sample drawn in every replicate, so the interval covers sampling noise in the data, uncertainty in the fitted parameters and Monte Carlo noise in the model. Over 200 replicates the 95% intervals are 0.0002–0.0036 wide and exclude zero everywhere, including $N=20$ at $\omega = 1$, [0.0027, 0.0030]. The growth with $1/\omega$ is resolved: at $N=6$ all five intervals are disjoint, and at $N=12$ and $N=20$ only the two strongest confinements touch.

The second test varies the analysis choices. Halving or doubling the number of histogram bins leaves $\Delta \cos$ unchanged at the fourth decimal, and swapping or interleaving the training and held-out halves moves it by at most 0.0003. The shell-detection threshold matters only where it changes the partition: lowered far enough to split the single $N=20$ ring into a spurious (1, 3, 16) at $\omega = 0.5$ and 0.1, or to give (1, 3, 8) at $N=12, \omega = 0.01$, it lowers $\Delta \cos$ by 10–16%, and in that variant the $N=20$ value at $\omega = 0.5$ drops just below the one at $\omega = 1$. The largest dependence is on the angular-jitter model: scaling the fitted jitter by 0.75 or 1.25 moves $\Delta \cos$ by up to +34% and –22%, but in both variants it stays positive and grows monotonically as the trap weakens, at every particle number. The size of $\Delta \cos$ is therefore model-dependent; its sign and its trend are not. The code, the replicates and every variant are in `scripts/wigner_null_bootstrap.py` and `results/analysis/2026-09-12.wigner_null_bootstrap/`.

Table F.5: Shell-model fit to the pair-distance distribution. The empirical $p_{\text{test}}(r_{ij})$ on held-out frames is compared with (i) a *polygon* model, in which each detected shell is a regular n_s -gon with a per-frame global rotation and fitted angular jitter, and (ii) a *uniform-angle* null with the same fitted radial distributions and no polygonal order. $\Delta \cos = \cos_{\text{poly}} - \cos_{\text{uni}}$ is what angular order buys; L_{poly}^2 is the polygon model’s residual.

N	ω	occ.	$\cos_{\text{poly}}$	$\cos_{\text{uni}}$	$\Delta \cos$	L_{poly}^2
6	1.0	(1, 5)	0.9988	0.9866	0.0122	4.69×10^{-3}
6	0.5	(1, 5)	0.9986	0.9836	0.0150	5.09×10^{-3}
6	0.1	(1, 5)	0.9974	0.9741	0.0234	6.83×10^{-3}
6	0.01	(1, 5)	0.9922	0.9536	0.0386	1.14×10^{-2}
6	0.001	(1, 5)	0.9942	0.9358	0.0585	1.04×10^{-2}
12	1.0	(1, 11)	0.9993	0.9937	0.0056	3.36×10^{-3}
12	0.5	(1, 11)	0.9990	0.9927	0.0063	4.01×10^{-3}
12	0.1	(1, 11)	0.9980	0.9896	0.0085	5.43×10^{-3}
12	0.01	(1, 11)	0.9929	0.9805	0.0124	9.92×10^{-3}
12	0.001	(3, 9)	0.9868	0.9604	0.0264	1.34×10^{-2}
20	1.0	(1, 19)	0.9992	0.9964	0.0028	3.43×10^{-3}
20	0.5	(1, 19)	0.9990	0.9958	0.0032	3.78×10^{-3}
20	0.1	(1, 19)	0.9982	0.9942	0.0040	4.95×10^{-3}
20	0.01	(1, 19)	0.9947	0.9900	0.0047	8.21×10^{-3}
20	0.001	(1, 7, 12)	0.9915	0.9820	0.0095	1.02×10^{-2}

F.4 The measured shell radius against the classical prediction

The classical minimum-energy configuration of six charges in a parabolic trap places five of them on a ring of radius $1.334\omega^{-2/3}$ [3, 4]. Nothing in the ansatz, the training objective or the sampler knows that number, so the fitted outer-shell radius of Table F.4 is an external check on the whole structural picture—and a sharper one than the exponent fits of §6.2.4, because it tests a prefactor as well as a power.

Table F.6: Fitted outer-shell radius of the $N=6$ (1, 5) molecule against the classical Wigner ring $r_{\text{cl}} = 1.334\omega^{-2/3}$. Radii in Bohr.

ω	r_{shell}	r_{cl}	ratio
1.0	1.66	1.33	1.247
0.5	2.52	2.12	1.188
0.1	6.80	6.19	1.098
0.01	29.65	28.74	1.032
0.001	135.67	133.40	1.017

The ratio falls monotonically toward unity, reaching 1.017 at $\omega = 10^{-3}$: the quantum ring sits 1.7% outside the classical one at the weakest trap, and the excess grows smoothly with confinement as zero-point motion pushes the shell outward. Fitting $r_{\text{shell}} \propto \omega^\alpha$ gives $\alpha = -0.637$ over the full grid and -0.650 over the weakest three confinements, against the classical $-2/3$.

Two remarks on why this quantity, rather than the r_{mode} of Table 6.3, is the right one to compare. First, r_{mode} is the mode of the *single-particle* radial density, which for a (1, 5) topology mixes the central electron with the ring and is therefore not the ring radius; fitting it gives $\alpha \approx -0.58$ at $N=6$ and -0.54 at $N=2$ over the full grid, and approaches $-2/3$ only on the weak-trap half. Second, the shell-resolved radius is the quantity the physics-informed proposal of §7.4 actually needs, so the agreement recorded here is what licenses building a sampler out of it.

Chapter G

The cost of a training step

The main text argues that removing the Markov chain from the gradient changes what a training step costs. This appendix times it. All numbers are wall-clock seconds per gradient step on one NVIDIA RTX 2080 Ti in double precision, on a node whose eight GPUs were shared with other users running at 67–100% utilisation throughout; they include that contention, and ratios are more reliable than absolute values. Networks are at initialisation, since the trained state does not change what a step costs, and $\omega = 1$. The grid of Table G.2 averages three steps after a discarded first one; the production collocation epochs vary by about 2% over eight epochs, and the production SR step is timed after its burn-in call, where consecutive steps agree to 0.1%. The scripts are `scripts/bench_production_sr.py`, `bench_walltime.py` and `bench_score_matrix.py`.

G.1 The two production pipelines

Table G.1 times the pipelines as they produced the results of this thesis. The SR-VMC step is the production routine with the settings of the example run: 1200 persistent walkers and five 20-sweep Metropolis segments per step, for 100 sweeps per walker and about 3.6×10^3 retained rows, with the exact Laplacian. Its phases are timed inside the routine without changing it. The collocation epoch is the robust recipe of §7.3.4: 16×4096 candidates from the Gaussian mixture, weighted by $|\Psi|^2/q$ and resampled to 4096, the exact-Laplacian local energy as a REINFORCE reward, and Adam. The two pipelines build their networks differently (108k parameters for the SR-VMC step, the ~ 49 k collocation ansatz of §4.5.3), so this table compares pipelines, not paradigms.

Table G.1: Wall time per step of the two production pipelines at $N=6$. The SR-VMC step’s first call adds a one-off burn-in of 1000 sweeps (its chain then takes 89 s); that call is excluded here.

	Markov chain	local energy	score matrix	total
SR-VMC step (100 sweeps $\times$ 1200 walkers)	8.5	12.2	495.2	516.4
collocation epoch (16 $\times$ 4096 candidates)	—	(not separated)		21.5

A production collocation epoch takes 3.1 s at $N=2$ and about 115 s at $N=12$ on the same hardware.

The production SR-VMC step is twenty-four times slower, and the Markov chain accounts for 1.6% of it. Ninety-six percent is the score matrix O that SR needs. The production routine builds it one row at a time by backpropagating each sample’s $\log |\Psi|$ through the graph of the whole walker batch, so each row costs roughly a backward pass over the entire batch.

G.2 Paradigm and optimiser, on one network

To separate the main timing contributions, Table G.2 runs three routes on one network—VMC with Adam, collocation with Adam, and VMC with SR—and times two SR implementations. All use a CTNN V-cycle correlator and CTNN backflow (108k parameters) with the same exact-Laplacian local energy. VMC uses

3072 walkers and 30 sweeps per step, collocation 4096 points resampled from 16×4096 candidates, SR 1024 samples. At $N=20$ both paradigms use 1024 configurations, because the shared cards did not have the memory for more. The SR score matrix is built either by the per-sample loop of the thesis’s code or by a vectorised per-sample gradient (`torch.func`), which returns the same matrix to a relative difference below 10^{-14} .

Table G.2: Seconds per step on one network. Entries marked * are composed from the Adam route’s sampling and local-energy times and the separately timed score matrix and solve; at $N=6$, where both were measured, composition and measurement agree to 1%.

N	Adam		VMC with SR	
	VMC	collocation	per-sample loop	vectorised
2	5.1	4.2	26.4	7.6*
6	16.3	17.6	62.1	18.6*
12	64.7	—	185*	66*
20	75.4	67.0	387*	79*

Three things follow. At the same optimiser the paradigm changes the cost of a step by about ten percent: at $N=6$ the chain takes 5.5 s of a VMC step and the importance weights 3.5 s of a collocation step, while the local energy takes 10–13 s of either. Collocation’s weights are not free, since every candidate needs $|\Psi|$ evaluated; per retained point they cost half of what the chain costs here (16 evaluations against 30 sweeps). The local energy is 62–84% of an Adam step from $N=6$ up, so a cheaper Laplacian would speed training more than any change of sampler. And the SR overhead is almost entirely the per-sample loop: building O for 1024 samples takes 19, 46, 121 and 312 s at $N = 2, 6, 12, 20$ by the loop and 2.0, 1.8, 2.4 and 3.8 s vectorised, with the Gram matrix and its solve another 1.2 s. Vectorised, SR costs 1.0–1.5 times an Adam step.

G.3 What per-step timing does not measure

A step’s wall time counts work, not information. The Markov chain’s real cost is statistical: the burn-in, paid once per run and again whenever the state moves far; the correlation between successive samples, which makes a batch of walkers worth fewer independent points than its size; and slow mixing between well-separated configurations, which is worst in the Wigner regime. None of that appears above, and none of it applies to independently drawn candidates. The measure that would capture it is time per *independent* sample, the per-sweep cost times the chain’s integrated autocorrelation time on a trained weak-confinement state, which we did not measure. Nor does the cost of a step say how many steps each route needs.

Chapter H

Code, data and reproducibility

Code. The source code used for the ansatz, both training routes, the structural and tangent-space diagnostics, the evaluation scripts and the figure scripts is public at <https://github.com/Im2q14u/Thesis>, together with retained run configurations, model metadata and the derived tables from which the numbers in Chapters 6 and 7 are quoted. The text alone, with the figures it includes, is at https://github.com/Im2q14u/Thesis_text.

Models and raw data. Trained weights and the $|\Psi|^2$ sample bundles from which every structural diagnostic is computed are too large for the repository. They are kept on the University of Oslo machine-learning nodes where the calculations ran and are available on request.

Seeds and initialisation. Every run records its seed in its metadata, and seeds are stated in the text wherever a claim rests on seed variation. The collocation runs are warm-started, and two facts limit how far they can be repeated from the current code. The canonical pretrained initialisation behind the continuation chains was lost in a repository rewrite in June 2026, so those runs can be re-evaluated from their saved checkpoints but not retrained bit for bit. And the Jastrow cusp convention changed in May 2026, from a fixed decay length to the oscillator length; checkpoints trained before the change must be evaluated with the convention they were trained with, which the evaluation script infers from the checkpoint metadata.

The importance-sampling error and what was checked after it. In March 2026 the collocation sampler was found to return, for each proposal draw, the log-density of the Gaussian component that generated it rather than the log-density of the whole mixture. The resulting error in the log-weights grows with the dimension Nd and with the spread of the component widths: negligible at $N=2$, $\omega = 1$, it reached a factor of order e^{12} in the importance weights at $N=6$, $\omega = 0.01$, which is why its signature—loss of accuracy at larger N and weaker confinement—was indistinguishable from an architectural limit. The error sat in the importance weights and therefore only in the collocation route; the production energies of Chapter 6 are evaluated by Metropolis sampling of $|\Psi|^2$, which does not use those weights. After the correction (26 March 2026) three things were checked. The collocation results reported here come from runs made after it, or, for $N=12$, from continuation runs whose final training stage used the corrected weights. Every $N=6$ and $N=12$ collocation energy in Tables 7.5 and 7.7 was independently re-evaluated for this thesis by Metropolis sampling, with four separately seeded chains per checkpoint and a blocking analysis (`scripts/eval_collocation_blocked.py`). And the structural analysis, whose original code had been lost with a deleted notebook, was recovered and restored (`src/wigner_structure.py`); it reproduces every published $\Delta \cos$ to six decimals.

Known gaps. Two quantities cannot be regenerated from the saved outputs: the cross-shell registration statistic $\hat{C}_{\text{global}}$ at $N=20$ (§6.2.5), whose analysis code was not retained, and the $N=20$, $\omega = 0.01$ entry of the backflow rank table (Table 7.3), whose per-run artefacts were not kept. The former is omitted from the quantitative interpretation because its phase convention cannot be verified. One inconsistency also remains: the single-particle radial diagnostics of Table 6.3 and Appendix F were computed in an earlier analysis pass, on runs that at the weakest traps are not the same trained states as those behind the shell-resolved diagnostics and the radial-density figures.

Chapter I

Declaration on the Use of Artificial Intelligence

I used generative AI tools (ChatGPT, Claude and OpenAI Codex) during this thesis, from March to September 2026, in three ways:

- to revise and edit the text, including drafting revised passages, which I then reviewed, rewrote where needed and approved;
- to debate results, possible explanations and next steps;
- to patch and debug code. Commits made through these tools are marked as co-authored in the repository history.

The research questions, the scientific decisions and the conclusions are my own. No numerical result was taken from generated text: reported values come from the cited literature or from my saved calculations. I checked suggestions against derivations, sources, code or saved results before using them, and rejected those I could not support. I take full responsibility for the content, claims and references of this thesis.

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